

MOH/P/PAK/391.18(GU)

CLINICAL PRACTICE GUIDELINES

Management of Hypertension

5TH EDITION (2018)



Malaysian Society of Hypertension



Ministry of Health Malaysia



Academy of Medicine of Malaysia

STATEMENT OF INTENT

This guideline was developed to be a guide for best clinical practice in the management of hypertension. All efforts were made to ensure references quoted were the most current at the time of printing. Specific attempts were made to use local data and publications to ensure local relevance. Adherence to this guideline may not necessarily lead to the best clinical outcome in individual patient care. Every health care provider is responsible for the care of his/her unique patient based on the clinical presentation and treatment options available locally. However, adherence to this guideline is strongly recommended as a starting point in managing patients as it constitute the best available evidence at the time of writing.

REVIEW OF THE GUIDELINES

This guideline was issued in 2018 and will be reviewed in 2023 or earlier if important new evidence becomes available. This is an update to the Clinical Practice Guideline on Management of Hypertension – 4th Edition (published 2013) and supersedes the previous. Electronic version will be made available on the following websites:

www.moh.gov.my
www.acadmed.org.my
www.msh.org.my

DISCLOSURE STATEMENT

The panel members had completed disclosure forms. None held shares in pharmaceutical firms or acted as consultants to such firms. Some may have been engaged as speakers in conferences or Continuing Professional Development activities mainly organised by the Malaysian Medical Association, the Malaysian Society of Hypertension, National Heart Association of Malaysia or similar professional non-governmental associations (NGOs). These events may or may not have received financial assistance from pharmaceutical companies as part of an educational grants (details are available upon request from the CPG Secretariat).

SOURCE OF FUNDING

The development of the CPG on Management of Hypertension (5th Edition) was supported via unconditional educational grant from Servier Malaysia Sdn. Bhd. The funding body was not involved in and has no influence on the development of the guidelines. An independent third party was engaged for all secretarial task and was appointed by and reported directly to the Malaysian Society of Hypertension.

Key Messages

1. Hypertension is defined as persistent elevation of systolic BP of 140mmHg or greater and/or diastolic BP of 90 mmHg or greater.
2. The prevalence of hypertension in Malaysians aged 18 years and above was 35.3% in 2015, a slight increase from 33.6% in 2011.
3. Hypertension is a silent disease; unfortunately, in 2015, for every two diagnosed patients in Malaysia there are 3 undiagnosed patients. This has not changed since 2011. Blood pressure should be measured at every chance encounter.
4. Untreated or sub-optimally controlled hypertension leads to increased cardiovascular, cerebrovascular and renal morbidity/ mortality and overall mortality.
5. A systolic BP of 120 to 139 mmHg and/or diastolic BP of 85 to 89 mmHg is defined as 'at risk blood pressure' and should be treated in certain high risk groups.
6. Healthy living should be recommended for all individuals with hypertension and 'at risk blood pressure'.
7. Decisions on pharmacological treatment should be based on global vascular risks and not on the level of blood pressure per se.
8. In patients with newly diagnosed uncomplicated hypertension and no compelling indications, choice of first line monotherapy includes ACEIs, ARBs, CCBs, diuretics and β -blockers.
9. Only 37.4% of Malaysian patients achieved blood pressure control (<140/90 mmHg) while on treatment. Although this is an improvement from 2011 (34.7%) every effort should be made to achieve target blood pressure in all patients. Target blood pressure depends on specific patient groups.
10. Combination therapy is often required to achieve target and may be instituted early in patients with stage II hypertension and in high risk stage I hypertension.
11. A patients whose BP is not controlled on three or more drugs (including a diuretic) is by definition having resistant hypertension.

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Foreword

In the Name of Allah, the Most Beneficent, the Most Merciful.

In 2015 the Ministry of Health released data from the National Health and Mortality Survey which focused on Non Communicable Diseases. Diseases of the heart and circulatory system (Cardiovascular diseases or CVD) still dominates the national health landscape being the number 1 cause of morbidity and mortality for the last few decades and is projected to do so for years to come. Of all the risk factors contributing to CVD, hypertension confer the greatest risk for both male and female based on the latest national burden of disease in 2008. It is thus pertinent that all health care providers directly or indirectly involved with CVD know what is latest in the management of hypertension.

I will like to record my utmost appreciation to all the members of the Working Group on Hypertension for their tireless effort in coming up with this latest edition of the Hypertension Clinical Practice Guideline. This is the fifth in the series since it was first launched in 1998. This reflects the rapid evolution of knowledge in hypertension driven by major outcome trials for which there were a few since the last edition 5 years ago. There are also more Malaysian studies quoted which is a testimony of the growing research interest in the topic nationally. I am happy to report that in some of the landmark multicentre clinical trials quoted, Malaysian researchers were actively involved. A special thanks to the Health Technology Assessment Unit of the Ministry of Health Malaysia for ensuring that the development of this CPG conforms to the high standards it had laid down.

Although the NHMS 2015 showed some positive developments in important key indicators on hypertension, there is still a lot of scope for improvement. It is hoped that this latest edition of the Hypertension CPG will continue to play an important role in controlling this major CVD risk factor. It is the hope of the Working Group that the release of this new edition will be followed by concerted effort by various stakeholders to implement the recommendations made. By so doing, we will have contributed in a significant way to combat the scourge of CVD particularly pre mature CVD. If that happens, this CPG will have served its purpose, God Willing.

Yours Sincerely,



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Rationale and Process of Guidelines Development

RATIONALE

The Clinical Practice Guideline on the Management of Hypertension was developed to provide a clear and concise approach to all health care providers on the current concepts in the management of hypertension. Since hypertension is managed by various levels of health care providers in Malaysia, attempts were made to ensure the different stakeholders will benefit from this CPG. This is reflected by the representation of the committee members who developed the guideline. There were four previous guidelines on hypertension; in 1998, 2002, 2008 and 2013. This edition is the fifth in the series and was deemed necessary due to new evidence which has emerged since the last edition. Prior to the publication of this edition, the National Health and Morbidity Survey 2015 was published. The results of the survey showed that the prevalence of hypertension has increased while the awareness has decreased compared to a similar survey done in 2011. However the rate of blood pressure control in the hypertensive population has increased by 10% (now 37.4% of treated patients are controlled). This rate of blood pressure control is still poor when one compares with a large nation like Canada where the control rate was more than 80%. This may reflect the fact that clinicians are still not clear of the target blood pressure to achieve in their patients. It is hoped that this CPG will contribute towards achieving the desired targets.

GUIDELINE DEVELOPMENT PROCESS

The members of the Development Group (DG) for this Clinical Practice Guideline (CPG) were from the Ministry of Health (MOH), Ministry of Higher Education (MOHE) and private healthcare providers. The membership of the DG was multidisciplinary and most specialties are represented by at least 2 experts. These are Cardiovascular Medicine, Nephrology, Geriatrics, Obstetrics and Gynaecology, Family Medicine and Clinical Pharmacology/Clinical Pharmacy. There are also specialties which are represented by 1 expert. These include Endocrinology, Neurology and Public Health. Where there are at least 2 members, 1 act as the principal author and the other reviewer of the initial draft. Each draft will then be reviewed collectively in every DG meeting.

The CPG update was done based on the CPG Management of Hypertension 4th edition of 2013. In the update, systematic review methodology was used and the scope covered include epidemiology and public health, definition and classification, blood pressure management, diagnostic criteria, investigations, global cardiovascular risk evaluation, general principle of management, non pharmacological and pharmacological management, management of patient sub groups and approach to resistant and refractory hypertension. Emerging areas in hypertension are also covered including Health Economics, Device Based treatment and scope for future research. A literature

search was carried out using the following electronic databases: mainly Medline via Ovid and Cochrane Database of Systemic Reviews and others e.g. Pubmed and Guidelines International Network. The search was done on Published English literature focusing mainly but not exclusively on Clinical Trials. Important observation studies, where relevant, were also looked at. Unlike some CPGs, the search was not limited to literature published in the last fifteen years. This is because hypertension management has been powered and driven by good evidence generated over the last thirty years. Some seminal and practice changing trials were conducted a few decades ago and the result so conclusive that repeating the trials will be an act of futility. In addition, the reference lists of all recent Hypertension Guidelines i.e. that written over the last 5 years only were retrieved and searched to further identify relevant studies. All searches were conducted from 24 May 2017 to 5 January 2018. Literature searches were repeated for all clinical questions at the end of the CPG development process allowing any relevant papers published after January 2018 to be included. Future CPG updates will consider evidence published after this cut-off date.

References were also made to the most recent CPG on Hypertension from the American College of Cardiology / American Heart Association released in full in December 2017. This CPG was evaluated using the Appraisal of Guidelines for Research and Evaluation (AGREE) II prior to them being used as references.

A total of 23 clinical questions were developed under 13 different sections. Members of the DG were assigned individual questions within these sections (refer to Appendix 5 for Clinical Questions). The DG members met 9 times throughout the development of these Guideline. All literature retrieved were appraised by at least two DG members using Critical Appraisal Skill Programme checklist, presented in evidence tables and further discussed in each DG meetings. All statements and recommendations formulated after that were agreed upon. Where evidence was insufficient, the recommendations were made by consensus of the DG. This CPG is based largely on the findings of systematic reviews, meta-analyses and clinical trials in particular trials where Malaysia participated. Where evidence are lacking or non-existent, local practices are taken into consideration.

The literature used in these guidelines were graded using the US/Canadian Preventive Services Task Force Level of Evidence (2001), while the grading of recommendation was done using the principles of Scottish Intercollegiate Guidelines Network (Version 1.0 updated July 2017). The writing of the CPG follows strictly the requirement of AGREE II.

On completion, the draft of the CPG was reviewed by external reviewers both nationally and internationally. It was also posted on the Malaysian Society of Hypertension (MSH) official website for feedback from any interested parties. The draft was finally presented to the Technical Advisory Committee for CPG, and the HTA and CPG Council MoH Malaysia for review and approval. Details on the CPG development methodology by MaHTAS can be obtained from Manual on Development and Implementation of Evidence-based Clinical Practice Guidelines published in 2015 (available at <http://www.moh.gov.my/index.php/pages/view/117>).

Objectives, Questions and Targets

OBJECTIVES

This guideline is intended to provide education and awareness on the proper ways to:

1. diagnose hypertension
2. assess and investigate a patient with hypertension

This guideline is intended to provide evidence on the:

1. optimal management of a patient with hypertension
2. latest therapeutics on subgroups of hypertensive patients

EXCLUSION

This guideline, however, does not cover:

1. strategies for hypertension screening
2. strategies to reduce population blood pressure

CLINICAL QUESTIONS

The two major clinical questions to be addressed in this guideline include:

1. What are the current best practices in the management of a patient with hypertension?
2. How can hypertension management be done in tandem with the overall strategy to manage global vascular risk of a patient?

For further detail, please refer Appendix 5.

TARGET POPULATION

This guideline is to be applied to adults (including the elderly and pregnant women) and children with hypertension. It is also applicable to hypertensive patients with various concomitant clinical conditions.

TARGET GROUP

This guideline is developed for all levels of health care providers involved in the management of hypertension in adults, elderly, pregnant women and children.

CLINICAL INDICATORS FOR QUALITY MANAGEMENT

Treatment setting:

Primary care / Secondary care

Name of indicator:

1. Rate of anti-hypertensive prescription for newly diagnosed cases of hypertension
2. Rate of blood pressure control among patients who are treated with antihypertensive drugs

Definition of control:

- <140/90 mmHg for all
- <140/80 mmHg for patients with diabetes
- <130/80 mmHg for patients with ischaemic heart disease/ cerebrovascular disease/renal impairment

Numerator:

1. Number of newly diagnosed cases of hypertension prescribed anti-hypertensive drugs
2. Number of patients on treatment who achieved blood pressure control

Denominator:

1. Total number of newly diagnosed cases of hypertension
2. Total number of patients who are diagnosed and on anti-hypertensive drug treatment

Rate of treatment = (Numerator/Denominator) x 100%

Rate of blood pressure control = (Numerator/Denominator) x 100%

LEVEL OF EVIDENCE

LEVEL	STUDY DESIGN
I	Evidence from at least one properly randomised controlled trial
II-1	Evidence obtained from well-designed controlled trials without randomisation
II-2	Evidence obtained from well-designed cohort or case-control analytic studies, preferably from more than one centre or group
II-3	Evidence from multiple time series with or without intervention. Dramatic results in uncontrolled experiments (such as the results of the introduction of penicillin treatment in the 1940s) could also be regarded as this type of evidence
III	Opinions of respected authorities based on clinical experience; descriptive studies and case reports; or reports of expert committees

Source: US/Canada Preventive Services Task Force

GRADES OF RECOMMENDATION

A	At least one meta analysis, systematic review, or RCT, or evidence rated as good and directly applicable to the target population
B	Evidence from well conducted clinical trials, directly applicable to the target population, and demonstrating overall consistency of results; or evidence extrapolated from meta analysis, systematic review, or RCT
C	Evidence from expert committee reports, or opinions and/or clinical experiences of respected authorities; indicates absence of directly applicable clinical studies of good quality

Source: Modified from the Scottish Intercollegiate Guidelines Network (SIGN)

Note: The grades of recommendation relate to the strength of the evidence on which the recommendation is based. It does not reflect the clinical importance of the recommendation.

Summary of Recommendations

ISSUES	RECOMMENDATIONS	GRADE
Measurement of Blood Pressure	Measure BP at every opportunity as a high number of Malaysians are undiagnosed.	B
	Check BP for every adult above age 18 years at least once as part of their annual health screening, and more frequently for those who are at risk (family history, obese and those at-risk of high blood pressure).	B
	Use electronic blood pressure measurement devices instead of mercury sphygmomanometers despite the latter being a gold standard for non-invasive measurement. (NEW)	C
	Before measuring BP, check if patients have stopped smoking, eating, drinking caffeinated drinks, have not been exercising for at least 30 minutes, be seated for at least 1 min in a quiet room, back & arm supported, (e.g. resting on the table), be seated with legs uncrossed, stopped talking and is relaxed.	C
	Check and use the correct bladder cuff size with placement at heart level.	C
	Encourage patient to do home BP monitoring (HBPM) which helps to empower them and may improve medication adherence.	C
	Use ambulatory BP monitoring (ABPM) only in selected clinical situations (e.g. to confirm isolated office, masked and labile hypertension).	C
Diagnosis and Initial Assessment	Assess initial BP measurement results and global CV risk before deciding on the appropriate follow-up required. (NEW)	C
Non-Pharmacological Management	BMI or Weight Achieve a weight loss of as little as 1kg from baseline to reduce blood pressure by 1 mmHg SBP. (NEW)	A
	Salt Intake Reduce salt intake to <2g of sodium or <5g of salt a day (equivalent to 1 teaspoonful of salt). (NEW)	A

ISSUES	RECOMMENDATIONS	GRADE
Non-Pharmacological Management (Continued)	Alcohol Refrain from alcohol intake. Advise patient who insists to continue drinking to consume $\leq$ two drinks per day.	A
	Exercise Advise patients to perform physical activity (e.g. moderate intensity aerobic exercise of at least 150 minutes per week).	A
	Diet Encourage diet rich in fruits, vegetables and dairy products with reduced saturated and total fat.	A
	Smoking Stop smoking to reduce overall cardiovascular risk.	C
	Relaxation Encourage patient to manage stress although evidence on relaxation interventions have not been convincing.	C
Pharmacological Management	Treat most patients with pharmacological agent life-long.	C
	Choose mono-therapy in patients with stage 1 hypertension and with no compelling indication from one of the 5 classes of drug (ACEIs, ARBs, CCBs, Diuretics or β -blockers) based on patient's individual clinical profile. (NEW)	C
	Choose combination therapy in patients with medium/high/very high risk stage 1 hypertension and stage 2 hypertension. (NEW)	A
	Treat BP to SBP <140 mmHg and DBP <90 mmHg for most hypertensive patients.	A
	Treat SBP to <130 mmHg and DBP <80 mmHg for high/very high risk patients.	A
	Use combination therapy (free or single pill) for most patients to achieve BP control.	A
	Arrange periodic scheduled visits to assess global CV risk, emerging new risk factors and organ damage/complications.	C
	Co-manage patients whose BP are controlled with primary care facilities (Klinik Kesihatan or private General Practice).	C

ISSUES	RECOMMENDATIONS	GRADE
Management of Severe Hypertension	Hypertensive Urgencies	
	Do not reduce BP rapidly (within minutes to hours) in hypertensive urgencies as it may precipitate ischaemic events.	C
	For patients whose BP responded with adequate rest (after 2 hours), discharge them with Hypertensive Urgency Discharge Plan. (NEW)	B
	For patients whose BP do not respond to adequate rest, start with combination oral pharmacotherapy targeting a BP reduction of 25% within 24 hours.	C
	Hypertensive Emergencies	
	Reduce BP by 10%-25% within minutes to hours but not lower than 160/100 mmHg. This is best achieved with parenteral drugs. (NEW)	B
	Reduce SBP to less than 140 mmHg during the first hour for patients with severe preeclampsia or eclampsia, and pheochromocytoma crisis. Reduce to less than 120 mmHg for patients with aortic dissection.	C
	Reduce BP by no more than 25% within the first hour; then, if stable, to 160/100 mmHg within the next 2 to 6 hours; and then cautiously to normal during the following 24 to 48 hours in all other situations.	C
Hypertension and Diabetes Mellitus	Initiate drug treatment if BP is consistently >140/80 mmHg.	A
	Use ACEIs in diabetes without proteinuria. Use ARB for ACEI intolerant patients.	A
	Use ACEIs or ARBs in patients with diabetes and proteinuria.	A
	Consider CCBs, diuretics or β -blockers if RAS blockers cannot be used.	C
	Aim for BP in the diabetic to be <140/80 mmHg.	A
	Consider to lower BP <130/80 mmHg in younger patients.	C
Hypertension and Renal Diseases	Patients with proteinuria of <1 g/24 hours, lower BP to <140/90 mmHg. (NEW)	A
	In patients with proteinuria of >1g/24 hours, lower BP to <130/80 mmHg. (NEW)	A

ISSUES	RECOMMENDATIONS	GRADE
Hypertension and Renal Diseases (Continued)	In patient >50 years, GFR >20 ml/min/1.73m ² and proteinuria <1g/day lower SBP <120 mmHg using Automated Self-measured Office BP to reduce cardiovascular event. (NEW)	B
	Choose RAS blockers as initial antihypertensive therapy for patients with micro- or macroalbuminuria.	A
	Add non-dihydropyridine CCBs if BP goal is still not achieved and there is persistent proteinuria.	A
	Consider concurrent diuretic therapy and dietary salt restriction as salt and water retention are important determinants of hypertension in CKD.	B
	Avoid dual RAS blockade in patients with CKD. (NEW)	A
Hypertension and Heart Diseases	Use β -blockers, ACEIs or ARBs in post myocardial infarction patients to reduce recurrent myocardial infarction and death.	A
	Initiate β -blockers, ACEIs and Aldosterone antagonists in patients with systolic heart failure (HFrEF) to reduce morbidity and mortality.	A
	Use ARBs or ACEIs and aldosterone antagonist in heart failure patients with preserved ejection fraction (HFpEF) to reduce morbidity including hospitalisation.	A
	Use RAS blockers in patients >75 years old with AF to reduce mortality. (NEW)	B
	Use any antihypertensive except β -blockers as first choice in patients with PAD.	C
	Give ACEI to patients with PAD to prevent vascular events. (NEW)	B
	Consider cilostazol in the elderly patients with symptomatic CAD and concurrent PAD. (NEW)	B
	Use ARBs as treatment of choice in hypertensive patients with LVH on ECG.	A
	Treat blood pressure to <140 / <90 mmHg in patients with concurrent IHD, peripheral arterial disease (PAD), PAD with/without AF. (NEW)	B
	Treat blood pressure to <130/80 mmHg in patients with LVH. (NEW)	B
	Prescribe antiplatelet agent unless contraindicated.	A

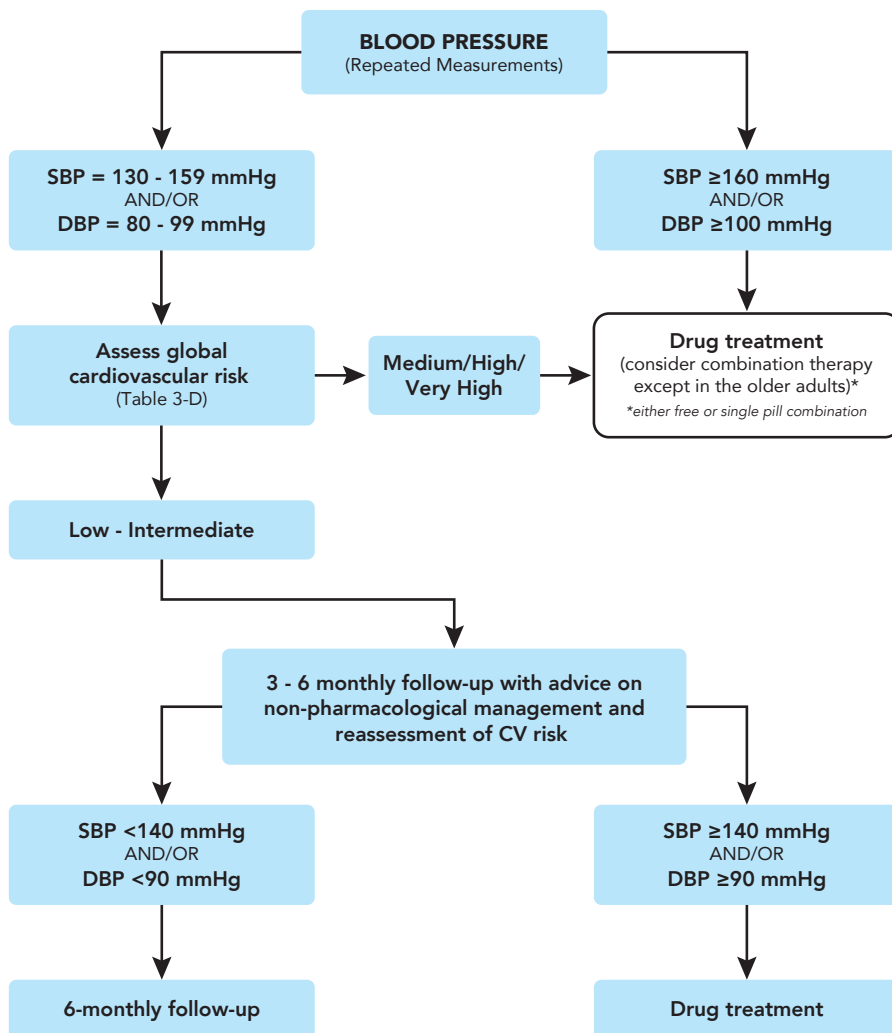
ISSUES	RECOMMENDATIONS	GRADE
Hypertension and Stroke	Treat BP to prevent both primary and secondary stroke.	A
	Treat BP to <140/90 mmHg for primary prevention.	B
	Lower BP to be <140/90 mmHg in both normotensive and hypertensive patients for secondary prevention. Combination of ACEI and diuretics is preferred for secondary prevention.	A
	Lower BP to <130/80 mmHg for secondary prevention in lacunar stroke. (NEW)	A
	Do not lower SBP <180 mmHg in the first 2 weeks in acute ischaemic stroke patients unless hypertensive emergencies co-exist. (NEW)	C
	Avoid lowering BP abruptly with sublingual nifedipine in acute stroke.	C
	Do not lower SBP to <140 mmHg in patients presenting within 6 hours of haemorrhagic stroke (HS) and presenting SBP of <220 mmHg.	C
	Consider aggressive reduction of BP in HS patients presenting with SBP ≥220 mmHg with continuous intravenous infusion of antihypertensive and frequent BP monitoring. (NEW)	C
Hypertension in The Older Adults	Measure standing BP and use it to guide treatment decision. (NEW)	C
	Assess comprehensively to confirm hypertension. (NEW)	C
	Assess for frailty, mobility, function, cognition, nutrition, postural hypotension and falls. (NEW)	C
	Individualised treatment based on clinical scenarios. (NEW)	C
	Target SBP <150 mmHg for >80 year olds.	A
	Target SBP <140 mmHg for 65-80 year olds. (NEW)	B
	Consider SBP <130 mmHg in fit 65-80 year olds. (NEW)	A
	Apply less strict targets for the frail, functionally and/or cognitively-impaired, those with multi-morbidities and those with adverse reactions from therapy. Consider de-prescribing in this group of patients. (NEW)	C

ISSUES	RECOMMENDATIONS	GRADE
Hypertension in Women	Use Korotkoff V to diagnose and monitor treatment of hypertension in pregnant women. (NEW)	C
	Consider automated BP device instead of mercury sphygmomanometer to diagnose and monitor treatment. (NEW)	A
	Provide counselling and appropriate management to women with chronic hypertension and who are planning for pregnancy.	C
	Avoid RAS blockers in all women of childbearing potential unless adequate precaution has been taken against pregnancy. (NEW)	A
	Refer pregnant women with hypertension to the obstetrician for further management.	C
	Provide low dose calcium supplementation (500-1000 mg daily) from early pregnancy to prevent pre-eclampsia. (NEW)	A
	Commence aspirin (100-150 mg and taken at bedtime) from 12-16 weeks and continue until delivery in pregnant women with one or more high risk factors or two or more moderate risk factors for pre-eclampsia. (NEW)	A
	The drugs of choice in pregnancy are still methyldopa (first line) and labetalol (alternative first line) with nifedipine as second line. (NEW)	C
	In an acute hypertensive crisis, use IV labetalol (20 mg slow bolus over 5 minutes followed by 40mg 10-20 minutes later) or continuous infusion of 1-2mg/minute. IV hydralazine (bolus or infusion) is an alternative but do not use it as first line treatment. (NEW)	C
	Use oral nifedipine 10 mg stat dose to rapidly control BP in acute hypertensive crisis prior to transfer to hospital.	C
	Administer parenteral magnesium sulphate as drug of choice for prevention of eclamptic fit.	A
	Hypertension and Oral Contraceptives	
	Advise woman who develop hypertension whilst on combined oral contraceptives (COC) to stop smoking and offer them alternative forms of contraception.	C
	Review BP at least every 6 months.	C

ISSUES	RECOMMENDATIONS	GRADE
Hypertension in Women (Continued)	Hypertension and Hormonal Therapy	
	Monitor BP in normotensive women taking HRT every 6 months	C
	Monitor closely hypertensive women on conjugated equine estrogen (CEE) every 3 months.	C
Hypertension in Neonates, Children and Adolescents	Measure BP at every encounter if the child have risk factors or annually for obese children >7 year old.	C
	Once a child is diagnosed with hypertension, he/she should be referred to a paediatrician for further evaluation and management.	C
	Start non-pharmacologic management especially weight reduction in obese children and in all children with BP of >90 th percentile.	C
	Once pharmacologic therapy is initiated, BP must be reduced to <90 th percentile (Systolic and Diastolic) and <130/80 mmHg in adolescents ≥13 years old. (NEW)	C
	In children and adolescent with CKD, lower BP to <50 th percentile. (NEW)	C
Economic Impact of Hypertension	Conduct more awareness programmes on clinical and economic benefits in prevention and early treatment of hypertension.	B
	Institute behavioral changes especially on medical treatment adherence to reduce complications and long-term healthcare cost. (NEW)	A
Resistant and Refractory Hypertension	Treat patients with at least 3 drugs (inclusive of a diuretic) before diagnosing resistant hypertension.	C
	Consider drug non-adherence and secondary hypertension before diagnosing resistant hypertension or refractory hypertension. (NEW)	C
	Add spironolactone as a fourth drug in resistant hypertension. (NEW)	A
	Consider referring for device based therapy in patients with true resistant and refractory hypertension.	C
Aspirin in Hypertension	Treat patients BP to target first before initiating aspirin therapy.	A
	Consider using aspirin in patients with higher baseline BP.	B

Summary

ALGORITHM FOR THE MANAGEMENT OF HYPERTENSION



Summary

RISK STRATIFICATION

Co-Existing Condition BP Levels (mmHg)	No RF No TOD No TOC	TOD or RF (1-2) No TOC	TOC or RF(≥3) or Clinical atherosclerosis or CKD	Previous MI / IHD, Previous stroke or Diabetes or CKD
SBP 130 - 139 and/or DBP 80 - 89	Low	Intermediate	High	Very High
SBP 140 - 159 and/or DBP 90 - 99	Low	Medium	High	Very High
SBP 160 - 179 and/or DBP 100 - 109	Medium	High	Very High	Very High
SBP >180 and/or DBP >110	High	Very High	Very High	Very High

Risk Level	Risk of Major CV Event in 10 years	Management
Low-Intermediate	<10%	Healthy living
Medium	10 - 20%	Drug treatment and healthy living
High	20 - 30%	Drug treatment and healthy living
Very high	>30%	Drug treatment and healthy living

TOD = Target organ damage (LVH, retinopathy, proteinuria).

TOC = Target organ complication (heart failure, renal failure).

RF = Additional risk factors (smoking, TC >6.5mmol/L, family history of premature vascular disease).

Clinical atherosclerosis = CHD, carotid stenosis, peripheral vascular disease, transient ischaemic attack, stroke.

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1 Epidemiology, Definition and Classification of Hypertension

Hypertension is defined as persistent elevation of systolic blood pressure (BP) of 140 mmHg or greater and/or diastolic BP of 90 mmHg or greater, taken at least twice on two separate occasions.

Although there is an attempt to redefine hypertension as >130 and/or 80 mmHg,¹ this Committee recommends that the old recommendation remains. The Committee is of the opinion the proposed lower definition will not change the way we treat our patients, particularly those with cardiovascular complications with BP equal to or more than 130/80 mmHg needs treatment to lower BP regardless.

Non-Communicable Diseases (NCDs) is already the main cause of death in Malaysia (Figure 1-A) and the biggest contributor in terms of disability life-years (DALYs), with high blood pressure the biggest contributor for both males and females² (Figure 1-B).

FIGURE 1-A Mortality Attributable to Risk Factors, Malaysia 2008²

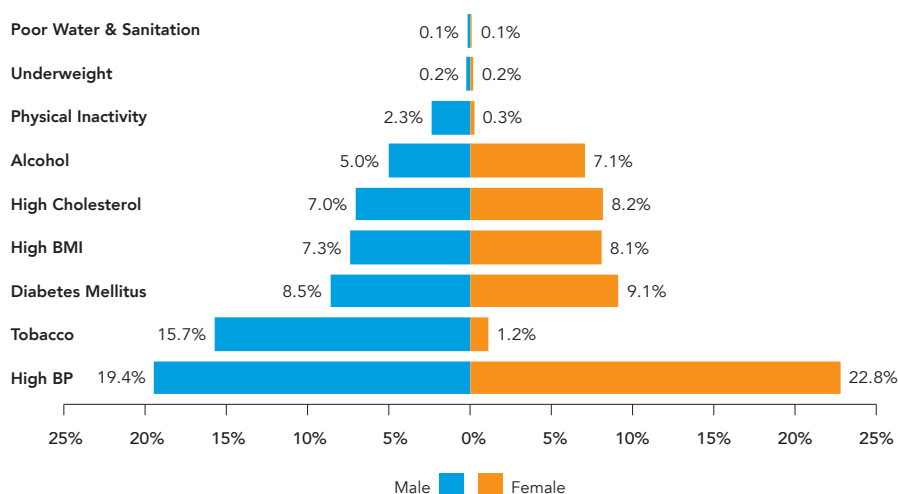
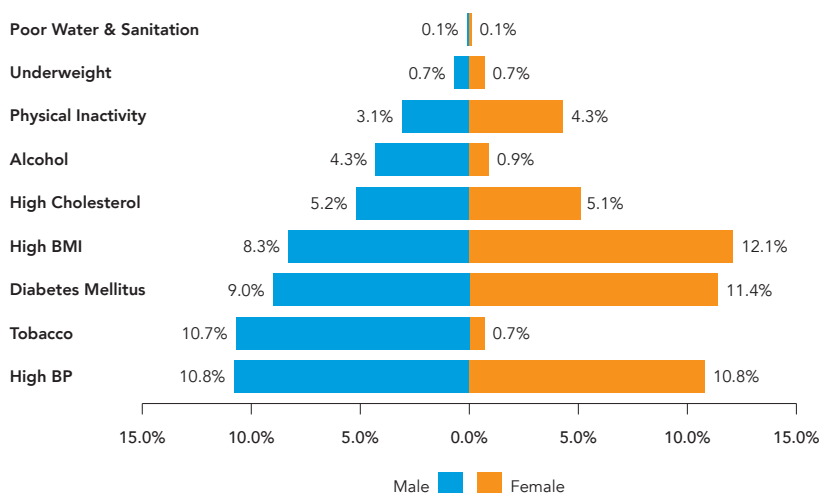


FIGURE 1-B DALYs Attributable to Risk Factors, Malaysia 2008²

The latest National Health and Morbidity Survey (NHMS) for NCD risk factors in 2015³ showed an overall prevalence of hypertension of 35.3% among adults 18 years and above. This is an increase from 33.6% in 2011 as compared to 34.6% in 2006. However, in terms of awareness, only 37.5% were aware in 2015 a drop from 40.7% in 2011. In 2006, the awareness rate was 35.6%.³

No significant difference between gender was observed. There was a general increasing trend in prevalence with age, from 6.7% in the 18-19 years age group, reaching a peak of 75.4% among the 70-74 years age group. Based on the 2015 survey, prevalence was the highest among other Bumiputras (37.3%) followed by Malays (36.4%), Indian (34.9%) and Chinese (34.2%).³

Hypertension was more prevalent in the rural area in all three NHMS: 36.9% versus 32.9% (2006), 36.5% versus 32.6% (2011) and 39.2% versus 34.1% (2015). In terms of gender differences, hypertension was more prevalent among males; 35.3% versus 33.9% (2006), 34.2% versus 33% (2011) and 35.9% versus 34.8% (2015).

The relationship between BP and risk of cardiovascular events is continuous, consistent and independent of other risk factors. The higher the BP, the greater the chance of myocardial infarction, heart failure, stroke and kidney diseases. The presence of each additional risk factor, such as dyslipidaemia, diabetes mellitus or smoking status, compounds the risk. Therefore, the main aim of identifying and treating high BP is to reduce these risks of end organ damage or end organ complications. The classification of clinic BP levels in adults is shown in Table 1-A.

TABLE 1-A Classification of Clinic Blood Pressure Levels in Adults

Classification*	Systolic (mmHg)		Diastolic (mmHg)	Prevalence in Malaysia ³
Optimal	<120	and	<80	30.7
Normal	120-129	and/or	80-84	25.3
At Risk	130-139	and/or	85-89	18.6
Hypertension				
Stage 1 (Mild)	140-159	and/or	90-99	17.3
Stage 2 (Moderate)	160-179	and/or	100-109	5.7
Stage 3 (Severe)	≥180	and/or	≥110	2.4
Isolated Systolic Hypertension	≥140	and	<90	11.2

Home and Ambulatory BP may be used to diagnose and classify elevated blood pressure (Table 1-B) (see section on chapter 2.2 and 2.3).

TABLE 1-B Criteria for Staging Hypertension Based on Clinic, Home and Ambulatory Blood Pressure Monitoring

Category	Clinic BP (mmHg)	Home BP Monitoring Average or Ambulatory BP Daytime Average (mmHg)
Stage I Hypertension	≥140/90	≥135/85
Stage II Hypertension	≥160/100	≥150/95
Severe Hypertension	SBP ≥180 or DBP ≥110	-

Note: Adapted from National Institute for Health and Clinical Excellence (NICE) Hypertension, 2011.⁴

1.1 Isolated Systolic Hypertension

Isolated systolic hypertension (ISH) is defined as SBP of ≥140 mmHg and DBP <90 mmHg. It is common after the age of 50, and carries with it a poor prognosis. Clinical trials have demonstrated that control of ISH reduces total mortality, cardiovascular mortality, stroke and heart failure events.^{5,6,7}

Changing patterns of BP occur with increasing age. The rise in SBP continues throughout life in contrast to DBP, which rises until approximately age 50, tends to level off over

the next decade, and may remain the same or fall later in life.^{8,9} Diastolic hypertension predominates before age 50, either alone or in combination with SBP elevation. The prevalence of systolic hypertension increases with age, and above 50 years of age, systolic hypertension represents the most common form of hypertension. DBP is a more potent cardiovascular risk factor than SBP until age 50; thereafter, SBP is more important.¹⁰

1.2 Isolated Office (“White-Coat”) Hypertension

Isolated office hypertension is characterised by an elevation in clinic blood pressure but normal home or ambulatory blood-pressure values. In these subjects the clinic BP is persistently above 140/90 mmHg but the home or daytime ambulatory systolic/diastolic BP measurements are lower than 135/85 mmHg.

1.3 Masked Hypertension

Patients with masked hypertension have normal clinic blood pressure but elevated daytime ambulatory or home blood-pressure level ($\geq 135/85$ mmHg). Prognosis of masked hypertension is worse than isolated office hypertension.¹²

For both isolated office and masked hypertension, once diagnosed, initial therapeutic interventions should be non-pharmacological and aim for adoption of healthy living. However, drug treatment is indicated, particularly when the patient’s cardiovascular risk profile is elevated or when target-organ damage (TOD) is detected.¹² (Refer to chapter 3 on Diagnosis and Initial Assessment).

SUMMARY

- Hypertension is defined as persistent elevation of systolic BP of 140 mmHg or greater and/or diastolic blood pressure of 90 mmHg or greater, taken at least twice on two separate occasions.

RECOMMENDATIONS

- Measure BP at every opportunity as a high number of Malaysians are undiagnosed.
- Check BP for every adult above age 18 years at least once as part of their annual health screening, and more frequently for those who are at risk (family history, obese and those at-risk of high blood pressure).

2 Measurement of Blood Pressure

Blood pressure should be measured under standardised condition (see section 2.1). It can be measured directly or indirectly. There are four common devices used for the indirect measurement of BP namely:

- electronic devices
- aneroid sphygmomanometer
- automated ambulatory BP devices
- mercury column sphygmomanometer

The mercury sphygmomanometer remains the gold standard for non-invasive measurement.^{13(Level III)} However, it is largely being replaced by the electronic blood pressure measurement devices due to environmental and health concerns.¹⁴

There are many calibrated electronic or ambulatory BP devices available in the market. Only models validated by professional bodies (www.bhsoc.org, www.aami.org) should be used.

An appropriate cuff size should be used. Both the length and width of the inflatable bladder are important. The bladder length should encircle at least 80% of the circumference whilst the width should be at least 40% of the circumference of the arm. The standard cuff size should be 13 cm x 24 cm. Too small a cuff size will give a falsely high reading. Too big a cuff will give a falsely low reading.

Blood Pressure should be measured in both arms on the first visit and the higher reading is taken as the systolic BP.¹³ Patient should be rested at least 1 minute before measurement. At least 2 readings preferably 1-2 minutes apart should be taken in the same arm with the patient in the same position. A third reading should be taken if the difference between the first two readings is greater than 10 mmHg. The last two readings should be averaged. In the elderly, BP should be taken sitting & standing. Blood pressure measurements should not be done on the arm with arterio-venous fistula in haemodialysis patients.

If the difference in BP between the two arms is >20/10 mmHg, further evaluation is required to look for the cause. If patients are at high risk of postural hypotension, blood pressure should be taken lying and one minute after standing.

A systolic drop of >20 mmHg after one minute of standing is considered a significant postural drop.¹³

2.1 Electronic BP Sets

As mentioned earlier, electronic BP sets are now preferred.

A technical committee assessment by MOH concluded that if electronic BP set is used, it must be confirmed by mercury Sphygmomanometer in patients with cardiac illness, atherosclerosis, renal disease and in children. (Digital Blood Pressure Measurement sets, Health Technology Assessment Section, Medical division, Ministry of Health Malaysia. 014-2017 available at <http://www.moh.gov.my>.)

These electronic machines are generally less accurate in patients with arrhythmias (e.g. atrial fibrillation).

2.2 Home BP Measurement (HBPM) Using Electronic Devices

Home BP measurement is a useful adjunct in the diagnosis and management of hypertension especially in selected patients. If properly performed, it has good prognostic value.^{15,16 (Level II-2)}

Systematic reviews have shown that HBPM is superior compared to office measurements in diagnosing hypertension, in uncontrolled hypertension, assessing antihypertensive treatment, improving patient's adherence (compliance) and provides potential cost saving.^{17 (Level I),18-20}

Additionally, some studies have shown that HBPM measurements can be an alternative to ABPM and may have similar prognostic value.^{21,22 (Level I)} Home devices that measure the blood pressure in the fingers or the wrists are not recommended.

Situations where HBPM is useful include:²³

- at initial assessment
- to diagnose hypertension
- to diagnose isolated office hypertension
- to diagnose masked hypertension
- to assess treatment effects
- to diagnose true resistant hypertension
- to encourage adherence to treatment
- to optimise blood pressure control

RECOMMENDATIONS

BP Measuring Technique

For Clinic BP, patients should:

- refrain from smoking, eating, caffeine intake or exercise for at least 30 minutes
- be seated for at least 1 min in a quiet room, back & arm supported (e.g. resting on the table)
- be seated with legs uncrossed, not talking and relaxed
- use the correct bladder cuff size placed at heart level

For home measurements, besides the above:

- a minimum measurement for 3 days (ideally 7 days) should be performed
- should be done at about the same time once in the morning (before drug intake if on treatment) and evening (before meal)
- two readings should be taken at each occasion (at least 1 minute apart)
- the readings must be immediately recorded in a specific logbook or stored in a device with memory

The following must be taken into consideration when interpreting HBPM:²³

- BP values measured on the first monitoring day should be disregarded
- Average the remaining BP measurements (at least 3 days)
- Mean home systolic BP >135 mmHg and/or diastolic BP >85 mmHg should be considered as elevated

2.3 Ambulatory Blood Pressure Monitoring (ABPM)

Most of the data upon which estimates of risk are based, as well as benefits of treatment have been accumulated from office BP readings and therefore ABPM is not essential for the diagnosis and management of most patients with hypertension.

The data provided by ABPM does not influence therapeutic decisions in the vast majority of patients. The current cost of ABPM devices will also limit its widespread use.

ABPM is useful in selected clinical situations. These include:^{24,25 (Level III)}

- diagnosis of isolated office hypertension
- diagnosis of masked hypertension
- patients with borderline or labile hypertension
- detection of nocturnal hypertension
- evaluation of suspected hypotensive symptoms, especially in the elderly
- fluctuating office BP readings
- confirmation of resistant hypertension

3 Diagnosis and Initial Assessment

Evaluation of patients with documented hypertension has three objectives:

1. To exclude secondary causes of hypertension. (Table 3-A)
2. To ascertain the presence of target organ damage or complication. (Table 3-B)
3. To assess lifestyle and identify other cardiovascular risk factors (Table 3-C) or coexisting condition that affect prognosis and guide treatment. (Table 3-D)

Such information is obtained from adequate history, physical examination, laboratory investigations and other diagnostic procedures.

TABLE 3-A Secondary Causes of Hypertension

- Parenchymal kidney disease
- Renovascular disease
- Sleep apnoea
- Primary aldosteronism
- Drug-induced or drug-related
 - » Oral contraceptives
 - » Steroids
 - » Non-Steroidal Anti-inflammatory Drugs / COX-2 Inhibitors
 - » Erythropoietin
- Cushing syndrome
- Pheochromocytoma
- Acromegaly
- Thyroid disease
- Parathyroid disease
- Coarctation of the aorta
- Takayasu Arteritis

TABLE 3-B Manifestations of Target Organ Damage (TOD) / Target Organ Complication (TOC)

Organ	Manifestations
Heart	• Left Ventricular Hypertrophy • Coronary Heart Disease • Heart Failure
Brain	• Transient Ischaemic Attack • Stroke • Dementia
Peripheral vasculature	• Absence of one or more major pulses in extremities (except dorsalis pedis) with or without intermittent claudication • Carotid bruit • Abdominal aortic aneurysm
Kidney	• GFR <60 ml/min/1.73m² • Proteinuria (1+ or greater) • Microalbuminuria* (2 out of 3 positive tests over a period of 4-6 months)
Retina	• Haemorrhages or exudates • Papilloedema

TOD = Target organ damage (LVH, retinopathy, proteinuria)

TOC = Target organ complication (heart failure, renal failure)

* defined as normal to mildly increased (UACR <30mg/g), moderately increased (UACR 30-300mg/g) and severely increased (UACR >300mg/g)

A complete history should include:

- duration and level of elevated BP if known
- symptoms of secondary causes of hypertension
- symptoms of target organ complications (i.e. renal impairment and heart failure)
- symptoms of cardiovascular disease (e.g. CHD and cerebrovascular disease)
- symptoms of concomitant disease that will affect prognosis or treatment (e.g. diabetes mellitus, heart failure, renal disease and gout)
- family history of hypertension, CHD, stroke, diabetes, renal disease or dyslipidaemia
- dietary history including salt, caffeine, liquorice and alcohol intake
- drug history of either prescribed or over-the-counter medication (NSAIDs, nasal decongestants, OCP/HRT)
- exposure to traditional or complementary medicine
- lifestyle and environmental factors including air pollution that will affect treatment and outcome (e.g. smoking, physical inactivity, substance abuse; recreational & doping, psychosocial stressors and excessive weight gain)
- presence of snoring and/or day time somnolence which may indicate sleep apnoea

Physical examination should include the following:

- General examination including height, weight and waist circumference
- Measure BP appropriately. (refer to Chapter 2)
- Fundus examination
- Examination for carotid bruit, abdominal bruit, presence of peripheral pulses and radio-femoral delay
- Cardiac examination for cardiomegaly, signs of heart failure and aortic regurgitation
- Abdominal examination for renal masses/bruit and aortic aneurysm
- Neurological examination to look for evidence of stroke
- Signs of endocrine disorders (e.g. Cushing syndrome, acromegaly and thyroid disease)
- Ankle brachial index (if available)

The minimum initial investigations aim to screen for presence of secondary causes of hypertension, determine the presence of CV risk factors, target organ damage (TOD) and target organ complication (TOC). They should include the following:^{26(Level III)}

- Full blood count
- Blood glucose
- Renal function tests (creatinine, eGFR, serum electrolytes)
- Lipid profile (total cholesterol, HDL cholesterol, LDL cholesterol and triglycerides)
- Uric acid
- Urinalysis (dip stick: albuminuria/microalbuminuria & microscopic haematuria)
- Electrocardiogram (ECG)

If the examination or investigations suggest the presence of a secondary causes, the patient should be referred for specialist evaluation. If there is evidence of TOD or TOC (Table 3-B), further tests should be considered.

TABLE 3-C Co-existing Cardiovascular Risk Factors for Risk Stratification

- Diabetes mellitus
- Dyslipidaemia
- Cigarette smoking
- Microalbuminuria/Proteinuria
- Estimated GFR <60 mL/min/m²

According to a study in Malaysia as many as 54% patients with essential hypertension did not have their cardiovascular risks adequately assessed.²⁷

Following initial clinical evaluation and investigations, the patient should be risk stratified. Many patients with hypertension have more than one other cardiovascular

risk factor. Each additional risk factor increases cardiovascular risk substantially. Hence, overall global cardiovascular risk of a patient with hypertension should be done.^{28,29} There are various ways to assess global cardiovascular risk and this includes using validated risk charts like the Framingham General Cardiovascular Risk Chart which has been validated locally and found to perform quite well^{30,31} or using risk stratification tables (Table 3-D) which stratifies the risk of developing major cardiovascular events, which includes stroke, myocardial infarction and total mortality.

TABLE 3-D Risk Stratification

Co-Existing Condition BP Levels (mmHg)	No RF No TOD No TOC	TOD or RF (1-2) No TOC	TOC or RF(≥3) or Clinical atherosclerosis or CKD	Previous MI / IHD, Previous stroke or Diabetes or CKD
SBP 130 - 139 and/or DBP 80 - 89	Low	Intermediate	High	Very High
SBP 140 - 159 and/or DBP 90 - 99	Low	Medium	High	Very High
SBP 160 - 179 and/or DBP 100 - 109	Medium	High	Very High	Very High
SBP >180 and/or DBP >110	High	Very High	Very High	Very High

Risk Level	Risk of Major CV Event in 10 years	Management
Low-Intermediate	<10%	Healthy living
Medium	10 - 20%	Drug treatment and healthy living
High	20 - 30%	Drug treatment and healthy living
Very high	>30%	Drug treatment and healthy living

TOD = Target organ damage (LVH, retinopathy, proteinuria).

TOC = Target organ complication (heart failure, renal failure).

RF = Additional risk factors (smoking, TC >6.5mmol/L, family history of premature vascular disease).

Clinical atherosclerosis = CHD, carotid stenosis, peripheral vascular disease, transient ischaemic attack, stroke.

TABLE 3-E Recommendations for Follow-Up Visit Based on Initial Blood Pressure Measurements for Adults

Initial BP (mmHg)			Recommended follow-up
Systolic		Diastolic	
<120	and	<80	Recheck in one year
120 – 139	and	80 – 89	Assess global CV risk & Recheck within 3 – 6 months
140 – 159	and/or	90 – 99	Assess global CV risk & Confirm within two months
160 – 179	and/or	100 – 109	Assess global CV risk & Evaluate within one month and treat if confirmed
180 – 209	and/or	110 – 119	Assess global CV risk & Evaluate within one week and treat if confirmed
≥210	and/or	≥120	Assess global CV risk & Initiate treatment after repeated measurement during the same encounter

Modified from JNC-VII³²(Level III)

4 Non-Pharmacological Management

Non-pharmacological management (healthy living) plays an important role in the management of hypertension and in improving overall cardiovascular health.³³⁻³⁵ When recommending healthy living, it is important to know that these interventions require a joint effort from patient, family and healthcare providers.

4.1 Weight Reduction

Dietary interventions to lower body weight are often recommended for overweight people with mild hypertension. In people with hypertension, weight reducing diet has been shown to reduce blood pressure and body weight.³⁶ A 4kg reduction in body weight would achieve a BP reduction of 4.5/3.2 mmHg. There is evidence that showed a reduction of 1kg in weight relates to 1 mmHg reduction in SBP.³⁷ However, the long term effect of weight loss on mortality and morbidity in people with hypertension is unknown.^{36,37}

4.2 Sodium Intake

High salt intake is associated with increased risk of stroke, stroke mortality, and coronary heart disease mortality.^{38,39} Reducing sodium intake significantly reduces blood pressure in adults.³⁸⁻⁴¹ WHO recommends a reduction of sodium intake to <2 g/day or <5 g/day of salt (about one teaspoon of salt) in adults.⁴² A recent Cochrane review has shown that a reduction of sodium intake from a high average of 201 mmol/day (11.6g of salt) to an average level of 66 mmol/day (3.8g of salt), resulted in a decrease in BP of 7.8/2.7 mmHg in Asian people with hypertension.⁴⁰ However there is inadequate evidence assessing the effect of reduced sodium intake on cardiovascular disease mortality and morbidity.^{38,39,43-45} In Malaysia, the estimated mean sodium excretion of normotensive people was 3.4 to 3.8 g, equivalent to 8.7 to 9.5 g of salt intake per day.^{46,47} This exceeds the recommended salt intake and hence salt reduction is recommended for most people especially the hypertensive population.⁴⁸

4.3 Alcohol Consumption

Alcohol consumption elevates BP. Previous meta-analysis has shown that reducing alcohol consumption reduced BP by 3.3/2 mmHg.^{49(Level I)} A recent meta-analysis has shown that reducing alcohol intake lowers BP in a dose-dependent manner with an apparent threshold.^{50(Level I)} People who drink are advised to limit alcohol consumption to < two drinks per day.^{50(Level I)}

4.4 Regular Physical Activity

Increased physical activity has been shown to reduce BP. However, there is a lack of data on its effect on cardiovascular events and mortality.⁵¹ Meta-analyses have shown that dynamic aerobic endurance,⁵²⁻⁵³ dynamic resistance,⁵⁴ and isometric resistance training⁵⁵⁻⁵⁷ lower BP. All patients with hypertension benefit from any form of these physical activities.⁵⁸ Cumulative moderate intensity aerobic exercise of at least 150 minutes per week is advised.^{59,60}

4.5 Healthy Eating

A diet rich in fruits, vegetables and low fat dairy products with reduced saturated and total fat can substantially lower BP (11/6 mmHg in hypertensive patients and 4/2 mmHg in patients with high normal BP).^{60(Level I)} A recent meta-analysis suggests that healthy dietary patterns such as the Dietary Approaches to Stop Hypertension (DASH), Nordic, and Mediterranean diet significantly lowered BP by 4.26/2.38 mmHg.⁶¹ These diets are rich in fruit, vegetables, whole grains, legumes, seeds, nuts, fish and dairy products and low in meat, sweets, and alcohol. A recent cohort study that included Malaysian population without cardiovascular disease has shown that higher fruit, vegetable, and legume consumption was associated with a lower risk of non-cardiovascular and total mortality.⁶²

4.6 Cessation of Smoking

Smoking can raise BP acutely. However the effect of chronic smoking on BP is less clear. Nevertheless smoking cessation is important in reducing global cardiovascular risk.

4.7 Relaxation Therapy

Stress management is useful but evidence on relaxation interventions on BP reduction has not been convincing.^{60,63} Yoga had been shown in a systematic review to reduce blood pressure by 4.2/3.6 mmHg but the quality of evidence is poor.⁶⁴

4.8 Dietary Potassium Intake

A meta-analysis has shown that increased dietary potassium intake reduces BP in adults with hypertension without adverse effect on blood lipid concentrations, catecholamine concentrations, or renal function.⁶⁵ For adults with normal renal function and not at risk of hyperkalaemia, increasing dietary potassium can reduce BP by 3.49/1.96 mmHg.⁶⁵ Higher dietary potassium intake was associated with a 24% lower risk of stroke.⁶⁵ This can be achieved by eating fruits, vegetables, nuts and legumes.

4.9 Others

Evidence for beneficial effect of micronutrient alterations, caffeine reduction and dietary supplementation with fish oil, calcium, magnesium, garlic and fibre on BP is limited.^{34,35,66-73} Meta-analyses have suggested that regular consumption of black tea and green tea can reduce BP but the sample size was small and quality of studies varied.^{74,75} (Level 1)

RECOMMENDATIONS

- Healthy Living must be instituted as an integral part in managing hypertension. (Grade A)
- Reduce salt intake, do regular physical activity, limit alcohol intake to < 2 drinks per day for those who drink, increase dietary potassium and lose weight to reduce BP. (Grade A)

5 Pharmacological Management

5.1 General Guidelines

All patients must be risk stratified to guide management. Decision to initiate pharmacologic treatment depends on the global cardiovascular risk (Table 3-D). It is the reduction of BP which provides the main benefits in the general hypertensive population.^{76 (Level I)} The choice of drug should be individualised.

5.1a Initiating Treatment

For patients with Stage I (mild) hypertension with low cardiovascular risk, advice should be given on healthy living for a period of three to six months. Pharmacological treatment has not been shown to prevent cardiovascular outcome in this group of patients.^{77,78} Patients should be seen at least twice (ideally monthly) during this period to assess the efficacy of the non-pharmacological intervention. Stage I patients with medium or higher risk⁷⁹⁻⁸⁴ should be offered drug treatment upon diagnosis (Figure 5-A).^(Level I)

5.1b Choosing Antihypertensive Drug Treatment

In patients with newly diagnosed uncomplicated hypertension and no compelling indications, choice of first line monotherapy includes ACEIs, ARBs, CCBs and diuretics which have all been shown to reduce cardiovascular morbidity and mortality.^{85-89 (Level I)} Beta-blockers are not recommended as first line monotherapy in this group of patients according to some guidelines.^{4 (Level III), 90, 91} This was based on an earlier meta-analysis which showed that it is not as effective in lowering blood pressure and in the prevention of stroke compared to the other anti-hypertensive agents.⁹²⁻⁹⁵ However more recent meta-analyses including updated versions of earlier meta-analyses^{95, 99 (Level I)} and an analysis¹⁰⁰ have suggested β -blockers (especially β_1 selective) can be given as first line agent. Other guidelines continue to recommend β -blockers as first line agent even in uncomplicated newly diagnosed hypertension.^{26, 101, 102} However, almost all guidelines recommend that β -blockers should be considered in younger patients in particular:

- those with an intolerance or contraindication to ACEIs and ARBs or
- women of child-bearing potential or
- patients with evidence of increased sympathetic drive.

Ideally, individualisation should be based on scientific evidence of reduction in clinical outcomes and co-morbidities (Table 5-C). Contraindications to the use of these drugs must also be considered.

In patients with stage I hypertension, treatment should be started with monotherapy at low dose. Monotherapy can lower BP to <140/90 mmHg in approximately 20–50% of patients with mild to moderate hypertension¹⁰³ If after a sufficient period of treatment (up to six weeks) with monotherapy BP is still not controlled, three options are available;

- the dose of the initial drug can be increased
- the drug can be substituted with another class of drug
- a second drug can be added

Choices of combination therapy is as shown in Table 5-A & 5-B.

If target BP is not achieved despite showing some blood pressure lowering effect with monotherapy, either increase the dose of the initial anti-hypertensive agent or add a second anti-hypertensive. The former may however give rise to dose-related adverse effects. Properly selected antihypertensive combinations may also mitigate the adverse effects of each other. If the patient does not show response or does not tolerate the initial drug, substituting with a drug from another class is recommended.

^(Level III) In patients presenting with stage II hypertension or beyond, combination therapy as first line is recommended.^(Level III) (Refer to Figure 5-A). Combination therapy can be considered as first line in high risk stage 1 hypertension especially for secondary prevention^{83,84 (Level I)}

Single Pill Combinations (SPC) is very convenient to use and promote treatment adherence by reducing pill burden and simplifying the treatment regimen.^{104,105,106}

^(Level I) In addition, it takes less time to achieve BP control using a combination than monotherapy.^{107,108,109 (Level I)}

It should be emphasized that simplification of the treatment regimen using SPC is only one strategy for improving adherence. For many patients, cost is a critical issue. In Malaysia, generic SPC are generally not available. Patented SPC are available but are more expensive. This may adversely affect adherence especially for self-paying patients. Free drug combination is the obvious choice in such circumstances. It is however worth emphasizing that available evidence showed SPC is associated with not only improved adherence, but also lower overall healthcare cost.^{110 (Level II-2)}

It is important to be reminded that the beneficial effects of BP lowering with pharmacotherapy is demonstrated from medium to very high CV risk. The absolute benefit in terms of CV event reduction is greater the higher baseline risk.^{111 (Level I)}

TABLE 5-A Effective Anti-Hypertensive Combinations Used in Outcome Trials

Effective combination	Patients studied
ACEI + thiazide-like diuretics	Post stroke ⁷⁹ , diabetes ⁸³
ARB + thiazide ^{82,112}	Hypertensive with Left Ventricular Hypertrophy. High risk hypertensives
CCB + ACEIs or β -blocker + thiazide ⁸⁰	Patients with Coronary Artery Disease
CCB + thiazide ⁸²	High risk hypertensives
CCB + ACEI ¹¹⁰	Medium risk hypertensives with no overt vascular diseases
ACEI + CCB ⁸⁴	High risk hypertensives
Thiazide-like diuretics + ACEI ¹¹³	Very elderly (>80 years old)
CCB + thiazide or thiazide diuretics ¹¹⁴	Medium risk hypertensives
CCB + ARB ¹¹⁴	Medium risk hypertensives
CCB + β -blocker ¹¹⁴	Medium risk hypertensives

TABLE 5-B Drug Combinations in Hypertension**Preferred** (based on outcome trials)^{79,80,82-84,112-115}

- ACEI / thiazide or thiazide-like diuretics
- ARB / thiazide diuretics
- ACEI / CCB
- β -blocker / thiazide diuretics
- CCB / thiazide diuretics
- Thiazide diuretics / K⁺ sparing diuretics
- CCB/ thiazide or thiazide-like diuretics
- CCB/ARB
- CCB / β -blocker

Acceptable (no outcome trial evidence yet)

- β -blocker / thiazide-like diuretics
- DRI/diuretic

ARB = angiotensin receptor blocker

ACEI = angiotensin-converting enzyme inhibitor

CCB = calcium channel blocker

DRI = direct renin inhibitor

5.1c Target Blood Pressure

Efforts must be made to achieve target BP. For patients <80 years old, the target SBP should be <140 mmHg and DBP <90 mmHg.^{88,100} For patients aged 80 years and above, aim for a target of <150/90 mmHg¹¹⁵ (Refer to chapter 7.5 on Hypertension in Older Adults). For high/very high risk individuals the target is <130/80 mmHg^{116,117} (Refer to chapter 7 on Hypertension in Special Groups).

If BP is still >140/90 mmHg with three drugs, including a diuretic at optimal tolerated doses, there is a need to exclude medication non-adherence and isolated office hypertension. After excluding these causes of uncontrolled hypertension, the patient is then defined as having resistant hypertension¹¹⁸ (Refer to chapter 10 on Resistant Hypertension). A quick check on the possible causes of resistant hypertension is required. These include:

- secondary hypertension
- excessive sodium intake, excessive liquorice intake, drugs and drug interactions. (see chapter 4 Non-pharmacological Management)
- complications of long standing hypertension such as nephrosclerosis, loss of aortic distensibility and atherosclerotic renal artery stenosis

5.2 Follow-Up Visits

Follow up intervals should be individualised based on global CV risk, pre-treatment BP levels and drugs used. For high and very high risk patients, it is advisable to bring the BP to target within 3 to 6 months.⁸² Once target BP is achieved, follow-up at three to six-month intervals is appropriate. As a rule, once the BP is controlled, most patients will require life-long treatment. Patients must be counseled to have at least six monthly follow ups even though the BP is well controlled and not to resort to merely going for repeat prescription without seeing a doctor. During these visits, doctors should assess persistence of BP control, adverse reaction to treatment, global vascular risk (including new onset and pre-existing CV risk factors) and complications of hypertension with may have developed since the last visit.

5.3 When To Refer

Most patients can be effectively managed by primary care practitioners. Patients with the following conditions should be referred to the appropriate specialists including Family Medicine Specialists for further assessment. Indications for referral to the appropriate specialists include:

- severe hypertension (>180/110 mmHg) - refer to chapter 6 on Management of Severe Hypertension
- suspected secondary hypertension
- resistant and refractory hypertension

- recent onset target organ damage
- pregnancy
- office hypertension with additional CV risk
- children and adults <30 years
- secondary prevention with multiple co-morbidities/risk factors

5.4 Step-Down Therapy

Step-down therapy is discouraged in the vast majority of patients. However in patients who insist on it, the following criteria must be considered first:

- Patients' BP must not be higher than stage I (mild) hypertension with low global CV risk
- BP well-controlled for at least 1 year on the same medication at the same dosage
- Must agree to be followed-up at least 3-6 monthly
- Must be motivated to adopt healthy living

A recent systematic review supports this recommendation.¹¹⁹ This review of studies done between 1975 to 2013 showed that a trial of treatment withdrawal in well controlled hypertensives followed by subsequent regular blood pressure monitoring is safe with minor adverse events. Predictors of successful withdrawals were patients on prior monotherapy and lower blood pressures before withdrawal.

RECOMMENDATIONS

- Treat most patients with pharmacological agent life-long. (Grade C)
- Choose mono-therapy in patients with stage 1 hypertension and with no compelling indication from one of the 5 classes of drug of agents (ACEIs, ARBs, CCBs, Diuretics or β -Blockers) based on patient's individual clinical profile. (Grade C)
- Choose combination therapy in patients with medium/high/very high risk stage 1 hypertension and stage 2 hypertension. (Grade A)
- Treat BP to SBP<140 mmHg and DBP<90 mmHg for most hypertensive patients. (Grade A)
- Treat SBP to <130mmHg and DBP <80 mmHg for high/very high risk patients. (Grade A)
- Use combination therapy (free or single pill) for most patients to achieve BP control. (Grade A)
- Arrange periodic scheduled visits to assess global CV risk, emerging new risk factors and organ damage/complication. (Grade C)
- Co-manage patients whose BP are controlled with primary care facilities (Klinik Kesihatan or private general practice). (Grade C)

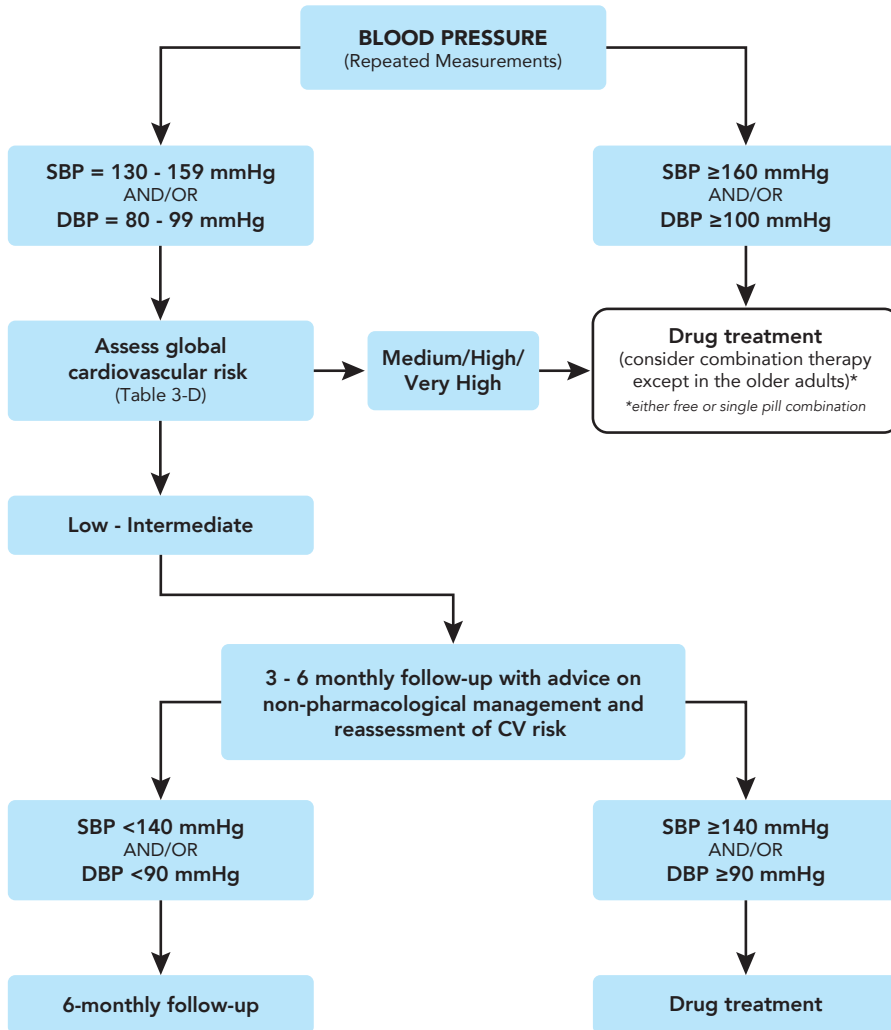
FIGURE 5-A Algorithm for the Management of Hypertension

TABLE 5-C Choice of Anti-Hypertensive Drugs in Patients with Concomitant Conditions

Concomitant Condition	Diuretics	β -blockers	ACEIs	CCBs	Peripheral α -blockers	ARBs
Diabetes mellitus (without nephropathy)	+	+/-	+++	+	+/-	++
Diabetes mellitus (with nephropathy)	++	+/-	+++	++	+/-	+++
Gout	+/-	+	+	+	+	++
Dyslipidaemia	+/-	+/-	+	+	+	+
Coronary heart disease	+	+++	+++	++	+	+++
Heart failure	+++	+++ [#]	+++	+ [@]	+	+++
Asthma	+	–	+	+	+	+
Peripheral vascular disease	+	+/-	++	+	+	+
Non-diabetic renal impairment	++	+	+++	++	+	++
Unilateral Renal artery stenosis	+	+	++ [§]	+	+	++ [§]
Older Adults with no co-morbid conditions	+++	+	+	+++	+/-	+
Very Elderly (>80 yrs) with no co-morbid conditions	+++	+	+++	++	+/-	++

The grading of recommendation from (+) to (+++) is based on increasing levels of evidence and/or current widely accepted practice.

+/- Use with care

– Contraindicated

[#] Metoprolol, bisoprolol, carvedilol, nebivolol – dose needs to be gradually titrated

[@] Current evidence available for amlodipine and felodipine only

[§] Contraindicated in bilateral renal artery stenosis

6 Management of Severe Hypertension

Severe hypertension is defined as persistent elevated SBP >180 mmHg and/or DBP >110 mmHg.

These patients may present with:

- incidental finding in an asymptomatic non-previously diagnosed patient
- treated hypertension on follow-up who are asymptomatic
- patients with symptoms which may include:
 - » non-specific symptoms like headache, dizziness, lethargy
 - » symptoms and signs of acute target organ damage/complication. These include acute heart failure, acute coronary syndromes, acute renal failure, dissecting aneurysm, subarachnoid haemorrhage, hypertensive encephalopathy and preeclampsia/eclampsia (Refer to chapter 7.4 Hypertension and Stroke, chapter 7.6 Hypertension in Women)

Patients are then categorised as having:

- a. hypertensive urgencies (urgency), or
 - b. hypertensive emergencies (emergency)
- (a) and (b) are also referred to as hypertensive crises.

In a recent large series, only a minority of patients admitted (5.1%) had hypertensive crises. Of those, more than three quarters (76.6%) constitute hypertensive emergencies.¹²⁰(Level II)

Management of these patients depends on the clinical presentation and laboratory investigations. The evaluation of these patients should include a thorough history and physical examination, particularly looking for signs of acute target organ damage/complication and causes of secondary hypertension. (Table 6-A)

The commonest reason of severe hypertension is long-standing poorly controlled essential hypertension. Other causes are as listed in Table 6-A.

TABLE 6-A Common Causes of Severe Hypertension

Causes	Example
Parenchymal renal disease	Chronic Kidney Disease Primary glomerulonephritis
Renovascular disease	Atherosclerotic disease Fibromuscular dysplasia Polyarteritis nodosa
Systemic disorders with renal involvement	Systemic lupus erythematosus Systemic sclerosis Vasculitides
Endocrine	Conn syndrome (primary hyperaldosteronism) Pheochromocytoma Cushing syndrome
Drugs	NSAIDs COX-2 inhibitors Oral Contraceptives Amphetamines / Metamphetamines Cyclosporin Cocaine Other Illicit Drugs Phencyclidine Clonidine withdrawal
Congenital disease	Coarctation of Aorta Polycystic kidney disease
Pregnancy related	Preeclampsia / eclampsia

6.1 Specific Management

The aim of management is to reduce BP in a controlled, predictable and safe manner, to avoid provoking or aggravating acute coronary syndrome, cerebral or renal ischaemia.

6.1.1 Hypertensive Urgency

Hypertensive urgency is defined as severe increase in BP which is not associated with acute end organ damage/complication and these include patients with grade III or IV retinal changes (also known as accelerated and malignant hypertension), but no overt symptoms and signs of acute target organ damage/complication. These patients may be admitted.

Blood pressure measurement should be repeated after 30 minutes of bed rest.^{121,122(Level II)} Initial treatment should aim for about 25% reduction in BP over 24 hours but not lower than 160/100 mmHg.^{123,124(Level II)} Oral drugs proven to be effective are outlined in Table 6-B. Combination therapy may be necessary. Importantly, there is no role for intravenous BP lowering drugs. Many of these patients have withdrawn from or are not adhering to antihypertensive therapy and do not have clinical or laboratory evidence of acute target organ damage.¹²⁵ Possible precipitating factors for hypertensive urgency include non-adherence to anti-hypertensive medications, less effective outpatient blood pressure control, acute pain, herbal supplement and emotional stress.^{125,126}

Therapeutic strategies for previously undiagnosed patients include (Figure 6-A: Flowchart in management of hypertensive urgency):

1. Rest in quiet room for at least 2 hours^{121,122,127}
2. Initiate oral anti-hypertensive agents if BP remains >180/110 mmHg¹²¹
3. Hypertensive urgency discharge plan (Figure 6-B)

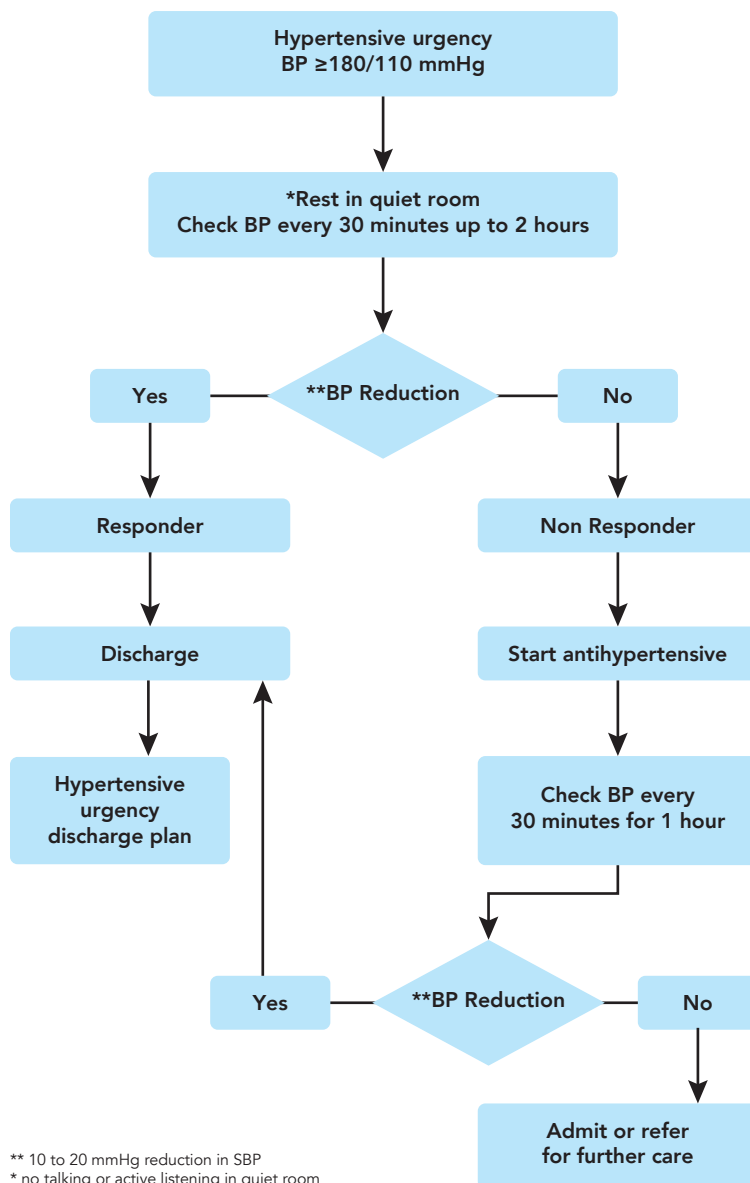
FIGURE 6-A Flowchart in Management of Hypertensive Urgency

TABLE 6-B Oral Treatment for Hypertensive Urgencies

Drug	Starting Dose (mg)	Onset of action (hr)	Duration (hr)	Frequency (prn)
1. Captopril	12.5 mg	0.5	6	1 - 2 hrs
2. Nifedipine	10 mg	0.5	3 - 5	1 - 2 hrs
3. Labetalol	200 mg	2.0	6	4 hrs

FIGURE 6-B Hypertensive Urgency Discharge Plan**Blood pressure monitoring**

- Home BP monitoring OR check by healthcare provider at least 3 times per week
- If BP >180/110 mmHg, repeat after 5 minutes; IF second BP higher or same as the first one OR have symptoms, seek medical help.

Medication

- Take anti-hypertensive as prescribed

Follow up care

- Adhere to clinic follow up appointment

When to call 999

- symptoms such as chest pain, difficulty in breathing or altered mental status occurs

6.1.2 Hypertensive Emergency

Hypertensive emergency is defined as severe elevation of blood pressure associated with new or progressive end organ damage/complication such as acute heart failure, dissecting aneurysm, acute coronary syndromes, hypertensive encephalopathy, acute renal failure, subarachnoid haemorrhage and/or intracranial haemorrhage. These may occur in patients with BP >180/110 mmHg, particularly if the BP has risen rapidly.

These patients:

- should be admitted for immediate intervention and monitoring.
- need to be reduce their BP rapidly based on clinical scenarios - refer also to chapter 7.4 Hypertension and Stroke, and chapter 7.6 Hypertension in Women.
- should have their BP reduced by 10%-25% within certain minutes to hours but not lower than 160/90 mmHg.^{128,129(Level III)}

This is best achieved with parenteral drugs. (Table 6-D)

TABLE 6-C Common Clinical Scenario of Hypertensive Emergencies with Treatment Goals

Clinical scenario	BP reduction	Additional consideration
Acute heart failure	BP lowering until symptom resolution. <25% within 1 hour, then ≤160/100 mmHg over 2 to 6 hours.	β-blocker or CCB use could cause exacerbation of symptoms.
Acute coronary syndrome	Reduce BP to reduce cardiac workload and improve coronary perfusion. <25% within 1 hour, then ≤160/100 mmHg over 2 to 6 hours. ¹²⁸ An alternative is to reduce DBP by 10% to 15% or to approximately 110 mmHg in 30 to 60 minutes, if the patient is clinically stable, further gradual reductions toward a normal BP can be implemented over the next 24 to 48 hours.	Consider type A aortic dissection as cause of acute coronary syndrome; avoid selective β-blockers if cocaine abuse suspected. ^{125,130,131}
Acute aortic dissection	Reduce SBP to ≤120 or BP ≤120/80 mmHg (lower if tolerable) and HR to <60 bpm within 1 hour.	Beta blockade should precede vasodilator (e.g., nicardipine or nitroprusside) administration, if needed for BP control or to prevent reflex tachycardia or inotropic effect; SBP ≤120 mmHg should be achieved within 20 min. ¹ Avoid β-blockers if severe aortic regurgitation is noted. ¹³¹

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Clinical scenario	BP reduction	Additional consideration
Hypertensive Encephalopathy	Reduce BP 20%–25% within 1 hour to reduce intracranial pressure.	Avoid nitroprusside because it can lead to intracranial oedema.
Acute renal failure	Reduce BP to around 25% within 3 to 24 hours. ¹³²	
Preeclampsia and Eclampsia	Reduce SBP to <140 mmHg within the first hour. <i>Refer Chapter 7.6 Hypertension in Women</i>	Definitive treatment is delivery of foetus. ACE inhibitors, ARBs, renin inhibitors, and nitroprusside contraindicated. ¹
Sympathetic crises	Rapid BP lowering until symptom resolution.	Avoid β -blocker monotherapy (except for labetalol).
Phaeochromocytoma	Rapid BP lowering until symptom resolution.	Avoid β -blocker monotherapy (except for labetalol).
Acute ischemic stroke	<i>Refer Chapter 7.4 Hypertension and Stroke</i>	
Haemorrhagic stroke	<i>Refer Chapter 7.4 Hypertension and Stroke</i>	

There has been very few head to head comparative trials on the management of hypertensive crises especially hypertensive emergencies. A recent meta-analysis showed that IV labetalol have comparable efficacy and safety compared to nicardipine with the later showing more predictable and consistent BP control.^{133,134} (Level 1)

Specific clinical scenarios requiring rapid lowering of SBP, usually to at least <140 mmHg, in the first hour of treatment include aortic dissection, severe preeclampsia or eclampsia, and pheochromocytoma with hypertensive crisis.^{1,131}

In summary, the selection of an antihypertensive agent should be based on the drug's pharmacology, pathophysiological factors underlying the patient's hypertension, degree of progression of target organ damage, the desirable rate of BP decline, and the presence of comorbidities. The therapeutic goal is to minimise target organ damage safely by rapid recognition of the problem and early initiation of appropriate antihypertensive treatment.^{1,129}

TABLE 6-D Treatment Options for Hypertensive Emergencies^{123,129,135}

Drugs	Dose	Onset of action	Duration	Remarks
Labetalol	<u>Adult*</u> 20 mg injected slowly for at least 2 min; followed by 40-80 mg every 10 min. Max: 200 mg <u>Children**</u> 1 month - 11 years: IV 0.25-0.5mg/kg (Max 20mg). IVI 0.5-1.0 mg/kg/hr initially. Maintenance: 0.25-3.0 mg/kg/hr.	≤5 min	3 - 6 hrs	Patient should remain supine during and 3 hr after the procedure. Caution in heart failure.
Nitroglycerine	<u>Adult*</u> Initial: 5-25 mcg/min. Usual range: 10-200 mcg/min; up to 400 mcg/min in some cases.	2 - 5 min	3 - 5 min	Preferred in acute coronary syndrome and acute pulmonary oedema.
Isosorbide Dinitrate	<u>Adult*</u> IV infusion 2-20 mg/hr, titrate based on target BP.	3 - 15 min	1 hour	Preferred in acute coronary syndrome.
Hydralazine*	<u>Adult*</u> Initial: 5-10 mg via slow inj, may repeat after 20-30 min. Alternatively, as a continuous infusion, initial dose of 0.2-0.3 mg/min. Maintenance: 0.05-0.15 mg/min. <u>Children**</u> 1 month - 11 years: IV 0.1-0.5 mg/kg (Max 10 mg) may be repeated after 4-6 hr. IVI 12.5-50 mcg/kg/hr Max 3 mg/kg/day.	10 - 30 min	3 - 8 hrs	Caution in acute coronary syndromes, cerebrovascular accidents and dissecting aneurysm. Unpredictable BP-lowering effects.

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Drugs	Dose	Onset of action	Duration	Remarks
Nicardipine	<u>Adult*</u> Slow IVI at an initial rate of 5 mg/hr. Increase infusion rate as necessary, up to max 15 mg/hr. Consider reducing to 3 mg/hr after response is achieved. <u>Children**</u> IV bolus 0.5-5 mcg/kg over 1 minute. IVI 1- 4 mcg/kg/min.	5 - 10 min	1 - 4 hrs	Caution in acute heart failure and coronary ischaemia.
Esmolol	<u>Adult*</u> Loading dose of 80 mg over 15-30 sec, followed by an infusion of 150 mcg/kg/min, may increase to 300 mcg/kg/min if necessary. <u>Children**</u> IV bolus 250-500 mcg/kg over 1 min. IVI 50-200 mcg/kg/min for 4 min. May repeat sequence.	1 min	10 - 20 min	Used in peri-operative situations and tachyarrhythmias.

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Drugs	Dose	Onset of action	Duration	Remarks
Sodium Nitroprusside	<u>Adult</u>* Initial: 0.3-1.5 mcg/kg/min, adjust gradually as needed. Usual: 0.5-6 mcg/kg/min. Max rate: 8 mcg/kg/min, discontinue if there is no response after 10 mins. May continue for a few hr if there is response. <u>Children</u>* IV 0.25-0.5 mcg/kg/min, may be repeatedly double at interval of 15-20 min. Max 6 mcg/kg/min.	seconds	1 - 5 min	Caution in heart failure. Require intra-arterial blood pressure monitoring. Lower dosing adjustment required for elderly and those already receiving antihypertensives.

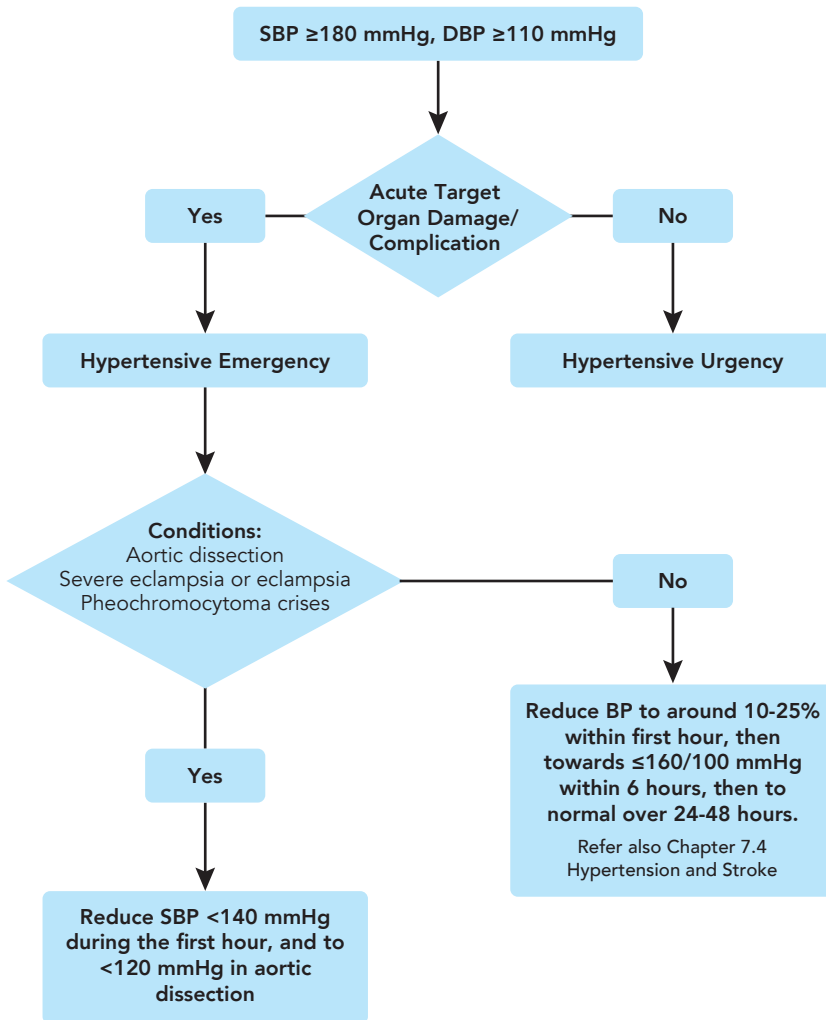
* Referenced from 153rd Edition, MIMS, 2018.

** British National Formulary for Children (BNFC) 2018-2019.

In pregnancy refer to chapter 7.6 Hypertension in Women.

TABLE 6-E Differences Between Hypertensive Emergency and Urgency

Variable	Emergencies	Urgencies
Symptoms	Yes	No or minimal
Acute target organ damage/complication	Yes	No
BP reduction rate	Minutes to hours	Hours to days
Evaluation for secondary hypertension	Yes	Yes

FIGURE 6-C Flowchart in Management of Hypertensive Emergency*

* Flowchart adopted from Whelton, et al 2017, pg 139 ¹

6.2 Dangers of Rapid Reduction in Blood Pressure

Rapid reduction of BP (within minutes to hours) in hypertensive urgencies should be avoided as it may precipitate ischaemic events.¹³⁶

Oral or sublingual drugs with rapid onset of action can result in an uncontrolled BP reduction. Several serious side effects have been reported with the administration of sublingual fast-acting nifedipine and therefore this is no longer recommended.¹³⁷(Level III) However oral nifedipine retard can be used and has been recommended as first line therapy for hypertensive urgencies.¹²⁴(Level III)

Following stabilisation of patient's BP, subsequent management is tailored towards achieving optimal control.

For management of patients with severe hypertension and stroke, refer to chapter 7.4 Hypertension and Stroke.

RECOMMENDATIONS

- In hypertensive urgencies, aim for 10-20 mmHg SBP reduction after 2 hours of rest. Failing this, pharmacotherapy should be initiated. (Grade B)
- Treat hypertensive urgencies with combination oral therapy targeting BP to reduce by around 25% within 24 hours. (Grade C)
- Treat hypertensive emergencies with intravenous drugs with specific targets based upon clinical scenarios. (Grade B)
- Reduce SBP to less than 140 mmHg during the first hour for patients with severe preeclampsia or eclampsia, and pheochromocytoma crisis. For patients with aortic dissection reduce SBP to less than 120 mmHg. (Grade C)
- Reduce SBP by no more than 25% within the first hour; then, if stable, to 160/100 mmHg within the next 2 to 6 hours; and then cautiously to normal during the following 24 to 48 hours in all other situations. (Grade C)

7 Hypertension in Special Groups

7.1 Hypertension and Diabetes Mellitus

From the Malaysian National Health and Morbidity Survey (NHMS) 2015, the prevalence of diabetes in those 18 years and above was 17.5%, and in the 18-19 years age group the prevalence was 5.5%, indicating that diabetics in Malaysia are generally younger.

Hypertension is common in patients with diabetes mellitus. Its presence increases the risk of morbidity and mortality. In 2016 the prevalence of hypertension in Malaysian diabetics was 76%.¹³⁸ Hypertension should be treated early in diabetes to prevent both microvascular and macrovascular complications and CV death.

7.1.1 Threshold for Treatment

Pharmacological treatment should be initiated in patients with diabetes when the BP is persistently ≥ 140 mmHg systolic and/or ≥ 90 mmHg diastolic.

The presence of microalbuminuria or overt proteinuria should be treated even if the BP is $< 140/90$ mmHg. An ACEI or ARB is preferred.^{139-147 (Level I)} In a proportion of patients, microalbuminuria may be normalised by high doses of ACEIs^{143 (Level I)} or ARBs^{144,145 (Level I)} even if the BP is already optimally controlled. Normalisation of microalbuminuria is associated with a reduction in the rate of decline in glomerular filtration rate.¹⁴⁸

7.1.2 Target Blood Pressure

Tight BP control should take precedence over the class of anti-hypertensive drug used.^{149,150 (Level I)} This will often require combination therapy.

Prior to the ACCORD study, it was generally accepted that for diabetics, the lower the BP the better the outcomes. However, in the ACCORD Study, diabetic patients at high risk of cardiovascular events who were randomised to a target systolic blood pressure (SBP) of < 120 mmHg, did not show a reduction in the composite outcome of CV death, myocardial infarct and stroke, as compared with < 140 mmHg.^{151 (Level I)} This could possibly be due to the J-curve phenomenon seen in this particular cohort of patients who had underlying cardiovascular disease at baseline. This is supported by a more recent population based analysis.¹⁵²

In diabetics with CKD, the recommended target BP is <130/80 mmHg.

Other reports^{153,154} emphasise that major trials like ACCORD incorporate elderly patients (mean age 62 years) with established CVD or with multiple CV risk factors. These trials may not be relevant to younger diabetics with few or no risk factors, in whom continued benefits are derived from lowering the SBP to below 140 mmHg. In Malaysia, diabetics are younger and more aggressive BP targets may confer greater benefits.

The targets recommended by Malaysian CPG have been consistent over the years; and the studies above confirm that generally the target BP should be aimed at <140/80 mmHg,^{83,155} with a target of <130/80 mmHg in younger patients and those at higher risk of cardiovascular disease.^{139,155,156} However, in diabetics with established CAD further lowering of the BP beyond 120/80 mmHg does not confer additional cardiovascular benefit.

7.1.3 Management

The management of the hypertensive diabetic will involve healthy living changes as well as drug treatment.

7.1.4 Principles of Pharmacological Management

The use of certain classes of anti-hypertensive drugs may be detrimental to the diabetic patient because of their modes of action or adverse effects. Diabetes control may be compromised and diabetic complications aggravated, for example:

- decreased insulin responsiveness with higher doses of diuretics
- masking of the early symptoms of hypoglycaemia with β -blockers and slower recovery from hypoglycaemia with non-selective β -blockers
- aggravation of the symptoms of peripheral vascular disease with β -blockers
- dyslipidaemia with most β -blockers and diuretics
- worsening of orthostatic hypotension with peripheral β -blockers or centrally acting drugs

Angiotensin-converting enzyme inhibitors (ACEIs) are the drugs of choice based on extensive data attesting to their cardiovascular benefits and reno-protective effects in patients with diabetic kidney disease.^{157 (Level I)} They have also been reported to prevent the onset of nephropathy in normoalbuminuric diabetic patients with or without hypertension.^{155,158,159 (Level II)} In addition they do not have adverse effects on lipid and carbohydrate metabolism. However, its routine use in normotensive

normoalbuminuric diabetic patients is currently not recommended. If an ACEI is not tolerated, an angiotensin receptor blocker (ARB) should be considered.

For type 1 diabetes with nephropathy, ACEIs are the recommended agent.¹⁴⁰ For type 2 diabetes, both the ACEIs & ARBs may be used.

Thiazide and thiazide-like diuretics can be added on when monotherapy is inadequate. Single Pill Combination of a thiazide-like diuretic (indapamide) with perindopril has been reported to reduce overall mortality.^{83(Level I)}

CCBs do not have significant adverse metabolic effects. They do not compromise glycaemic control in diabetic patients. They can be effectively combined with a RAS blocker to lower blood pressure in hypertensive diabetics. The combination of benazepril with amlodipine was superior to the combination of benazepril with hydrochlorothiazide, leading to a reduction of cardiovascular events in the overall study population as well as the diabetic subgroup.^{84(Level 1)}

Beta-blockers may be used when ACEIs, ARBs or CCBs cannot be used or when there are concomitant compelling indications. However, they should be used with caution, especially in patients with type 1 diabetes.^{160(Level III)}

The SGLT2 inhibitors are a new class of oral anti-diabetic agents. In addition to lowering blood sugar levels, they lower the BP modestly but consistently.¹⁶¹ However, the mechanism by which BP is lowered is unknown. In the EMPA-REG trial^{162(Level 1)} and the CANVAS Program,^{163(Level 1)} the SBP was lowered by a mean of 5.0 and 3.9 mmHg respectively. In the two trials, the composite endpoints of CV death, myocardial infarct and strokes was also reduced. The reduction in the composite endpoints may however, not be related to BP reduction and the underlying mechanism in improving CV outcome is also unknown.

RECOMMENDATIONS

- Initiate ACEIs in diabetes without proteinuria. Use ARB for ACEI intolerant patients. (Grade A)
- Initiate ACEIs or ARBs in patients with diabetes and proteinuria. (Grade A)
- Consider CCBs, diuretics or β -blockers if RAS blockers cannot be used. (Grade B)

7.2 Hypertension and Renal Diseases

7.2.1 Hypertension and Non-Diabetic Chronic Kidney Disease

Hypertension may be a cause or consequence of chronic kidney disease (CKD).^{164,165} CKD is one of the commonest causes of secondary hypertension. Hypertension in CKD is often associated with an elevated serum creatinine, proteinuria and/or haematuria. The prevalence of hypertension increases with increasing levels of renal impairment, and approximately 50-75% of individuals with GFR <60 ml/min/1.73m² (CKD stages 3–5) have hypertension.¹⁶⁶ Hypertension accelerates the progression of CKD and may lead to end stage renal disease (ESRD). In addition, CKD is associated with an increased risk of cardiovascular disease. There are proven benefits of blood pressure lowering for prevention of cardiovascular events in patients with moderately reduced kidney function¹⁶⁷ as well as those on dialysis.^{168,169}(Level 1)

Tight control of BP in patients with CKD is therefore important. BP goals depend on urinary protein excretion. The target BP should be <140/90 mmHg for patients with CKD¹⁷⁰⁻¹⁷² (Level 1) and <130/80 mmHg for those with proteinuria ≥1g/24hours.¹⁷³(Level 1) Where well tolerated, aiming towards an Automated Office Systolic BP of 120mmHg in patients aged >50 years with nondiabetic nephropathy, GFR>20 ml/min/1.73m² and proteinuria <1g/day has shown cardiovascular benefits.^{174,175}(Level II-2) When aiming towards <120 mmHg systolic, close monitoring is recommended to detect treatment-related adverse effects including hypotension, syncope, electrolyte abnormalities and acute kidney injury.¹⁷⁵ All anti-hypertensive drug classes can be used to achieve this goal.¹⁶⁷

In the management of hypertension in CKD, control of BP and proteinuria are the most important factors in terms of retarding CKD progression. Anti-hypertensive agents that reduce proteinuria have an advantage in patients with nondiabetic proteinuric nephropathy. Meta analyses of comparative trials concluded that ACEI conferred an anti-proteinuric effect greater than other anti-hypertensive drugs.¹⁷⁶(Level 1) Overall 30% reduction in incidence of ESRD with ACEI can be expected.¹⁷⁷ The anti-proteinuric effect and reduction in ESRD was beyond that attributable to the BP lowering effect.^{172,178}(Level 1) This anti-proteinuric effect of ACEI was most prominent in patients on a low sodium diet or those treated with diuretics. Patients with proteinuria >3g/24 hours benefit the most.^{172,178} The advantage of ACEI is most apparent in patients with rapid progression of renal disease associated with proteinuria. ARBs are similar to ACEI in lowering BP and reducing proteinuria.

Combined RAS blockade can reduce proteinuria more than monotherapy.¹⁷⁹ However this is associated with an increased risk of hyperkalaemia, hypotension and renal failure.¹⁸⁰(Level 1) Hence, this approach is not recommended in patients with CKD.

Renal insufficiency should not be a contraindication to starting ACEI or ARB therapy, nor should it be a reason for discontinuing therapy. Serum creatinine level should be checked within the first two weeks of initiation of therapy and also after every increase in dose. If there is a persistent rise (at least 2 occasions) of serum creatinine

of >30% from baseline within two months, ACEIs^{181(Level II-3)} or ARBs should be reduced or stopped after excluding other precipitating factors. These patients should be referred to a nephrologist or physician.

In patients with GFR <30 ml/min/1.73m², thiazide diuretics may not be effective antihypertensive agents and therefore loop diuretics are preferred.^{182(Level III)} Concurrent diuretic therapy will often be necessary in patients with renal insufficiency since salt and water retention is an important determinant of hypertension in this setting.

Calcium channel blockers may be used in renal disease. In those with proteinuria, the non-dihydropyridine group of CCBs namely diltiazem or verapamil are preferred, as they have an additional anti-proteinuric effect.^{176(Level I)} Dihydropyridine CCBs can be considered if optimal BP is not achieved but should not be used as monotherapy in patients with proteinuria. The combination of an ACEI and a non-dihydropyridine CCB is more antiproteinuric than either drug alone.^{183(Level II-2)}

SUMMARY

- BP goals depend on urinary protein excretion.
- Anti-hypertensive agents that reduce proteinuria have an advantage in patients with nondiabetic proteinuric nephropathy.
- ACEI and ARBs confer an anti-proteinuric effect greater than other anti-hypertensive drugs.

RECOMMENDATIONS

- Patients with proteinuria of <1g/24 hours, lower BP to <140/90 mmHg. (Grade A)
- In patients with proteinuria of >1g/24 hours, lower BP to <130/80 mmHg. (Grade A)
- In patient >50 years, GFR >20 ml/min/1.73m² and proteinuria <1g/day lower SBP <120 mmHg using Automated Self-measured Office BP to reduce cardiovascular event. (Grade B)
- Choose RAS blockers as initial antihypertensive therapy for patients with micro- or macroalbuminuria. (Grade A)
- Consider concurrent diuretic therapy and dietary salt restriction as salt and water retention are important determinants of hypertension in CKD.
- Add non-dihydropyridine CCBs if BP goal is still not achieved and there is persistent proteinuria. (Grade A)
- Avoid dual RAS blockade in patients with CKD. (Grade A)

7.2.2 Renovascular Hypertension

Renovascular hypertension (RVH) is defined as a rise in arterial pressure attributable to reduced perfusion of the kidney(s).¹⁸⁴ It is important to diagnose renovascular hypertension as it is potentially reversible. Treatment also has the potential to restore or preserve renal function. The aetiology of renovascular hypertension includes the following:

- Atherosclerotic renovascular disease
- Fibromuscular dysplasia
- Takayasu's arteritis
- Transplant renal artery stenosis

Atherosclerotic renal artery stenosis (ARAS) is an important cause as it can lead to ESRD.¹⁸⁵ It is also associated with coronary heart disease, cerebrovascular disease and peripheral vascular disease. In patients with ARAS older than 60 years, the five-year-survival is 45% in patients with bilateral ARAS and 18% in those requiring dialysis therapy.¹⁸⁶

The presence of a stenotic renal vessel in a patient with hypertension does not necessarily equate to RVH. Some clinical features suggestive of RVH include:

- onset of hypertension before 30 years, especially without family history
- recent onset of hypertension after 55 years or deterioration in BP control in a previously well-controlled patient
- resistant hypertension
- abdominal bruit; particularly if associated with a unilateral small kidney
- flash pulmonary oedema
- renal failure of uncertain cause in the presence of normal urine sediment
- renal failure induced by ACEIs or ARBs
- coexisting diffuse atherosclerotic vascular disease

Renal angiography including measurement of the pressure gradient remains the gold standard in the diagnosis of RVH.¹⁸⁷ Non-invasive investigations include doppler sonography, captopril-enhanced isotope scan, spiral CT angiography (CTA), and magnetic resonance angiography (MRA) in patients with normal renal function.¹⁸⁸

All patients with ARAS require intensive medical therapy. ACEIs or ARBs are recommended to control blood pressure and reduce clinical events in those with known cardiovascular disease. Medical treatment includes statins, low dose aspirin, cessation of smoking and management of diabetes when present. This approach can be considered for patients with stenosis less than 70% or those with stable renal function and good BP control despite radiological evidence of stenosis >70%. These lesions should be monitored for progression using colour duplex sonography and the renal function of patients must be carefully monitored. A persistent rise in creatinine

of >30% over 2 months warrants cessation of ACEI/ARB drug therapy. This is best done under specialist's supervision.

If revascularisation is required, it is usually achieved by percutaneous angioplasty with stenting or surgical revascularisation in patients with complex anatomic lesions. Revascularisation should be considered under the following circumstances:¹⁸⁸

- A short duration of blood pressure elevation prior to the diagnosis of RVH
- Recurrent flash pulmonary oedema or refractory heart failure
- Resistant hypertension
- Intolerance to optimal medical therapy e.g. deterioration of renal function during antihypertensive therapy
- Otherwise unexplained progressive deterioration in renal function

Where indications for revascularisation are uncertain, 3 prospective randomised trials have not demonstrated compelling benefits either with endovascular stents or surgery when added to effective medical therapy.^{189-191(Level 1)}

Patients with fibromuscular dysplasia (FMD) rarely have excretory dysfunction, and hypertension in these patients generally responds to ACEIs.^{192(Level II-2)} Given the typical patient with FMD (young female with lower angioplasty-related risks, the need for many years of anti-hypertensive treatment plus limitations of RAS blockers during pregnancy), most clinicians would probably favour angioplasty for patients with FMD.^{193(Level III)} However, the benefits of angioplasty may be limited. The chance of achieving normal BP without anti-hypertensive agents is less than 30%, although some improvement in BP may be expected in an additional 50% or more.¹⁹³

SUMMARY

- Optimal medical management of ARAS includes ACEI or ARB for blood pressure control and reduction of CV events, statins, low dose aspirin and glycaemic control.

RECOMMENDATIONS

- Patients with RVH due to ARAS should be primarily medically managed because renal angioplasty and stenting has not shown any advantage over optimal medical therapy alone. (Grade B)
- Patients with RVH due to FMD should be considered for revascularisation. (Grade B)

7.3 Hypertension and Heart Disease

7.3.1 Hypertension and Coronary Heart Disease

Hypertension is the major risk factor for atherosclerosis, driving overall CV risk, thus sustained good control is important. This is especially important in the presence of other risk factors. Multiple CHD risk factors combine as multipliers, to increase CHD risk that is greater than the sum of its individual components. Management of patients with hypertension should consider the individual's absolute CHD risk. (Refer Table 3-D Risk Stratification) The decision to initiate drug treatment should take this into consideration.

Although the recommended target is <140/90 mmHg, in those at high CV risk in whom it is deemed safe on clinical grounds, and in whom drug therapy is well tolerated, aiming for a lower blood pressure may be considered.^{116,117,194(Level 1)}

Clinical studies have also shown that coronary events in hypertensive patients with CHD are reduced in those whose blood pressure is controlled.^{195(Level 1)} Based on many studies using different groups of antihypertensives, the benefits are achieved predominantly by lowering the blood pressure rather than the use of any specific class of antihypertensive agent.^{96,196,197(Level 1)}

There are clinical trials showing morbidity and mortality benefits of anti-hypertensive agents such as β -blockers, ACEIs and ARBs,^{198-200(Level 1)} following myocardial infarction. Following any coronary event, patients will be at high risk of subsequent events, especially if the hypertension is not controlled. In the first 2 weeks after an MI, β -blockers have been shown to reduce re-infarctions and mortality in the short term (30 days).^{201-204(Level 1)}

Cardio-selective β -blockers are preferred. In CHD patients with symptomatic & stable angina, the treatment of choice should be a β -blocker or a CCB. Short-acting nifedipine should not be used.

Combination therapies, with suitable drugs should be considered to reduce adverse effects and improve adherence (refer Chapter 5 Pharmacological Management).

The blood pressure target in patients with hypertension and CHD is <130 / <80 mmHg.^{116(Level 1)}

7.3.2 Hypertension and Heart Failure

Hypertension is the most frequent underlying cause of heart failure. Chronic, uncontrolled hypertension can cause heart failure with reduced ejection fraction ($\leq 40\%$) (HFrEF).²⁰⁵

In hypertensive patients with HFrEF, hypertension aggravates heart failure by increasing left ventricular afterload, promoting left ventricular remodeling and progression of myocardial damage. Hypertension treatment is important for improving the long-term prognosis.

Anti-hypertensive agents including β -blockers,²⁰⁶⁻²⁰⁸ ACEIs,²⁰⁹ and aldosterone antagonist,²¹⁰ have shown mortality benefits and reduction in the number of hospitalisations, in patients with HFrEF. In these patients, ACE inhibitors and β -blockers are recommended for initial therapy. β -blockers are contraindicated in the presence of acute heart failure. Careful monitoring for hyperkalaemia is recommended when combining an aldosterone antagonist with an ACEI or ARB. The evidence for ARB is less convincing²¹¹ but they may be used for ACEI intolerant patients.²¹²⁻²¹⁸

In patients with heart failure with preserved ejection fraction $>50\%$ (HFpEF), blood pressure control is important. For these patients, results with ARBs have been mixed.^{213,215} Several randomised control trials evaluating the efficacy of ARBs found no effect on prognosis including all-cause mortality. In these patients spironolactone has been shown to reduce hospitalisation with HF.²¹⁹

A meta-analysis with ACEI has shown a modest effect on HFpEF.^{220(Level 1)} However, a large-scale, prospective study (Swedish Heart Failure Registry) indicated that the total mortality rate was lower in patients taking RAS blockers.²²¹

Should hypertension be persistent in spite of ACEI / ARB, aldosterone antagonist and/or β -blocker, CCBs which are not negatively inotropic, such as amlodipine and felodipine, can be added. These patients should also be on loop diuretics for symptomatic relief.

RECOMMENDATIONS

- Use β -blockers, ACEIs or ARBs in post myocardial infarction patients to reduce recurrent myocardial infarction and death. (Grade A)
- Initiate β -blockers, ACEIs and Aldosterone antagonists in patients with systolic heart failure to reduce morbidity and mortality. (Grade A)
- Use ARBs or ACEIs and aldosterone antagonist in heart failure patients with preserved ejection fraction to reduce morbidity including hospitalisation. (Grade A)
- Treat blood pressure to <140 / <90 mmHg.

7.3.3 Hypertension and Atrial Fibrillation

Hypertension is the most important risk factor for new onset of atrial fibrillation (AF)^{222,223} of which it increases the risk of cardiogenic cerebral embolism. In the presence of AF, the incidence of cardiovascular events and mortality increases by 2.5-fold²²⁴⁻²²⁶. In particular, left ventricular hypertrophy and left atrial enlargement are independent risk factors for new onset of atrial fibrillation. When antihypertensive treatment leads to the regression of left ventricular hypertrophy, the incidence of atrial fibrillation decreases.²²⁷

Hypertension further increases the risks of stroke and arterial embolism in patients with chronic atrial fibrillation.^{228,229}

Although anticoagulants are used to reduce the risk of stroke in patients with AF, it also increases the incidence of hemorrhagic complications, especially intracranial hemorrhage.²³⁰ Strict blood pressure control is necessary in patients taking antithrombotic drugs.²³¹

A few small studies^{232,233} and subgroup analyses of larger trials^{234,235} have reported that ARB can reduce the incidence of recurrent atrial fibrillation or help maintain patient in sinus rhythm.²³⁶

For elderly patients (>75 years old) who are on anticoagulation for atrial fibrillation, both ACEi and ARB reduce mortality.^{237(Level II-2)}

For rate-control of permanent atrial fibrillation, β -blockers or non-dihydropyridine CCBs (verapamil and diltiazem) should be considered.²³⁸

The blood pressure target is <140/90 mmHg.²³⁹

7.3.4 Hypertension and Peripheral Arterial Disease

Hypertension and peripheral arterial disease (PAD) can co-exist. The risk factors for PAD include hypertension, diabetes, current smoking and dyslipidaemia.²⁴⁰ As atherosclerosis is a systemic vascular disease; diffuse atherosclerosis, CAD, and renovascular disease often coexist in these patients. 2-5% of patients with hypertension have intermittent claudication and 25-55% of patients with peripheral arterial disease present with hypertension.²⁴¹

Patients with PAD have almost three times the risk of a cardiovascular event and death.²⁴² They should be screened for atherosclerotic disease of the other systems. Control of hypertension in patients with PAD is poor.²⁴³ The aim of treatment in PAD is

both symptom relief and prevention of cardiovascular events. There is no consensus on the treatment of choice for hypertensive patients with PAD,²⁴⁴ although sub-analysis of major trials showed benefits of ACEI in patients with PAD.²⁴⁵ β -blockers may cause vasoconstriction and worsen frequency of intermittent claudication. They may be used with caution in patients with compelling indications (CHD and/or HF). Should patients present with Raynaud's phenomenon, consider CCBs which have vasodilating properties. Cilostazol has been shown to be useful especially in the elderly with disabling peripheral arterial disease.²⁴⁶ In addition to these medications, patients should stop smoking. Other therapies including LDL-cholesterol lowering and better control of diabetes are also recommended.

The blood pressure target is <140/90 mmHg²⁴⁷(Level II-2)

RECOMMENDATIONS

- Treat blood pressure in hypertensives with peripheral arterial disease (PAD) to <140/90 mmHg. (Grade B)
- Use any antihypertensive except β -blockers as first choice. (Grade C)
- Give ACEI to patients with PAD to prevent vascular events. (Grade B)
- Consider cilostazol in the elderly patients with symptomatic CAD or PAD. (Grade B)
- Counsel patients to stop smoking. (Grade B)
- Ensure other concurrent risk factors (especially diabetes, dyslipidaemia) are optimally managed.
- Prescribe antiplatelet agent unless contraindicated. (Grade A)

7.3.5 Hypertension and Left Ventricular Hypertrophy (LVH)

Left Ventricular hypertrophy is caused by pressure load and often regresses through long-term antihypertensive treatment. Those with LVH are at risk of premature cardiovascular events or death.

The most important factor in the regression of cardiac hypertrophy is good BP control. Hence all antihypertensives currently used can reduce cardiac hypertrophy through sustained control of blood pressure.^{248,249} Regression of LVH can also be achieved by weight reduction and salt restriction. Echocardiography is more sensitive than ECG for detection of LVH.²⁵⁰ Several studies have suggested that BP lowering leads to regression of LVH.^{251,252} To reduce clinical outcome, ARBs are preferred in hypertension with LVH on ECG.¹¹²(Level 1)

RECOMMENDATIONS

- Target BP <130/80 mmHg. (Grade A)
- Use ARBs as treatment of choice in hypertensive patients with LVH on ECG. (Grade A)

7.4 Hypertension and Stroke

Hypertension is the most important modifiable risk factor for ischaemic stroke (IS) and haemorrhagic stroke (HS).²⁵⁴ Blood Pressure levels are consistently shown to be associated with the risk for stroke.^{255,256} Although both SBP and DBP are associated with stroke, SBP is more predictive.²⁵⁷ Data from the Malaysian National Stroke Registry (NSR) showed that more than 75% out of the 1018 patients included in the registry has hypertension as its major risk factor.²⁵⁸

Worldwide, 15 million people suffer from stroke annually. Of these, 5 million die and another 5 million are left permanently disabled.²⁵⁴ It is presently among the top four leading causes of death in ASEAN countries.²⁵⁹ In Malaysia, stroke is the leading cause of death and disability in adults. Its incidence has alarmingly increased annually by 29.5% (ischaemic stroke) and 18.7% (haemorrhagic stroke).²⁵⁸

7.4.1 Primary Prevention of Stroke

Systematic reviews of 17 primary prevention trials involving a total of 47,000 participants showed that lowering SBP by 10–12 mmHg and DBP by 5–6 mmHg leads to a 38% reduction in the risk of stroke.^{260(Level 1)}

The benefits have been shown in both systolic-diastolic hypertension and in isolated systolic hypertension.^{261,262, (Level 1)} All classes of antihypertensives have the potential to prevent stroke. Calcium channel blockers in particular, provided significantly better protection against stroke compared with diuretics and/or β -blockers in Asian^{263(Level 1)} and Caucasian^{115(Level 1)} populations. In the elderly^{264(Level 1)} and very elderly^{113(Level 1)} hypertensives, diuretics has been shown to prevent stroke.

7.4.2 Treatment of Hypertension in Acute Stroke

Treatment of elevated BP in acute stroke is controversial.²⁶⁵ Stress-related high BP values (>140/90 mmHg) are present in up to 80% of patients with acute stroke while almost 25% of patient presents with markedly raised SBP values >180 mmHg.^{266,267} In a majority of patients, a decline in blood pressure without any specific medical treatment will occur within days or weeks.^{268,269} A slightly higher systemic BP is required to maintain the cerebral perfusion in the situation of increased intracranial pressure, partial thrombosis and disturbed cerebral perfusion (See Figure 7.4-A).

7.4.2.1 Ischaemic Stroke (IS)

Current guidelines recommend that treatment of hypertension in acute IS should be delayed for several days or up to 2 weeks after an IS unless there is hypertensive encephalopathy, severe left ventricular failure, acute renal failure, acute myocardial infarction, aortic dissection, acute pulmonary oedema or repeated BP readings reveal SBP values >220 mmHg and DBP >120 mmHg.²⁷⁰ (Table 7.4-A). Existing antihypertensive medications during the acute phase of stroke should be deferred until patients have suitable enteral access and are medically and neurologically stable.²⁶⁵ (Level III)

In cases where acute BP reduction is indicated, BP lowering should be done cautiously targeting BP reduction of 10 to 20% from the baseline BP over 24 hours. More profound BP reductions ($>20\%$) have been associated with neurological and functional worsening.²⁷⁰ BP management in patients for thrombolysis/thrombectomy will not be discussed in this guideline (Discussed in the ischaemic stroke CPG).

7.4.2.2 Haemorrhagic Stroke (HS)

Current recommendations for treatment of elevated BP levels in patients with acute HS are more aggressive than those with IS²⁷¹(Level III) (Table 7.4-A). Both the Antihypertensive Treatment of Acute Cerebral Hemorrhage (ATACH) trial,²⁷¹(Level 1) a 4-tier dose-escalation study of intravenous nicardipine-based BP lowering in 80 patients within 3 hours of ICH, and the pilot phase Intensive Blood Pressure Reduction in Acute Cerebral Hemorrhage (INTERACT1)²⁷²(Level 1) trial in 404 mainly Chinese patients within 6 hours of ICH found rapid reduction of SBP to <140 mmHg to be safe. The INTERACT2 trial has shown no increase in death or serious adverse events from early intensive BP lowering in eligible patients with elevated SBP. INTERACT2²⁷³(Level 1) trial and subsequent meta-analysis²⁷⁴ have suggested that acute aggressive lowering of SBP to ≤ 140 mmHg within 3-6 hours of onset provide some evidence to indicate improved functional recovery. The more recent ATACH-II trial enrolled patients within 4.5 hours of ICH onset. They were randomly assigned to blood pressure reduction with intravenous nicardipine to achieve systolic pressures in the range of 140 to 179 mmHg (standard care) or 110 to 139 mmHg (intensive blood pressure lowering). This trial took more aggressive and faster approach in BP lowering. Unfortunately, it did not show any significant outcome in mortality and functional recovery. There were significantly more renal adverse events within 7 days after randomisation in the intensive-treatment group.²⁷⁵

Parenteral agents, such as labetalol or nicardipine that are easily titrated and have minimal vasodilatory effects on cerebral blood flow are preferred. Otherwise, easily titratable intravenous medications can also be used. The use of sublingual nifedipine should be avoided because of the risk of abrupt BP reduction and possible worsening ischaemia.^{270,273}(Level III)

TABLE 7.4-A Current Guideline for the Management of Blood Pressure in Acute Phase of Ischaemic and Haemorrhagic Stroke*

Acute phase of ischaemic stroke ^{270,273,276}	
BP level, mmHg	Treatment
SBP ≤220 or DBP ≤120	Defer anti-hypertensive therapy
SBP >220 or DBP >120	i. Labetolol. 20 mg injected slowly for at least 2 min; followed by 40-80 mg every 10 min. Max: 200 mg. ii. Nitroglycerine. Initial: 5-25 mcg/min. Usual range: 10-200 mcg/min; up to 400 mcg/min in some cases. iii. Nicardipine. Slow IVI at an initial rate of 5 mg/hr. Increase infusion rate as necessary, up to max 15 mg/hr. Consider reducing to 3 mg/hr after response is achieved. iv. Sodium Nitroprusside. Initial: 0.3-1.5 mcg/kg/min, adjust gradually as needed. Usual: 0.5-6 mcg/kg/min. Max rate: 8 mcg/kg/min, discontinue if there is no response after 10 mins. May continue for a few hours if there is response. v. Target: 10–20% reduction from baseline BP over 24 hours.
Acute phase of haemorrhagic stroke ²⁷⁶	
SBP 150 – 220 mmHg	Avoid aggressive SBP lowering to <140 mmHg.
SBP >220 mmHg	Consider aggressive BP lowering within 6 hours with continuous intravenous infusion and close BP monitoring. Target: SBP lowering towards 140 mmHg.

* Referenced from 153rd Edition, MIMS, 2018.

7.4.3 Secondary Prevention of Stroke

Patients who have had a stroke or a Transient Ischaemic Attack (TIA) are at increased risk of future stroke, especially in the following few months.²⁷⁷ Annual recurrence rate is 12.5% per year.²⁷⁸ Survival rates decreased from 63.7%, 42.8% and 24% at 1, 5 and 10 years respectively. Of those who survived at 10 years, almost a third had poor range of clinical outcomes.²⁷⁹

Lowering BP has been shown to reduce the risk of subsequent strokes.^{280,281} Meta analyses of randomised controlled trials confirm approximately 30–40% reduction in stroke risk with blood pressure lowering.²⁸²

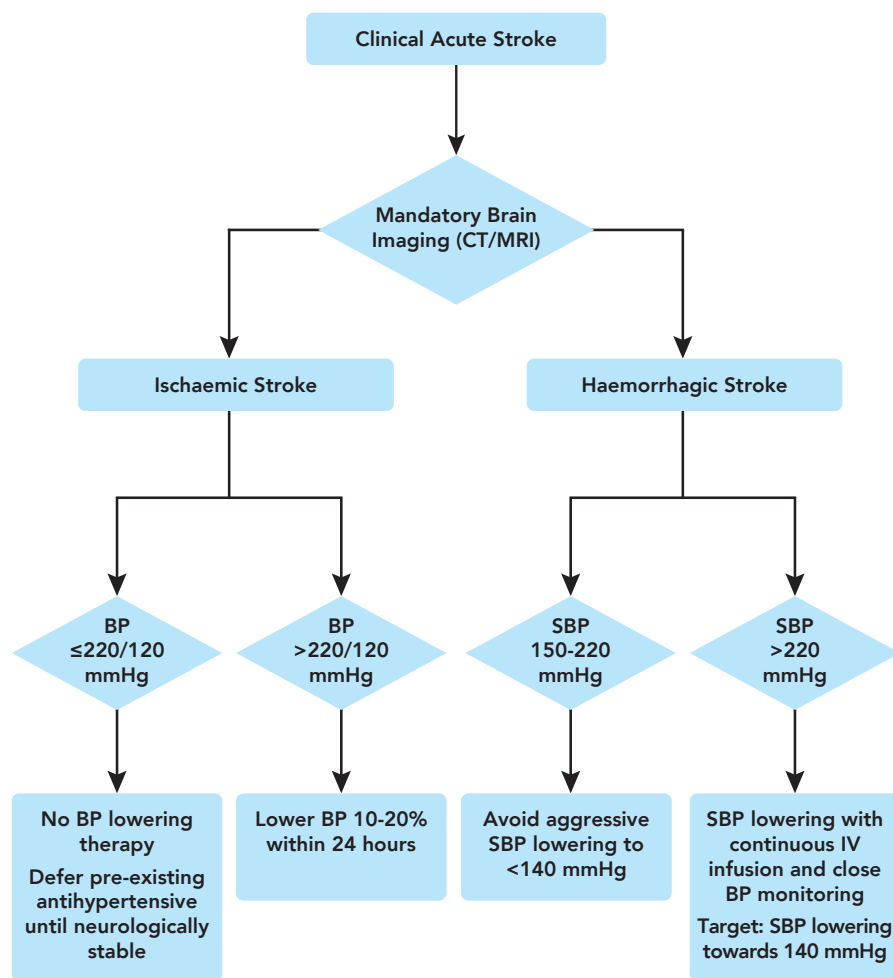
Combination of ACEI and thiazide-like diuretic has been shown to reduce stroke recurrence in both normotensive and hypertensive patients when treatment was started at least two weeks after the stroke.^{79(Level 1)}

Three additional large-scale randomised trials of antihypertensive medications after stroke have been published. In one such trial, patient with hypertension and a stroke or TIA within 2 years of the event were randomised to an ARB or CCB. Despite similar BP reductions, recurrent total strokes and TIAs were less frequent among those randomised to ARB. There was a reduction in primary composite events which were significantly lower with ARB, with reduction in TIAs accounting for most of the benefit of ARB.^{283(Level 1)} However in a bigger trial, patients with Ischaemic stroke were randomised to ARB or placebo within 90 days of an event with no clear benefits of ARB in preventing recurrent stroke after 2.5 years of follow-up.^{284(Level 1)} In another placebo control trial involving ARB, patients with IS were randomised within 30 hours following onset of symptoms.^{285(Level 1)} At 6 months follow-up, there were no significant difference in the composite primary endpoint (stroke, myocardial infarction, or vascular death). Taken together, a specific role for ARB in secondary stroke prevention cannot be confirmed.^{286(Level 1)}

The target BP after a stroke is less clear. More recent guidelines suggested a target of <140/90 mmHg^{26(Level III)} but the most recent major outcome trial suggest that for patients with recent lacunar stroke, a target of <130/80 mmHg is beneficial^{287(Level 1)} especially for prevention of intracranial haemorrhage.

RECOMMENDATIONS

- Treat blood pressure to prevent both primary and secondary stroke. (Grade A)
- Do not lower SBP <180 mmHg in the first 2 weeks in acute ischaemic stroke patients unless hypertensive emergencies co-exist. (Grade C)
- Do not lower SBP to <140 mmHg in patients presenting within 6 hours of haemorrhagic stroke (HS) and BP of <220 mmHg. (Grade C)
- Consider aggressive reduction of BP in HS patients presenting with SBP >220 mmHg with continuous intravenous infusion of antihypertensive and frequent BP monitoring. (Grade C)
- Avoid lowering BP abruptly with sublingual nifedipine in acute stroke. (Grade C)
- Lower BP to be <140/90 mmHg in both normotensive and hypertensive patients for secondary prevention. (Grade A)
- Lower BP to <130/80 mmHg for secondary prevention in lacunar stroke. (Grade A)

FIGURE 7.4-A Treatment Algorithm for Acute Stroke

7.5 Hypertension in the Older Adults

Hypertension is a very common modifiable risk factor for cardiovascular morbidity and mortality in older people. The definition of hypertension in the older adult (>65 years old) is the same as that of the general adult population. Hypertension is an increasingly important public health concern as our population ages. The prevalence of hypertension in adults >65 in Malaysia has been reported to be 71.7%,²⁸⁸ and the proportion of older adults in Malaysia is expected to increase from 9.0% in 2017 to 15% of the total population by 2030.²⁸⁹

Hypertension magnifies the risk for cardiovascular disease, with each 20 mmHg increase in systolic blood pressure and 10 mmHg increase in diastolic BP associated with a doubling in risk of death from stroke and coronary artery disease.²⁵⁵ In older adults, the risk of cardiovascular events and death is twice as that observed in younger individuals at same levels of BP.⁴ SBP increases linearly with age, leading to an increase in prevalence of isolated systolic hypertension in the older adult. SBP is a better predictor of cardiovascular events than DBP.²⁹⁰

Treatment of hypertension in the older adult, particularly of high SBP, significantly reduces the risk of all-cause mortality, cardiovascular mortality, and cardiovascular events (myocardial infarction, heart failure and stroke).^{6,79,83,113,117,151,253,262,263,287,291-295}(Level 1) There is evidence that this benefit extends to even the very elderly (>80 years old).¹¹³(Level 1)

Despite the profusion of published data, there remains ongoing debate regarding optimal management of BP in the older adults, with various guidelines espousing different targets and treatment recommendations. The term 'hypertension in older adults' itself causes some difficulty as it encompasses the age groups of the 'young-old' (65-70 years) to the 'old-old' (>80 years). This group of patients is extremely heterogeneous with regards to comorbidities, frailty, and physical and cognitive functioning. Hypertension trials are also widely variable in terms of age cut-offs, inclusion and exclusion criteria, patient cardiovascular risk profiles, treatment regimens and method of BP measurement.

Current evidence supports treating the older adult to a target SBP of <150 mmHg to improve all cardiovascular outcomes.^{6,113,262,293}(Level 1) There is some evidence that targeting a SBP of <140 mmHg may be beneficial, especially in reducing risk of stroke,^{296,297} whilst a recent trial supports even stricter targets (SBP <130 mmHg).¹¹⁷(Level 1)

7.5.1 Considerations in the Older Adults

Hypertension management in older adults are often complicated by the various pathologies associated with aging. There are numerous challenges including multiple comorbidities, postural hypotension, falls, functional and cognitive impairment and frailty. These conditions frequently overlap. The absolute benefit of aggressive BP treatment in older adults with multiple comorbidities and frailty is not well known.

7.5.1.1 Multiple Comorbidities

Multiple morbidity increases with age and its prevalence is estimated to be about 65% and 82% for those aged 65-84 and 85 years and above respectively.²⁹⁸ Multimorbidity is defined as the presence of 2 or more long term conditions.²⁹⁹ At least two-thirds of hypertensive patients have another chronic disease.

There is limited evidence from randomised controlled trials to guide management of hypertension in this group of patients.³⁰⁰ Observational studies imply that a SBP of 140-160 mmHg is associated with better mortality outcomes in older adults with impaired physical or cognitive functioning.³⁰¹⁻³⁰⁷(Level II-2) Observational data have also found that treating to lower targets (SBP <120 mmHg and DBP <70 mmHg) may increase the risk of death and cardiac events in older high-risk individuals (those with CAD, LVH and diabetes mellitus),³⁰⁸⁻³¹⁴(Level II-2) supporting a J-curve association between both SBP and DBP and adverse outcomes in older persons.

7.5.1.2 Polypharmacy and Adverse Drug Reactions

Polypharmacy is another feature associated with older adults and has been shown to be related to poor outcomes including postural hypotension, falls, electrolyte disturbances, heart failure, hospitalisation and mortality.³¹⁵ Antihypertensives commonly contribute to polypharmacy in older adults.³¹⁶ Adverse drug reactions are more frequent and often more severe in older adults due to physiological changes affecting drug pharmacokinetics and pharmacodynamics, presence of multiple comorbidities, and polypharmacy. Therefore, it is important to critically evaluate the need for each medication.

7.5.1.3 Postural Hypotension and Falls

Advancing age is associated with an independent increase in prevalence of postural hypotension. Postural hypotension is defined as a sustained reduction in SBP of at least 20 mmHg or DBP of 10 mmHg from lying to standing position.³¹⁷ In symptomatic older adults it is recommended to check BP up to 3 minutes of standing. Both uncontrolled hypertension, especially ISH, and aggressive BP treatment have been associated with postural hypotension.³¹⁸⁻³²²(Level II-1) Postural hypotension, symptomatic or not, is associated with falls in older adults.^{318,323,324}(Level II-2) Less strict BP targets may therefore be acceptable in the very elderly, the frail, those with multimorbidities and previous fallers.³²²(Level III)

7.5.1.4 Cognition

In addition to cardiovascular disease, another pertinent health concern for the ageing Malaysian population is cognitive decline and dementia. Dementia has a devastating impact on quality of life, and is associated with significant escalation

of healthcare expenditure. Hypertension in midlife (40-65 years old) has been found in longitudinal studies to be a risk factor for developing cognitive decline in later life. Hypertension predisposes mainly to development of vascular cognitive impairment, but has also been found to be a risk factor for Alzheimer's pathology.^{325,326}(Level II-2)

Evidence from randomised controlled trials on the effect of antihypertensive treatment on incident cognitive decline and dementia in older adults has largely been negative, due to most trials being of insufficient duration,^{6,295,327-329} although observational studies of up to 20 years suggest a benefit.³⁰¹ Conversely, observational data suggest that aggressive treatment of hypertension (to <130 mmHg SBP and <70 mmHg DBP) in older adults is associated with increased risk of new-onset cognitive impairment,^{301,325,326} and more rapid cognitive decline in individuals with established dementia.³³⁰

7.5.1.5 Frailty

Frailty increases in the older adult, especially in those over 80.³³¹ It is defined as increased vulnerability to physical stressors as a result of reduced physiological reserve.³³² The HYVET trial showed benefit of treating hypertension in adults >80 years of age and did not find any effect of frailty on the benefit of hypertensive treatment. However, it should be noted that HYVET excluded the very frail. Post-hoc analysis of the SHEP trial³³³(Level II-1) found that the benefit of treatment on cardiovascular and all-cause mortality disappeared in those with functional impairment. Observational studies have also demonstrated the importance of frailty status on hypertension and outcomes.^{301,303}(Level II-2) Therefore, results from the HYVET and SPRINT trials which support treatment in the very elderly and intensive therapy respectively need to be interpreted with caution. Goals should be driven by patients' functional status and comorbidities.

7.5.2 Assessment

Recommendations for BP measurements in older adults are similar to those for the general population. If there is presence of postural hypotension, the standing BP is used to guide treatment decisions.

It is very important to ascertain hypertension and the true level of BP before commencing or adding pharmacological therapy. We advocate multiple readings using an automated office blood pressure monitor after up to 3 minutes of quiet rest.

Evaluation of older patients with hypertension should not differ from that of younger adult populations. In cases of resistant hypertension, secondary causes such as atheromatous renal artery disease should similarly be ruled out.

Where appropriate, one should consider a formal frailty assessment using one of the validated tools.^{334(Level II)} At a minimum, one should observe for a reduction in mobility, decreased functional ability and impaired cognition.

7.5.3 Treatment

The >65's are the most medically heterogeneous population and a simple 'one size fits all' approach would not be appropriate. Individualised decision in the context of comorbidities and patient tolerance to medication(s) is important.

Therapy should be started cautiously with monotherapy at a low dose and titrated upwards slowly. The patient should be reviewed frequently in the initial stage (2 – 4 weeks). Initiation with combination therapy is not encouraged and considered only after failure of initial therapy.

In the presence of ADRs such as postural hypotension and falls, de-prescribing should be considered. There is evidence that de-prescribing does not result in an increase in mortality in this group of patients.³³⁵ This will also improve pill burden for the patient and address polypharmacy.

Treat the older adult when SBP is >160 mmHg. Treatment targets are as stated below:

TABLE 7.5-A Treatment SBP Targets for Older Adults

Older Adult Population	Target SBP (mmHg)
>80 years old	<150
65-80 years old	<140
Multiple comorbidities Functional and cognitive impairment Frail Institutionalized Experiencing ADRs	Consider less strict targets Limit number of antihypertensive agents

For fit* 65-80 years old patients consider target SBP <130 mmHg.

* free from health conditions that limit mobility and/or functional ability with good nutrition and cognitive status.

Non-pharmacological Management (refer to chapter 4)

Non-pharmacological interventions, particularly sodium restriction and weight loss, have been proven to be efficacious in the older adults.^{336(Level 1)} Refer to chapter 4 for more on non-pharmacological methods.

7.5.4 Conclusion

There are many challenges in treating hypertension in the older adult. Managing blood pressure in isolation is not conducive to achieving a patient-centered approach. BP targets depend on many factors unique to each patient. Healthcare providers must be mindful that in older adults, additional aspects must be considered before starting therapy. These include frailty, physical and cognitive functioning, and tolerance to treatment. Less strict BP targets may be considered in certain situations.

SUMMARY

- Treatment of hypertension in the older adult, particularly of high SBP, significantly reduces the risk of all-cause mortality, cardiovascular mortality, and cardiovascular events (myocardial infarction, heart failure and stroke), even in the very elderly.
- There is strong evidence of benefit for treating to target SBP <150 mmHg.
- There is some evidence for treating to even lower target SBPs in certain subgroups of older adults.
- Older patients with multimorbidities, cognitive or functional impairment and frailty are under represented in randomised control trials for hypertension. Observational studies suggest worse outcomes in these patients.

RECOMMENDATIONS

- Measure standing BP and use it to guide treatment decision. (Grade C)
- Assess comprehensively to confirm hypertension. (Grade C)
- Assess for frailty, mobility, function, cognition, nutrition, postural hypotension and falls. (Grade C)
- Individualised treatment based on clinical scenarios. (Grade C)
- Target SBP <150 mmHg for >80 year olds. (Grade A)
- Target SBP <140 mmHg for 65-80 year olds. (Grade B)
- Consider SBP <130 mmHg in fit 65-80 year olds. (Grade A)
- Apply less strict targets for the frail, functionally and/or cognitively-impaired, those with multi-morbidities and those with adverse reactions from therapy. Consider de-prescribing in this group of patients. (Grade C)

7.6 Hypertension in Women

7.6.1 Hypertension in Pregnancy

Hypertension in pregnancy is defined as systolic blood pressure (SBP) ≥ 140 mmHg and/or diastolic blood pressure (DBP) ≥ 90 mmHg.³³⁷ An increase of SBP of 30 mmHg and DBP of 15 mmHg above baseline BP is no longer recognised as hypertension if absolute values are below 140/90 mmHg. Nevertheless, this warrants close observation, especially if proteinuria and hyperuricaemia are also present.³³⁸ Korotkoff V should be used as the cut-off point for DBP, and Korotkoff IV utilised only when Korotkoff V is absent.^{337(Level III)} The gold standard tool to measure blood pressure in preeclampsia is still the mercury sphygmomanometer.³³⁷ In the absence of a mercury sphygmomanometer, an automated blood pressure measuring device can be used provided it is calibrated annually against the mercury sphygmomanometer. Automated devices are acceptable for BP measurement in pregnancy based on a prospective study, which showed that there was no difference in the maternal and fetal outcomes despite lower DBP reading using the automated device.³³⁹

7.6.1.1 Proteinuria

Significant proteinuria in pregnancy is defined as ≥ 300 mg protein in a 24-hour urine sample, or a spot urine protein-creatinine ratio ≥ 30 mg/mmol.³³⁷ If the dipstick is the only test available, 2+ is approximated to ≥ 300 mg/day proteinuria.³³⁷ Significant proteinuria reflects advanced disease and is associated with poorer prognosis.

7.6.1.2 Classification

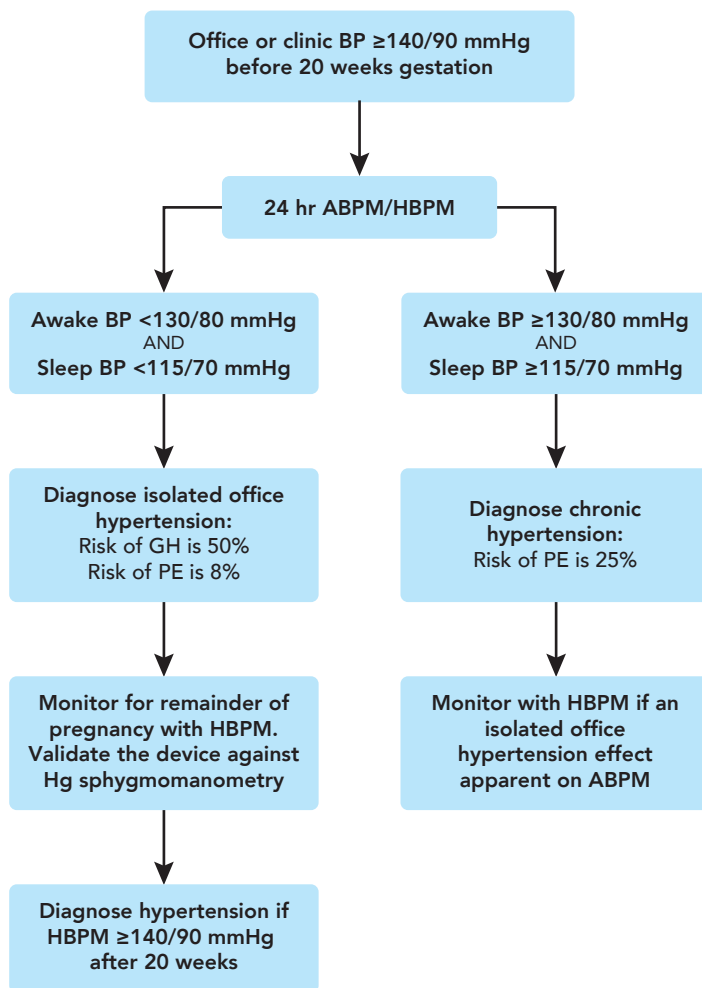
There are various classifications for Hypertension in Pregnancy. The most recent is by the International Society for the Study of Hypertension in Pregnancy (ISSHP).³³⁷

1. Preeclampsia (PE): de novo or superimposed on chronic hypertension
 - a. PE is clinically diagnosed in the presence of de novo hypertension after 20 weeks gestation, with one or more of the following:
 - i. Significant proteinuria
 - ii. Renal insufficiency: serum creatinine ≥ 90 micromol/l or oliguria
 - iii. Liver disease: raised transaminases and/or severe right upper quadrant or epigastric pain
 - iv. Neurological problems: convulsions (eclampsia), hyperreflexia with clonus or severe headaches, persistent visual disturbances (scotoma)

- v. Haematological disturbances: thrombocytopenia, coagulopathy, haemolysis
 - vi. Fetal growth restriction
- b. PE superimposed on chronic hypertension is diagnosed in the presence of any of the following, in a woman with chronic hypertension:
- i. De novo proteinuria after 20 weeks gestation
 - ii. A sudden increase in the severity of hypertension
 - iii. Appearance of features of PE-eclampsia, and
 - iv. Worsening proteinuria in a woman with pre-existing proteinuria early in gestation

This is followed by normalisation of the BP by three months postpartum. Oedema is no longer part of the definition of PE.³⁴⁰ Excessive weight gain or failure to gain weight in pregnancy may herald the onset of PE.³⁴¹

2. Gestational hypertension is defined as hypertension detected for the first time after 20 weeks gestation. Although it usually runs a benign course, it can progress into PE in 25% of cases, more so if it presents before 34 weeks.³³⁷
3. Isolated office hypertension is defined as elevated BP of 140/90 mmHg only in the clinic with normal BP demonstrated by ambulatory BP monitoring (ABPM) either awake or during sleep. In the absence of ABPM device, HBPM can be used. Studies in non-pregnant population showed that they are comparable.³⁴² Women in this group should not be considered low risk as they may progress to gestational hypertension (50%) or PE (8%).^{337,343}
4. Chronic hypertension is hypertension diagnosed prior to 20 weeks gestation or presence of hypertension preconception, or de novo hypertension in late gestation that fails to resolve three months postpartum.

FIGURE 7.6-A ABPM to Diagnose and Manage Isolated Office Hypertension in Pregnancy.*

ABPM = Ambulatory Blood Pressure Monitoring

HBPM = Home Blood Pressure Monitoring

* adapted from Brown MA 2014.³⁴⁴

7.6.1.3 Key Points in Primary Care Practice

The primary care physician plays an important role in the prevention and early detection of PE and its complications. An obstetrician should lead the joint management of women with hypertensive disorders in pregnancy.

1. Preconception counseling and adjustment of treatment in women with chronic hypertension.

Women with chronic hypertension may require a change in the type of anti-hypertensive agent used pre-pregnancy.^{345(Level III)} The drugs of choice in pregnancy are methyldopa and labetalol (Table 7.6-A). Atenolol has been shown to lead to fetal growth restriction. The use of ARBs, ACEIs and thiazide diuretics are associated with fetal anomaly and are therefore contraindicated in pregnancy.³⁴⁶ Women in the reproductive age group requiring these drugs should be on effective contraception. In the event of unplanned pregnancy, the drugs must be stopped.³⁴⁷

It should be noted that the treatment of hypertension in pregnancy is solely for maternal safety particularly for prevention of intracranial bleeding. It does not reduce the risk of development of preeclampsia or perinatal mortality, nor improve fetal growth.³⁴⁸ There is still no clear evidence on the target BP that should be achieved prior to or during pregnancy. A recent study comparing less tight (target DBP of 100 mmHg) against tight (target DBP 85 mmHg) control of BP in women with non-proteinuric chronic hypertension or gestational hypertension showed no significant differences between these two groups with regard to both maternal and perinatal complications.³⁴⁹

2. Recognition of women at risk of preeclampsia for commencement of prophylaxis.

Risk	Risk Factors
Moderate ³⁴⁹	• primigravida • age >40 years • pregnancy interval >10 years • body mass index of >35 kg/m² at first visit • family history of PE • multiple pregnancy
High ³⁴⁹	• hypertensive disease during previous pregnancy • chronic kidney disease • autoimmune disease such as Systemic Lupus Erythematosus (SLE) or anti-phospholipid syndrome (APS) • type 1 or type 2 diabetes mellitus, and • chronic hypertension

3. Prophylactic therapy

a. Aspirin

Women with ≥ 2 moderate or one high risk factor should be started on low dose aspirin from 12 weeks up to 16 weeks of gestation until delivery.³⁵⁰ The dosage should be 100-150 mg and taken at bedtime in order to significantly reduce the incidence of PE.³⁵¹⁻³⁵³

b. Calcium

A systematic review showed that low dose calcium supplement (generally 500-1000mg daily) commenced before 20 weeks gestation reduces the risk of PE.³⁵⁴

c. Vitamin D

There is no proven role of vitamin D in reducing the risk of PE.³⁵⁵

d. Others

Other supplements in pregnancy such as marine oil, garlic, and pyridoxine have no proven benefits.³⁵⁶⁻³⁵⁸ Combined vitamin C and E (i.e. tocopherol from soybean) should be avoided because they significantly increase the incidence of low birth weight without any preventive effect against PE.³⁵⁹

4. Prediction of the incidence of preeclampsia

A predictive test that is available locally, measuring serum sFlt-1/PlGF ratio from 20 weeks of gestation onwards has good negative predictive value for a week. It is useful in identifying which patients require admission and close monitoring. It will also help to decide on the need for antenatal corticosteroids in anticipation of pre-term delivery.³⁶⁰

5. Fetal anomaly screening

Women with chronic hypertension have about 20-30% increased risk for fetal congenital cardiac anomaly.³⁶¹ These women are to be referred to the Maternal-Fetal Medicine (MFM) specialist in the tertiary centre to be recommended to undergo nuchal translucency (NT) scan at 12-14 weeks followed by a detailed ultrasound scan at 22-24 weeks of gestation. If a cardiac anomaly is detected, cardiology referral is recommended.

6. Prevention of eclampsia and other complications of preeclampsia

Patient and healthcare provider education on the importance of signs and symptoms of preeclampsia for early diagnosis and referral for further management may prevent progression to eclampsia.^{362,363}(Level II-2)

7.6.1.4 Severe Preeclampsia

Severe preeclampsia must be promptly identified so that the patient can be urgently admitted to hospital for close observation and timely delivery. The American College of Obstetricians and Gynecologists defines severe preeclampsia based on the following features:³⁶⁴

- a. Systolic BP ≥ 160 mmHg or diastolic BP ≥ 110 mmHg on two occasions at least 4 hours apart while the patient is resting
- b. Thrombocytopenia – platelet count below $100,000/\text{cm}^3$
- c. Abnormal liver enzymes (elevated AST/ALT), severe persistent right upper quadrant or epigastric pain unresponsive to treatment
- d. Pulmonary oedema
- e. New onset of cerebral or visual disturbances

Diagnosis of severe preeclampsia should not depend solely on the above criteria. If in doubt, it is better to over rather than under diagnose. This will prevent delay in referral. Any patient with preeclampsia should be closely monitored, as the progression to severe preeclampsia is unpredictable and rapid.

In the event of an acute hypertensive crisis, IV hydralazine, IV labetalol, or oral nifedipine, may be used to lower the BP.^{365,366}(Level I) Sublingual nifedipine is no longer recommended (Table 7.6-B).³⁶⁷(Level III) Diuretics are generally contraindicated as they reduce plasma volume, may cause Intrauterine Growth Restriction (IUGR) and may possibly increase perinatal mortality. Their only use is in the treatment of acute pulmonary oedema.³⁴¹(Level III) In order to reduce the risk of maternal stroke, the blood pressure should be reduced within 30-60 minutes.³⁶⁸

7.6.1.5 Anticonvulsants in Preeclampsia-Eclampsia

Parenteral magnesium sulphate is currently the drug of choice for the prevention of eclampsia and to abort an eclamptic fit (Table 7.6-C).^{367,369}(Level 1) The alternative is intravenous diazepam, bearing in mind that it is inferior in efficacy compared to magnesium sulphate. Magnesium sulphate also provides fetal neuroprotection following preterm birth with a significant reduction in the incidence of cerebral palsy.³⁷⁰

7.6.1.6 Postpartum Care

Postpartum, women with hypertensive disorders in pregnancy are advised to have their BP checked regularly at local clinics if there is a significant delay in their scheduled hospital follow-up. In these patients, the dose of anti-hypertensive should be tailed down gradually and not stopped suddenly.

On average, anti-hypertensive agents are required for longer in women with preeclampsia (approximately two weeks) compared with those with gestational hypertension (approximately one week) although there is substantial variability among women that cannot be predicted reliably.³⁷¹

De novo onset of hypertension or aggravation of BP levels during the postpartum period can occur.³⁷² These patients should be promptly referred to hospital especially if there is significant proteinuria.^(Level III) Eclampsia may occur in the postpartum period. Chronic hypertension is diagnosed when the hypertension and/or proteinuria persist after three months postpartum.^{337,338}

7.6.1.7 Long Term Follow-Up

Evidence suggests that up to 13% of women with preeclampsia will have underlying essential hypertension that was not suspected antenatally.³⁷³ In addition, following severe preeclampsia, there is an increased risk of ischaemic heart disease, thromboembolism and stroke.³⁷⁴ Long-term follow-up of patients with a history of hypertension in pregnancy is therefore advisable.^(Level III)

7.6.1.8 Reducing Mortality

A substantial reduction in preeclampsia/eclampsia related mortality could be achieved by widespread screening for hypertension and proteinuria. Early referral and delivery is indicated for severe PE.³⁷⁵

TABLE 7.6-A Anti-Hypertensive Drugs Commonly Used in Pregnancy

Drug	Remarks
Methyldopa (first line)	Oral 250 mg tds, doubling every 48 hrs (up to 1 gm tds) until BP well controlled. Oldest anti-hypertensive agent used in pregnancy, with best safety profile.
Labetalol (alternative first line)	Oral 100 mg bd, doubling every 48 hrs (up to 400mg bd) until BP well controlled.
Nifedipine (second line)	Oral 10 mg tds, up to 20 mg tds, when BP poorly controlled despite maximum doses of methyldopa ± labetalol.

TABLE 7.6-B Anti-Hypertensive Drugs for Severe Preeclampsia with Acute Hypertensive Crisis³⁷⁶

Drug	Administration	Remarks
Labetalol	In IV bolus: • 20 mg then 40 mg 10–20 mins later • 80 mg every 10–15 mins up to 200 mg Infusion: continuous infusion of 1–2 mg/min until BP stabilises, then stop or reduce to 0.5 mg/min.	May cause fetal bradycardia.
Nifedipine	Oral 5–10 mg stat (repeat in 30 mins if necessary). After the initial emergency dose, 10–20 mg can be given every 3–6 hrs until BP stabilises.	Especially prior to transferring a patient from a peripheral clinic to hospital.
Hydralazine	Initial: 5–10 mg via slow inj, may repeat after 20–30 min. Alternatively, as a continuous infusion, initial dose of 0.2–0.3 mg/min. Maintenance: 0.05–0.15 mg/min.	No longer recommended as first line treatment for acute hypertensive crisis in pregnancy. ³⁷⁴

TABLE 7.6-C Anti-Convulsant for Eclampsia (and Severe Preeclampsia)

Drug	Administration	Remarks
Magnesium Sulphate^{377,378}	IV: • 4g slow bolus over 10 mins • followed by 1–2 g/hr maintenance infusion given via a controlled infusion pump IM (deep): • 10g loading dose • Followed by 5 g every 4 hrs in alternate buttock*	Clinical monitoring is important looking for signs of toxicity: • loss of deep tendon reflexes • respiratory depression with rate <16/min • renal impairment (hourly urine output <30 ml/hr)
Diazepam³⁷⁸	• 10 mg IV bolus, followed by 40 mg in D5% slow infusion so that patient remains sedated	• Only when magnesium sulphate is contraindicated or not available

RECOMMENDATIONS

- Use Korotkoff V to diagnose and monitor to treatment of hypertension in pregnant women. (Grade C)
- Consider automated BP device instead of mercury sphygmomanometer to diagnose and monitor treatment. (Grade A)
- Provide counselling and appropriate management to women with chronic hypertension and who are planning for pregnancy. (Grade C)
- Avoid RAS blockers in all women of childbearing potential unless adequate precaution has been taken against pregnancy. (Grade A)
- Refer pregnant women with hypertension to the obstetrician for further management. (Grade C)
- Provide calcium supplementation from early pregnancy to prevent PE. (Grade A)
- Commence aspirin from 12-16 weeks and continue until delivery in pregnant women with one or more high risk factors or two or more moderate risk factors for PE. (Grade A)
- Use oral nifedipine 10 mg stat dose to rapidly control BP in acute hypertensive crisis prior to transfer to hospital. (Grade C)

7.6.2 Hypertension and Oral Contraceptives

Combined oral contraceptives (COC) can induce significant increases in BP with chronic use, which is nearly always reversible after 4 weeks of discontinuation.³⁷⁹ Hypertension has been reported even with low-dose-oestrogen monophasic pills.³⁸⁰ A woman who develops hypertension while using COC should be advised to stop taking them and should be offered alternative forms of contraception.^{381(Level III)} Low dose combined hormonal contraceptives should only be used if no other method is suitable, even for women with controlled hypertension.³⁸²

Drospirenone (a progestin), has anti-mineralocorticoid diuretic effects, and can lower BP when combined with oestrogen in COCs.³⁸³ It is a recommended alternative for patients with hypertension or who developed hypertension but wish to continue oral contraception. All progestogen-only methods are appropriate except in women whose BP is higher than 160/100 mmHg. In these patients, the injectable depot medroxyprogesterone acetate (DMPA) is contraindicated, along with all oestrogen-containing contraceptives.³⁸¹

Baseline BP must be assessed before initiating hormonal contraceptives. Blood pressure should then be measured at least every six months.^(Level III) The same applies to usage of the combined contraceptive patch and the vaginal ring.

7.6.3 Hypertension and Menopausal Hormonal Therapy

The presence of hypertension is not a contraindication to oestrogen-based menopausal hormonal therapy (MHT). It is recommended that all women treated with MHT should have their BP monitored every six months.^(Level III) The decision to continue or discontinue hormonal therapy in these patients should be individualised.

Two large trials on women aged 50-79 years, concluded that the use of MHT increased cardiovascular events.^{384,385 (Level I)} Conjugated equine estrogen (CEE), alone or in combination with medroxyprogesterone acetate, was used in the study. In view of this, greater caution and closer monitoring is required for hypertensive patients on CEE. Drospirenone when used as progestin in HRT, showed improvement in BP control.^{379,386}

TABLE 7.6-D COC and Hormonal Therapy Preparations Containing Drospirenone*

Hormonal Preparation	Trade Name	Active Ingredients	
		Oestrogen	Progestin
COC	Yasmin® / Liza®	Ethinyl oestradiol 0.03 mg	Drospirenone 3 mg
	Yaz® / Liz® / Lizelle®	Ethinyl oestradiol 0.02 mg	Drospirenone 3 mg
MHT	Angeliq®	Estradiol 1 mg	Drospirenone 2 mg

* Referenced from 153rd Edition, MIMS, 2018.

7.7 Hypertension in Neonates, Children and Adolescents

Hypertension in Neonates and Infants

The incidence of hypertension in neonates admitted to Neonatal Intensive Care Unit was 1.3%.³⁸⁷

It is more common in neonates and infants with antenatal steroids, maternal hypertension, postnatal acute renal failure, chronic lung disease, patent ductus arteriosus or in those with indwelling umbilical arterial catheters.³⁸⁷ Catheter related hypertension is related to thrombus formation at the time of line placement.³⁸⁸

Measurement of BP

Healthy term neonates rarely have hypertension. Routine BP measurements are not advocated in this group. The gold standard of BP measurement in neonates is by direct measurement of arterial pulse pressure wave form.

Standardised Protocol for BP Measurement in Neonates:^{389,390}

- measure by oscillometric device
- lie prone or supine
- use appropriate sized BP cuff
- use right upper arm
- measured when infant is asleep or in quiet awake state
- 3 successive BP reading at 2 min intervals

A reference table for BP values after two weeks of age in infants from 26 to 44 weeks has been derived after taking into consideration gestational age at birth, postconceptional age and size for gestational age. The 95th and 99th percentile values are intended to serve as reference to identify infants with persistent hypertension that may require treatment.³⁸⁹ (Refer to Appendix 1)

Treatment is recommended when BP is consistently above the 99th percentile. There are few published case series that used diuretics, ACEI, β -blockers and CCB.^{389(Level II-2)}

There is concern over the use of ACEI in preterm neonates.³⁹¹ It has been reported to cause an exaggerated fall in BP and may impair the final stages of nephron maturation and its use is best avoided until 44 weeks postconceptional age.^{389,392(Level III)}

Hypertension in Children and Adolescents

In Malaysia, the prevalence of hypertension in primary school children in one study was 13.4%.³⁹³

Prevalence of hypertension in children and adolescents is increasing in tandem with the increasing prevalence of obesity in this group.^{394,395}

The NHMS 2016 has shown that the prevalence of obesity among children below 5 years was 6% (Weight for Height $>+2SD$).³⁹⁶ In 2011, the prevalence of obesity among children below 10 years was 5.3%.³⁹⁷

Measurement of BP: Who and when³⁹⁸

1. Children ≥ 7 years old

- Healthy children: Measure annually if obese
- Children with diabetes, renal disease, aortic arch obstruction, coarctation or on medications known to increase BP: Measure at every medical encounter

2. Children < 7 years who are at risk of developing hypertension

Measure at every medical encounter for those with:

- history of complications requiring neonatal intensive care
- congenital heart disease
- recurrent urinary tract infections, hematuria or proteinuria
- known renal disease or urological malformation
- family history of congenital renal disease
- solid-organ transplant
- treatment with drugs known to raise BP
- other systemic illness associated with hypertension (neurofibromatosis, tuberous sclerosis)
- evidence of raised intracranial pressure

Although the latest guidelines from the US Task force recommends the age cut off to be 3 years, we recommend 7 years taking into consideration the current state of resources in the primary health centre.^(Level III)

BP Measurement Technique

1. The initial BP measurement may be oscillometric.³⁹⁹ (On a calibrated machine that has been validated for use in the pediatric population.)
2. If BP level >90th percentile on oscillometric devices, confirmatory measurement should be obtained by auscultation. Re-measure BP twice by using auscultatory technique and average these two.
3. Measurement of BP in children follows the same principles as set out in the section on BP measurement. Special attention needs to be paid in selection of an appropriate cuff size in relation to the child's right upper arm.

SUMMARY

- The diagnosis of hypertension in children and adolescents is made when the auscultated BP values on three repeated and different visits are greater than the 95th percentile for age, sex, and height of the patient, or is $\geq 130/90$ mmHg (whichever is lower).³⁹⁸

Height and gender are important determinants of pediatric BP. BP levels are interpreted based on gender, age and height. In the 2017 American Academy of Pediatrics guidelines, normative table were revised by using data from-normal-weight children only. (Refer to Appendix 2 & 3)

TABLE 7.7-A Definitions of BP Categories, Stages, Patient Evaluation and Management (0-18 years)

Category	Children Aged 1-13 years	Children Aged ≥13 years	Frequency of BP measurement	Patient Management
Normal	<90 th percentile	<120 / <80 mmHg	Opportunistic	Lifestyle counselling
Stage 1 Hypertension	≥95 th to <95 th percentile +12 mmHg or 130/80 mmHg to 139/89 mmHg (whichever lower)	130/80 to 139/89 mmHg	Initial Recheck in 1-2 week Check upper & lower extremity BP Recheck in 3 months	Lifestyle counselling Lifestyle counselling Diagnostic evaluation Initiate treatment Specialist referral
Stage 2 Hypertension	≥95 th percentile +12 mmHg or ≥140/90 mmHg (whichever is lower)	≥140/90 mmHg	Initial Check upper & lower extremity BP Recheck within 1 week	Lifestyle counselling Diagnostic evaluation Initiate treatment Specialist referral in 1 week

If the patient BP is symptomatic or >30 mmHg above the 95th percentile (or >180/120 mmHg in an adolescent), send to an emergency department.

Once a child is diagnosed with hypertension, he should be referred to a paediatrician for further evaluation and management.⁴⁰⁰

The healthcare provider should obtain a perinatal, nutritional, physical activity, psychosocial and family history and perform a physical examination to identify finding suggestive of secondary causes of hypertension.

Primary Hypertension

Children and adolescents ≥ 6 years of age do not require an extensive evaluation for secondary causes of hypertension if they have a positive family history of hypertension, are overweight or obese, and/or do not have history or physical examination findings suggestive of a secondary cause of hypertension.

Secondary Hypertension

Causes of secondary hypertension in children:

- Renal parenchymal disease and renal structure abnormality (most common)
- Renovascular disease
- Coarctation of the aorta
- Endocrine hypertension
- Drug induced (corticosteroids)

Screening test and relevant population

1. Routine investigations to be performed in all patients
 - Urinalysis
 - Chemistry panel (electrolytes, urea, and creatinine)
 - Lipid profile
 - Renal ultrasonography in those < 6 years of age or those with abnormal urinalysis or renal function
2. Investigations to assess comorbidities (the obese child)
 - Fasting blood sugar
 - Haemoglobin A1c
 - AST, ALT
3. Optional tests to be obtained on the basis of history and initial study

Isolated Office Hypertension

A patient with BP levels $> 95^{\text{th}}$ percentile in a doctor's office but who is normotensive outside a clinical setting has "Isolated Office Hypertension".

Ambulatory blood pressure measurement is necessary to confirm hypertension in otherwise healthy children. ABPM levels should be interpreted with appropriate paediatric normative data for children > 5 years of age or height of ≥ 120 cm. Isolated

office hypertension is diagnosed by ABPM when the mean SBP and DBP <95th percentile and SBP and DBP load <25%.⁴⁰¹

Isolated office hypertension does not require treatment but may need repeat ABPM in one- to two-year intervals to detect development of sustained hypertension.

Treatment of Paediatric Hypertension

Goals of therapy for children with hypertension:

- To achieve a BP level that reduces the risk of target organ damage.
- To reduce risk of premature atherosclerosis and early development of cardiovascular disease.
- To reduce risk of developing adult hypertension and metabolic syndrome.⁴⁰²⁻⁴⁰⁴

Goals of Therapy^{398(Level III)}

Children and adolescents with hypertension	BP (Systolic and Diastolic) to <90 th percentile and <130/80 mmHg in adolescents ≥13 years old
Children and adolescents with both chronic kidney disease and hypertension	BP <50 th percentile

Lifestyle and Non-Pharmacologic Treatment

Non-pharmacologic management including dietary changes, exercise and weight reduction (if obese) is recommended in all children with hypertension.³⁹⁸

Pharmacologic Treatment

Definite indications for initiating pharmacotherapy include:³⁹⁸

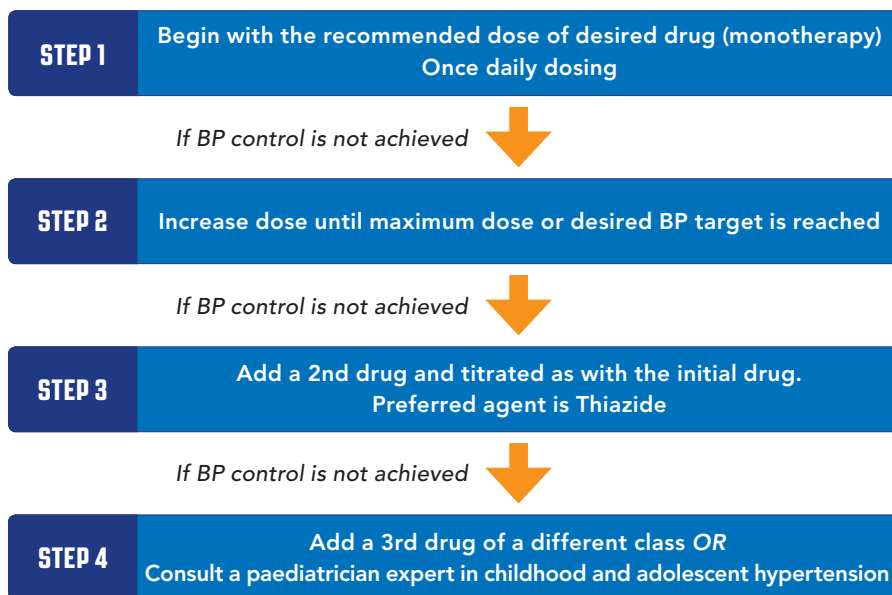
- hypertension with failed lifestyle modification
- stage 2 hypertension without a clearly modifiable factor (e.g. obesity)
- any stage of hypertension associated with chronic kidney disease or diabetes mellitus
- hypertension with target organ damage

Clinicians should initiate pharmacologic treatment with an ACEI, ARB, long-acting calcium channel blocker, or thiazide diuretic.

Stepped Care Approach

An individualised stepped care approach to the use of anti-hypertensive drugs has been recommended by the Management of High Blood Pressure in Children Clinical Practice Guideline.³⁹⁸

Stepped care approach:



Appendix 4 Dosing recommendations for the initial prescription of antihypertensive drugs for outpatient management of chronic hypertension in children and neonates.

Proteinuric Chronic Kidney Diseases³⁹⁸

ACEI or ARBs are preferred in children with proteinuric CKD.

Obese Hypertensive Children³⁹⁸

Diuretics and β -blockers are potentially diabetogenic and hence should be avoided as initial therapy in children who are obese and hypertensive.

RECOMMENDATIONS

- For children with risk factors: measure at every encounter.
- For obese children >7 years old: measure annually.
- Once a child is diagnosed with hypertension, he should be referred to a paediatrician for further evaluation and management.
- At the time of diagnosis of hypertension, clinician should provide advice on diet and recommend moderate to vigorous physical activity to help to reduce BP.
- Once pharmacologic therapy is initiated, BP must be reduced to <90th percentile (Systolic and Diastolic) and <130/80 mmHg in adolescents ≥13 years old.

8 Economic Impact of Hypertension

Hypertension and its sequelae results in a huge healthcare burden. It is the major risk factor for stroke, MI, HF and end-stage renal disease.^{258,406-408} More than half of hypertensives are unaware and almost two thirds of patients on treatment are not controlled.²⁸⁸

The cost of treating hypertension consists of direct and indirect cost. Direct cost includes the cost of treatment (drugs, investigations, healthcare providers time and transportation) whereas indirect cost measures the income lost due to hypertension and its complications. A difficult to measure indirect cost would be the cost to society in managing patients with sequelae of hypertension (e.g. managing patients with a hypertensive bleed, hypertensive heart and renal failure).

A study in a primary care setting showed that more than half of treatment cost is driven by antihypertensive medications. This cost increases as the severity of hypertension increases.^{409,410} The direct cost to the Ministry of Health for antihypertensive medication has steadily increased from RM570.3 million in 2014 to RM608.8 million in 2016.⁴¹¹

The direct and indirect cost for hypertensive complications is not available for Malaysia. A study in Malaysia showed that the estimated direct cost for complications in patients with diabetes for the outcomes of stroke, myocardial infarction and heart failure to be as high as RM12,685 per admission. The estimated cost of diabetes complication to the country annually may be as high as RM3.52 billion based on the highest estimate sensitivity analysis. It is likely that hypertension complication with its higher prevalence would cost more than this.⁴¹²

Patients with hypertension have a significantly higher risk of developing CKD which is associated with a high socioeconomic burden. The annual cost of chronic haemodialysis was RM40,557 and peritoneal dialysis was RM38,138 per patient per year in 2009. The average cost of adult living kidney transplant was RM29,482.⁴¹³ The total estimated cost for Renal Replacement Therapy (RRT) was RM1.7 billion. Hypertension contributed to at least 20% of this cost (RM340 million).

Treating hypertension is cost effective.⁴¹⁴ A recent study showed that screening and treating hypertension even for primary prevention has a high health impact (equivalent to RM50,925 per QALY adjusted for purchasing parity).⁴¹⁵ The impact on treating hypertension for secondary prevention is expected to be even higher.

Community-based interventions such as education, screening and self-monitoring have been shown to be cost effective in preventing and controlling hypertension. These initiatives targeted health behavioural changes and medication adherence. They may even reduce the cost of complications and long-term healthcare cost.^{415,416}

The vast majority of antihypertensive drugs are now off patent, making generics more affordable. For example, amlodipine became more affordable when its generic form was introduced in 2009. This was evident in the public sector as the expenditure for amlodipine dropped 10-fold, from RM85.8 million in 2008 to RM8.2 million in 2010 making hypertension treatment more cost effective.⁴¹⁷

Hypertension is responsible for at least 45% and 51% of deaths from heart disease and stroke respectively.⁴¹⁸ A more concerted effort should be taken for the early diagnosis and better control of hypertension. This will reduce the direct and indirect cost of treating hypertension and its complications to the patients, family, society and the government as a whole.

SUMMARY

- Treating hypertension is cost effective especially with the widespread availability of generic drugs.

RECOMMENDATIONS

- Conduct more awareness programmes on clinical and economic benefits in prevention and early treatment of hypertension.

9 Types of Antihypertensive Agents

9.1 DIURETICS

Diuretics, specifically thiazide diuretics, have been the mainstay of hypertension treatment, alone or in combination with other anti-hypertensive agents. Diuretics work via inducing natriuresis which alter long term sodium balance, leading to reduced peripheral vascular resistance and sustained blood pressure reduction. Diuretics provide synergistic effect to almost all anti-hypertensive agents, particularly renin-angiotensin system (RAS) blockers. There were also outcome data that supported the potential cardiovascular benefits of thiazide diuretics, particularly in those with preserved renal function.⁴¹⁹ (Level 1)

Thiazide diuretics are classified into thiazide (e.g. hydrochlorothiazide) and thiazide-like diuretics, e.g. chlorthalidone (CTD) and indapamide. Locally, hydrochlorothiazide (HCTZ) is the most commonly used thiazide diuretic. Most positive outcome studies used thiazide-like diuretics, i.e. chlorthalidone^{5,89} and indapamide.¹¹³ Even though there were no head to head trials comparing HCTZ and CTD, a systematic review and network meta-analyses reported superiority of CTD to HCTZ in preventing cardiovascular events. This difference might be attributed to the pleomorphic effects of CTD or to the shorter duration of action of HCTZ.⁴²⁰ (Level II-1)

HCTZ is effective in the range of 12.5mg – 50mg daily dose.⁴²¹ (Level II-1) However a dose above 25mg per day is more likely to cause electrolyte and metabolic adverse effects. The major out come data for the use of thiazide-like diuretics (CTD) was from ALLHAT study.⁸⁹ (Level II-1) Chlorthalidone-based regimen was equally effective in reducing clinical outcome as lisinopril and amlodipine. In chronic kidney disease with eGFR<30ml/min/1.73m², thiazide diuretics are less effective and a switch to loop diuretics is recommended.

A recent case-control study highlighted a significant increase in the risk of skin and lip squamous cell carcinoma among thiazide diuretics users.^{422,423} (Level II-2) There was a clear dose-response effect with the highest cumulative dose of HCTZ having the highest risk. There was more than 7-fold increased risk of squamous cell carcinoma for a cumulative use of ≥200,000 mg HCTZ (equivalent to 50mg daily for a duration of more than 11 years). Even though this is an observational study, the risk is not negligible.

TABLE 9.1-A Recommended Dosing for Diuretics

Diuretics	Starting Dose*	Recommended Maximum Daily Dose*
Hydrochlorothiazide	12.5 mg od	25 mg od
Amiloride/hydrochlorothiazide 5 mg/50 mg	1 tablet od	1 tablet od
Indapamide SR	1.5 mg od	1.5 mg od

* Referenced from 153rd Edition, MIMS, 2018.

9.2 Beta-Blockers (β-Blockers)

Beta-blockers have long been used in the treatment of hypertension. They are particularly useful in hypertensive patients with effort angina, tachyarrhythmias or previous myocardial infarction where they have been shown to reduce cardiovascular morbidity and mortality. Certain β-blockers have been shown to be beneficial in patients with heart failure. (Table 5-C)

Beta-blockers are absolutely contraindicated in patients with uncontrolled asthma and relatively contraindicated in other forms of obstructive airways disease (including controlled bronchial asthma). It is also absolutely contraindicated in patients with severe peripheral vascular disease and heart block (2nd and 3rd degree).

They are generally well tolerated. Adverse effects reported include dyslipidaemia, masking of hypoglycaemia, and increased incidence of new onset diabetes mellitus. Despite that, a long-term follow-up of a study in newly diagnosed type 2 diabetes showed that the benefit of β-blocker persisted and is even better than an ACEI.⁴²⁴(Level II-2) Other reported adverse events include erectile dysfunction, cold extremities and nightmares (especially for lipophilic β-blockers), increased triglyceride levels and reduced HDL levels (especially for non-selective β-blockers). Use of β-blockers during pregnancy is cautioned.

In a major landmark study, an ARB was shown to be superior than β-blocker in patients with high risk hypertension and ECG LVH¹¹² (Level 1) This prompted a meta-analysis on the use of β-blockers in the treatment of hypertension.⁹² Beta-blocker therapy did not reduce the risk for first myocardial infarction compared to other drugs but was associated with a significant 16% higher risk for stroke when compared to non-β-blocker therapy and that atenolol in particular was associated with a significant 26% increase in the risk of stroke when compared to other anti-hypertensive agents. Beta-blockers lower brachial systolic blood pressure but not the aortic pressure compared to other drugs. Heart rate is reduced but peripheral resistance is increased, thus increasing the arterial wave reflection during systole rather than diastole.⁴²⁵ (Level II-1) Similarly, another meta-analysis⁹⁴

and a systematic review also showed that β -blockers were associated with a significant increase in their withdrawal due to side effects.⁴²⁶

Caution is necessary in the interpretations of negative findings from earlier analysis on β -blocker as:

- most of the studies involved atenolol (hence the comparative outcomes of other newer vasodilating β -blockers are not well established as there has been no comparative studies between the β -blocker sub-classes.
- almost all the studies were carried out in the West, hence the comparative outcomes of other ethnic groups are not well established.

However, more recent meta analysis showed that β -blocker is as effective as other drugs in improving clinical outcome.⁹⁵⁻⁹⁸ The latest Cochrane Review⁹⁵ however indicates that as first line treatment, they are:

- inferior to CCB for total mortality outcomes.
- better than placebo for total CVD (primarily driven by decrease strokes) but no better than other classes of anti-hypertensive agents.
- better than placebo for strokes but worse than CCB and no better than other classes of anti-hypertensive agents for total coronary heart disease outcome.

It is thus reasonable for β -blockers to be used as single first line therapy to initiate anti-hypertensive therapy for patients with hypertension especially if there are specific compelling needs for its use such as those with post-MI or heart failure. Some guidelines like The National Institute for Clinical Excellence (NICE) UK Guideline,⁴ JNC VIII⁹⁰ and the ACC/AHA¹ did not recommend β -blockers as first line anti-hypertensive agent. It is however still recommended as first line by other guidelines.^{26,101} including guidelines from this region.^{102,427,428}

TABLE 9.2-A Recommended Dosing for β -blockers

β-blockers	Starting Dose*	Recommended Maximum Daily Dose*
Acebutolol	200 mg bd	1.2 g in divided doses
Atenolol	50 mg od	100 mg od
Betaxolol	10 mg od	20 mg/day
Bisoprolol	5 mg od CrCl <40, 2.5 mg/day	20 mg/day
Metoprolol	50 mg bd	200 mg bd
Nebivolol	5 mg od	40 mg od
Propranolol	40 mg bd	320 mg bd

* Referenced from 153rd Edition, MIMS, 2018.

RECOMMENDATIONS

- Consider β -blocker as single first-line anti-hypertensive agent especially when there are compelling indications for their use. (Grade A)
- Choose vasodilating over non-vasodilating β -blockers as a preference. However there had not been extensive head-to-head comparison between β -blockers. (Grade C)

9.3 Calcium Channel Blockers

Calcium channel blockers (CCBs) are a structurally and functionally heterogeneous class of drug. The main mechanism of action is vasodilation, which decreases peripheral resistance. Certain subtypes of dihydropyridine (DHP) CCBs, e.g. T-type CCBs have been shown to dilate both the afferent and efferent arterioles, reduce glomerular capillary pressure and proteinuria. This may play a role in prevention of kidney damage and preservation of renal function.

In view of the effective BP lowering property and excellent safety profile, CCBs especially dihydropyridine (DHP) type have been recommended as first-line anti-hypertensive agents. A metaanalysis showed that CCBs reduced stroke in hypertensive patients more than placebo and β -blockers but were not different than ACE inhibitors and diuretics.⁴²⁹ (Level 1) However, there is no evidence that dihydropyridine CCBs are superior to other antihypertensive agents in Asian populations for the treatment of hypertension in reducing cardiovascular death, major cardiovascular events, stroke, congestive heart failure, and coronary revascularisation.⁴³⁰ (Level 1)

Metanalyses have shown that RAS blockers and CCBs combinations are superior to other combinations in lowering cardiovascular events, in addition to a better safety profile.^{431,432} (Level 1)

TABLE 9.3-A Recommended Dosing for CCBs

Dihydropyridines	Starting Dose*	Recommended Maximum Daily Dose*
Amlodipine	5 mg od	10 mg od
Felodipine	5 mg od	10 mg od
Isradipine	2.5 mg bd	10 mg bd
Lercanidipine	10 mg od	20 mg od
Nifedipine	5 mg tid	20 mg tid
Non-dihydropyridines		
Diltiazem	90 mg bd	180 mg bd
Diltiazem SR	100 mg od	200 mg od
Verapamil	80 mg tid	160 mg tid
Verapamil SR	120 mg od	480 mg od

* Referenced from 153rd Edition, MIMS, 2018

9.4 Renin-Angiotensin-System (RAS) Blockers

The RAS is implicated at all stages of the “CV continuum” that links hypertension with other risk factors and major CV events. Therefore, it represents a rational and established therapeutic target when lowering blood pressure.⁴³³

9.4.1 Ace Inhibitors (ACEIs)

ACEIs are effective antihypertensive agents which can lower cardiovascular risk, reducing mortality and morbidity in hypertensives and those at high cardiovascular risk.^{245,434 (Level 1)} They are more effective in preventing coronary artery disease in patients with hypertension.^{435 (Level 1)} ACEIs are generally well tolerated and do not have adverse effects on lipid and glucose metabolism. Their safety profile is good. ACEIs have also been shown to reduce mortality and morbidity in patients with congestive heart failure^{436-438 (Level 1)} and in post myocardial infarction patients with reduced left ventricular ejection fraction.^{439-444 (Level 1)}

In patients with established vascular disease but normal left ventricular function, ACEIs reduce mortality, myocardial infarction, stroke and new-onset congestive heart failure.²⁴⁵

In the diabetic patient, ACEIs have been shown to reduce cardiovascular mortality.^{155 (Level 1)} These agents prevent the onset of microalbuminuria, reduce proteinuria and retard progression of diabetic and non-diabetic renal disease.^{445,446 (Level1)}

Adverse effects include cough and, rarely, angioedema. In patients with renovascular disease or renal impairment, deterioration in renal function may occur. Serum creatinine and potassium should be checked before initiation and within 2 weeks after starting. If there is hyperkalemia (>5.6 mmol/L) or a persistent rise of serum creatinine of more than 30% from baseline within two months, the dose of the ACEI should be reduced or discontinued.

This class of drug may increase foetal and neonatal mortality and therefore are contraindicated in pregnancy and breast feeding. Counselling should be given to women of child bearing age before initiation of RAS blockers. Pregnant patients should seek immediate medical advice.

TABLE 9.4-A Recommended Dosing for ACEIs

ACEIs	Starting Daily Dose*	Recommended Maximum Daily Dose*
Captopril	25 mg bd	50 mg tds
Enalapril	10 mg od	40 mg daily
Lisinopril	10 mg od	80 mg daily
Perindopril	4 mg (as erbumine) or 5 mg (as arginine) od	8 mg (as erbumine) or 10 mg (as arginine) od
Ramipril	2.5 mg od	10 mg daily
Imidapril	5 mg od	20 mg daily

* Referenced from 153rd Edition, MIMS, 2018.

Combination Therapy with ACEI

The combination of an ACEI and a dihydropyridine CCB is preferred over the combination of an ACEI and a thiazide diuretic in patients with hypertension and high CV risk.^{84 (Level 1)}

9.4.2 Angiotensin Receptor Blockers (ARBs)

ARBs are drugs which specifically block angiotensin II receptors. Unlike ACEIs, persistent dry cough is less and as such ARBs are recommended for risk reduction in ACEI intolerant patients.^{447,448 (Level 1)}

ARBs are effective in preventing progression of diabetic nephropathy^{145,449 (Level 1)} and may reduce the incidence of major cardiac events in patients with heart failure,^{450,451 (Level 1)} hypertensive LVH^{452 (Level 1)} and diastolic heart failure.^{213 (Level 1)} In patients with LV dysfunction post MI, ARBs have also been shown to be non-inferior to ACEIs.^{453 (Level 1)}

The cardioprotective effects of ARBs when compared to ACEIs especially for prevention of myocardial infarction, CV and all cause mortality were recently called into question.^{454 (Level 1)} An earlier large meta-analysis of ARBs showed that although it did not increase the risk of myocardial infarction compared to placebo or active control, unlike ACEIs, it seem not to have special cardio protective effects.^{455 (Level 1)} A more recent meta-analysis however concluded that that ARBs do reduce CV events including the risk of myocardial infarction.^{456 (Level 1)}

Despite conflicting findings from various meta analyses, it is important to look at the original studies, especially “head to head” trials on these drugs. In high risk CV patients with or without hypertension, the evidence showed that ARB is non-inferior to ACEI for CV protection.^{457 (Level 1)} However in patients with left ventricular dysfunction, ACEI have more evidence including reducing mortality and ARB is used for ACEI intolerant patients.^{451,458 (Level 1)} As for diabetics patients with or without hypertension, ACEI improves CV outcome including total mortality especially in combination with thiazide-like diuretics.^{83 (Level 1)} The same is true for non-diabetic nephropathy^{178 (Level 1)} and type 1 diabetes mellitus with nephropathy.^{140,141 (Level 1)} However for type 2 diabetic nephropathy, both ACEI^{155 (Level 1)} and ARB^{144-146 (Level 1)} improve renal outcome although only ACEI has the added advantage in improving CV and renal outcomes. In hypertensives with ECG left ventricular hypertrophy, CV protection (especially stroke reduction) have been demonstrated with ARB.^{112 (Level 1)} On the other hand for secondary stroke prevention, the evidence favour ACEI especially in combination with thiazide-like diuretics.^{79 (Level 1)} Table 9.4-C summarises the available evidence on the therapeutics of RAS blockers in patients with various comorbidities.

TABLE 9.4-B Recommended Dosing for ARBs

ARBs	Starting Dose*	Recommended Maximum Daily Dose*
Candesartan	8 mg od	32 mg od
Irbesartan	150 mg od	300 mg od
Losartan	50 mg od	100 mg od
Telmisartan	40 mg od	80 mg od
Valsartan	80 mg od	320 mg od
Olmesartan	20 mg od	40 mg od

* Referenced from 153rd Edition, MIMS, 2018.

The safety profile of ARB is very similar to ACEI except for a lower incidence of cough.

Combination of ACEI and ARB

The combination of ACEI and ARB is not recommended and is to be avoided.

TABLE 9.4-C RAS Blockers Use in Co-Morbidities 116,133,136,137,140-147,459,460

Condition	ACEIs	ARBs
Diabetes mellitus (CV protection)	Preferred	If ACEI intolerant
Diabetes mellitus (eGFR>60) + proteinuria (Renal protection)	Either	Either
Diabetes mellitus type 1 (eGFR <60) +/- proteinuria (Renal protection)	Preferred	If ACEI intolerant
Diabetes mellitus type 2 (eGFR <60) +/- proteinuria (Renal protection)	Either	Either
Non-diabetic proteinuria/renal impairment	Preferred	If ACEI intolerant
Heart failure (HFrEF)	Preferred	If ACEI intolerant
Stroke	Preferred	If ACEI intolerant
Coronary heart disease High CV risk patients	Either	Either
Coronary heart disease Post MI	Preferred	If ACEI intolerant
Left ventricular hypertrophy		Preferred

9.5 Miscellaneous Drugs

9.5.1 The α -Blockers and the Combined α , β -Blockers

The peripheral α 1-adrenergic blockers lower BP by reducing peripheral resistance. They also reduce prostatic and urethral smooth muscle tone and provide symptomatic relief for patients with early benign prostatic hyperplasia (BPH).⁴⁶¹ Unless there are other compelling reasons, they should be the treatment of choice for hypertensive patients with BPH. The use of non-specific α -blockers like phentolamine and phenoxybenzamine has been restricted to the treatment of pheochromocytoma.

In addition, α -blockers have favourable effects on lipid metabolism. However postural hypotension is a known side effect, especially at initiation of therapy.^{462,463} They should be used with care in the elderly.

Combined α and β -blockers offer enhanced neurohormonal blockade. Labetalol has been in use for over 20 years and is safe in pregnancy (Refer to chapter 7.6 on Hypertension in Pregnancy). The intravenous formulation is useful in hypertensive emergencies, including pre-eclampsia and eclampsia.⁴⁶⁴

Carvedilol has been shown to be effective in hypertension and also to improve mortality and morbidity in patients with heart failure.⁴⁶⁵⁻⁴⁶⁷ (Level 1) In addition, it has no adverse effects on insulin resistance and lipid metabolism.⁴⁶⁸ However, its safety in pregnancy has not been established.

TABLE 9.5-A Recommended Dosing for α -blockers

α -blockers	Starting Dose*	Recommended Maximum Daily Dose*
Doxazosin	1 mg od	16 mg od
Prazosin	0.5 mg bd-tds	20 mg in divided doses
Terazosin	1 mg nocte	20 mg od

* Referenced from 153rd Edition, MIMS, 2018.

Table 9.5-B Recommended Dosing for α , β -blockers

α , β -blockers	Starting Dose*	Maximum Dose*
Labetolol **	100 mg bd	2.4 gm per day in 2-4 divided doses
Carvedilol ***	12.5 mg od	50 mg od or in divided doses if necessary

* Referenced from 153rd Edition, MIMS, 2018.

** In the elderly start with 50 mg bd.

*** The dosage of carvedilol for patients with heart failure and angina pectoris is different from the doses indicated.

9.5.2 Centrally Acting Agents

The centrally acting agents available in this country are alpha-methyldopa, clonidine and moxonidine. The common side effects of the centrally acting agents include drowsiness, dry mouth, headache, dizziness and mood change. Moxonidine is less likely to cause these reactions. The side-effects may decrease after a few weeks of continued treatment. In general, treatment should begin with the lowest possible dose to minimise the side-effects.

Alpha-methyldopa has been in use for many years. It is the drug of choice for hypertension in pregnancy. It may be considered for resistant hypertension in combination with other classes of anti-hypertensive agents.⁴⁶⁹

Clonidine should **NOT** be withdrawn suddenly because rebound hypertension may occur.⁴⁷⁰ The use of clonidine is discouraged because safer and more potent drugs are available.

Moxonidine is an orally administered imidazoline compound with selective agonist activity at imidazoline II receptors. It can be used as monotherapy in patients with mild to moderate hypertension or in combination with other anti-hypertensive agents. Studies have suggested that it may improve the metabolic profile of patients with impaired glucose tolerance or diabetes. Rebound hypertension on cessation of the drug is less likely compared to clonidine but abrupt withdrawal is not recommended. It has been shown to increase mortality in patients with heart failure and is therefore contraindicated in patients with heart failure or at risk of heart failure.⁴⁷¹

TABLE 9.5-C Recommended Dosing for Centrally Acting Agents

Drug	Starting dose	Maximum dose
α-methyldopa*	250 mg bd-tds	3000 mg daily
Clonidine	50 mcg tds	2400 mcg daily
Moxonidine**	• 200 mcg od • To be avoided if GFR <30	• 600 mcg in 2 divided doses • 400 mcg daily (GFR 30–60) • To be avoided if GFR <30

* For dosage in pregnancy, refer to chapter 7.6 on Hypertension in Pregnancy.

** Referenced from 153rd Edition, MIMS, 2018.

9.5.3 Direct Vasodilators

The only direct vasodilators available in Malaysia are hydralazine and minoxidil. Hydralazine is only available in parenteral formulation for hypertensive emergencies. Minoxidil may be considered for refractory hypertension. The usefulness of this class of drugs is limited by their side-effects, including headache, compensatory tachycardia, salt and water retention. Hirsutism is a troublesome side-effect with long-term use of minoxidil. These drugs should only be prescribed by physicians familiar with their usage.

TABLE 9.5-D Recommended Dosing for Minoxidil

Drug	Starting Dose	Maximum dose
Minoxidil*	5 mg per day	100 mg per day

* Referenced from 153rd Edition, MIMS, 2018.

9.5.4 Drugs In Development

As the pathophysiology of hypertension becomes clearer, compounds are being developed to interrupt the pathways leading to an elevated blood pressure and to normalise it.

Among the compounds being tested are:

1. Those targeting aldosterone (anti-aldosterone agents). The strategy is to either block aldosterone production (aldosterone synthase inhibitors) or to block its receptor (mineralocorticoid receptor antagonists).

The prototypes of mineralocorticoid receptor antagonists would be spironolactone and eplerenone. Finerenone is the latest in this class and is currently undergoing phase III trials, not only for hypertension but also for diabetic nephropathy and heart failure. Because it has low affinity for steroid receptors, there is a lower incidence of gynaecomastia, erectile dysfunction, and reduced libido.

2. Those targeting the classical RAS. These would include angiotensin converting enzyme 2 activators, angiotensin type 2 receptor agonists and vaccines against the angiotensin type 1 and type 2 receptors.
3. Centrally acting aminopeptidase inhibitors to inhibit excessive sympathetic outflow from the brain.
4. Vasopeptidase inhibitors which degrade natriuretic peptides.
5. Dual-acting angiotensin receptor–neprilysin Inhibitors. The combination of valsartan and sacubitril is already in clinical use. However this combination has found its niche in the management of heart failure rather than hypertension.

6. Dual-Acting Endothelin Converting Enzyme–Nepriylsin Inhibitors
7. Natriuretic Peptide Receptor Agonists
8. Vasoactive Intestinal Peptide Receptor Agonist
9. Intestinal Na⁺/H⁺ Exchanger 3 Inhibitor
10. Dopamine β -hydroxylase (D β H) Inhibitor

TABLE 9.5-E New Drugs for Hypertension (adapted from Oparil S. and Schmieder RE. 2015)⁴⁷²

Drug	Mechanism of action	Status
BAY 94–8862 (finerenone)	Mineralocorticoid receptor antagonist	Phase III
LCZ696	Dual-acting angiotensin receptor-nepriylsin inhibitor	Phase III
SLV-306 (Daglutril)	Dual acting endothelin-converting enzymes-nepriylsin inhibitor	Phase II
PL-3994	Natriuretic peptide A agonist	Phase II
Vasomera (PB1046)	Vasoactive intestinal peptide receptor 2 (VPAC2) agonist	Phase II
Vaccines CYT006-AngQ β	Vaccine against angiotensin II	Phase II
Preeclampsia drugs DIF	Anti-digoxin antibody fragment	Phase II expedited
ATryn	Recombinant antithrombin	Phase III

DIF = Digoxin-immune Fab.

9.6 Traditional Herbal Medicine and Hypertension

9.6.1 Traditional Medicine for Hypertension

In 2000, the WHO in a key paper, (<http://who.int/medicines/areas/traditional/definitions/en/>) clearly outlined the definition and scope of traditional medicine as follows:

Traditional medicine

Traditional medicine is the sum total of the knowledge, skills, and practices based on the theories, beliefs, and experiences indigenous to different cultures, whether

explicable or not, used in the maintenance of health as well as in the prevention, diagnosis, improvement or treatment of physical and mental illness.

Complementary/alternative medicine (CAM)

The terms “complementary medicine” or “alternative medicine” are used interchangeably with traditional medicine in some countries. They refer to a broad set of health care practices that are not part of that country’s own tradition and are not integrated into the dominant health care system.

Herbal medicines

Herbal medicines include herbs, herbal materials, herbal preparations and finished herbal products, that contain as active ingredients parts of plants, or other plant materials, or combinations.

In most countries including Malaysia, the traditional treatment of hypertension will consist of:

- oral, usually herbal medication
- various forms of relaxation exercises, including yoga and qigong

Herbs used in hypertension

There are few publications in the English literature on the treatment of hypertension with herbs. The most quoted is by Tabassum and Ahmad⁴⁷³, the “Role of Natural Herbs in the Treatment of Hypertension”. They comprehensively describe almost 50 herbal remedies for hypertension, including garlic, ginger, roselle, black plum, mistletoe, wheat bran, cocoa, wild tomato, sesame, radish, pomegranate, basil, cork wood, tomato, linseed, flaxseed, black mangrove, French lavender, pima cotton, soybean, carrot, swamp or river lily, Chinese hawthorn, black bean, coffee weed, tea, green oat, breadfruit, celery and prickly custard apple.

An extensive online review on Medscape⁴⁷⁴ showed that herbal medicines for hypertension have not undergone the rigorous testing for efficacy and safety that is expected of modern drugs. None have undergone the gold standard of randomised placebo controlled trials. On top of that, there are reports that various herbal preparations (usually in capsule form) have been adulterated with modern drugs. For example, glibenclamide has been found in anti-diabetic preparations, and sildenafil has been found in compounds to increase male virility. Corticosteroids have also been added to traditional herbal medicine, with prolonged usage causing Cushing’s Syndrome and hypertension. Hence, anecdotal reports on the efficacy of certain herbs cannot be relied on.

SUMMARY

- There is no conclusive evidence that traditional medicine produces sustained reductions in BP with good clinical outcomes.

RECOMMENDATIONS

- Traditional medicines are not recommended for the treatment of hypertension.

9.6.2 Relaxation Exercises for Hypertension

Since the 1970s there have been reports on the efficacy of yoga in the treatment of hypertension (SBP -17 mmHg and DBP -10 mmHg). These trials, albeit small, have been compared against a control group and are credible.^{64,475,476} Qigong has also been demonstrated to lower the BP (SBP -17 mmHg and DBP -10 mmHg). But when compared against conventional exercise there was no significant difference.^{477,478}

The American Heart Association,⁴⁷⁹ in their Scientific Statement on Meditation and Cardiovascular Risk Reduction, noted that BP may be lowered but the results were variable and inconsistent.

The regimes are postulated to work by reducing excessive sympathetic outflow from the brain, and in the case of qigong, to improve blood flow to various organs in the body.

Relaxation regimes as part of a healthy living regime can be useful in the holistic management of hypertension. Yoga and qigong or any other form of relaxation exercises may usefully complement drug treatment.

SUMMARY

- Relaxation therapy may complement non-pharmacological and pharmacological treatment of hypertension however it is not recommended as primary treatment for hypertension.

10 Resistant and Refractory Hypertension

10.1 Resistant Hypertension

Resistant hypertension is defined as uncontrolled hypertension ($>140/90$ mmHg) with good medication adherence while on three^{26,480} or four anti-hypertensive agents (including a diuretic) in adequate doses. In a study of a primary care center involving 1,217 hypertensives in Malaysia, the prevalence of resistant hypertension (as defined above) was reported to be 8.8%.⁴⁸¹ Similar prevalence was observed in national surveys in the United States (National Health and Nutritional Examination Survey from 2003-2008), the estimated prevalence was 8.9%.⁴⁸² It is however worth noting that the prevalence may be an overestimate if white coat resistance is excluded. Ambulatory Blood Pressure Monitoring studies on treatment resistant patients have demonstrated that between 30-35% of patients have pseudo resistant or office resistant hypertension.⁴⁸³ This is consistent with a study from a tertiary center in Malaysia.⁴⁸⁴

Before labeling a patient as having resistant hypertension, it is important that the practitioner ascertain that:

- a. the patient adheres to medication (by definition at least 80%)
- b. the blood pressure is measured appropriately
- c. the patient does not have 'office resistant hypertension'
- d. an appropriate combination and dosage of drugs is prescribed, namely 3 drugs including a RAS blocker, a CCB and a diuretic
- e. the patient is not taking any substances which may antagonise the hypertensive effects of the drugs taken (e.g. NSAID, sympathomimetics, liquorice, oral contraceptives, corticosteroids)

It is therefore important that a thorough review of the patient's history, physical examination and investigations be done including estimation of renal function glomerular filtration rate (eGFR). A home or ambulatory blood pressure measurement should be done to exclude isolated office hypertension. (Refer to chapter 2 on Measurement of Blood Pressure).

Once a patient is confirmed to have true resistant hypertension, consider referral for exclusion of secondary causes (Refer to chapter 3 on Diagnosis and Initial Assessment).

Excluding Secondary Hypertension

Although the prevalence of secondary hypertension is around 5%, its prevalence is higher in patients with resistant hypertension. Depending on series, prevalence of secondary hypertension among patients with resistant hypertension can be as high as 66%, with obstructive sleep apnoea, accounting for most of it.^{485,486} In two large series, primary aldosteronism was diagnosed in 11% of patients with resistant hypertension.^{487,488} Subsequent investigations arranged should be guided by symptoms present, examination findings elicited and results from preliminary investigations. It is prudent that any investigations to be ordered or arranged must be rational with cost effectiveness in mind.

Treatment options in resistant primary hypertension

a. Non-pharmacological Management

Non-pharmacological approaches (healthy living) must be re-emphasised. (Refer to chapter 4 on Non-Pharmacological Management)

b. Pharmacological Management

A fourth drug should be added to the combination of RAS blocker, CCB and diuretic. Two recent metaanalyses of randomised and non-randomised controlled trials showed that spironolactone is superior to active controls (which includes alpha blockers, β -blockers, candesartan, frusemide or alpha methyl dopa) in reducing office, home and ambulatory blood pressures.^{489,490} (Level 1) These meta analyses of disparate studies was strengthened by a multicenter randomised controlled double blind trial which confirmed that spironolactone is the drug of choice as the fourth drug in resistant hypertension.⁴⁹¹ (Level 1)

10.2 Refractory Hypertension

The definition has been proposed to be used on patients whose BP are not controlled after ≥ 5 antihypertensives.⁴⁹²

If blood pressures are still not controlled with four drugs, a fifth drug may be considered. Subsequent therapeutic options include a β -blocker, an alpha blocker or a centrally acting drug. The prevalence of refractory hypertension among patients referred to a specialty clinic for resistant hypertension was reported to be 2.7%⁴⁸⁹ in one series and 9.5% in another.⁴⁹³

Both resistant and refractory hypertensives are candidates for device-based intervention.

RECOMMENDATIONS

- Treat patients with at least 3 drugs (inclusive of a diuretic) before diagnosing resistant hypertension. (Grade C)
- Consider drug non-adherence and secondary hypertension before diagnosing resistant hypertension. (Grade C)
- Add spironolactone as a fourth drug in resistant hypertension. (Grade A)
- Consider referring for device based therapy in patients with true resistant and refractory hypertension. (Grade C)

11 Aspirin in Hypertension

Although the benefits of aspirin in secondary CV prevention is incontrovertible, that for primary prevention remains controversial.⁴⁹⁴ A large meta analysis suggested that for primary prevention, the risk of significant bleeding outweigh the benefits of CV protection.⁴⁹⁵ In patients with hypertension a large RCT showed that low dose aspirin (75 mg daily) reduced major CV events especially for MI but had no effect on the incidence of stroke. Non-fatal major bleeds were however twice as common with aspirin.²⁹⁹ Subgroup analysis of this large trial showed that patients who benefited most are those with well treated hypertensive at higher baseline CV risk or higher baseline BP.²⁹⁹ The benefits of low dose aspirin were also most convincing in patients with well controlled BP and moderate rise in serum creatinine (>114 $\mu\text{mol/L}$).⁴⁹⁶

A recent large cohort study in Asia showed that aspirin given to uncomplicated hypertensive patients for primary prevention significantly reduced all cause and cardiovascular mortality.⁴⁹⁷ However, since it is associated with an increased risk of major bleed, careful evaluation of risk/benefit analysis must be made by the doctor before initiating aspirin.

RECOMMENDATIONS

- Consider using aspirin in patients with higher baseline BP. (Grade B)
- Treat patients BP to target first before initiating aspirin therapy. (Grade A)

12

Device and Procedure Based Therapy in Hypertension

One of the potential approaches in treating true resistant hypertension and severe newly diagnosed hypertension is device therapy. This includes renal denervation therapy (RDN) and carotid sinus stimulation. In these patients, blood pressure rise is initiated and sustained by sympathetic over activation⁴⁹⁸ and BP reduction can result from its inhibition.^{499,500}

Renal denervation therapy gained popularity following early blood pressure reduction result seen in patients with resistant hypertension on medical therapy in the SIMPLICITY HTN-1 and 2 trials. However, the 24-hour ambulatory blood pressure was not significantly reduced. SIMPLICITY HTN-3, which had a sham control arm showed a neutral result suggesting a lack of benefit with the single electrode radiofrequency catheter.^{501(Level 1)}

Since 2014, several sham controlled randomised trials employing variations in improved techniques (multi-electrode catheter, ultrasound and alcohol based denervation) have been initiated. The recently published SPYRAL HTN-OFF MED examined the effect of RDN in 80 newly diagnosed hypertensive patients. They included patients with SBP of 150-180 mmHg without any anti-hypertensive medications and found a significant reduction (-5.5 mmHg for SBP) when compared to sham control at 3 months follow up.^{502(Level II-1)} This may suggest a potential for this intervention in newly diagnosed hypertensives. Further confirmatory results from other trials are awaited to confirm the clinical indication for multi-electrode RDN for newly diagnosed hypertension.⁵⁰³

Baroreceptor activation therapy (BAT) is based on sympathetic inhibition by carotid sinus stimulation. The arterial stretch baroreceptors respond with a higher discharge rate, to lower blood pressure, in the setting of rising blood pressure. In chronic hypertension, this firing is blunted, rendering it less sensitive to respond to changes in blood pressure.⁵⁰⁴ Carotid stimulation via surgical implantation of electrodes onto the carotid bulbs have resulted in lowering the blood pressure.⁵⁰⁵ This invasive procedure is still limited by technical, safety issues and cost.

Clinical development in this area should be accompanied by investigations identifying predictors for good treatment response. Device based therapy should not be part of routine medical care until further evidence is available.^{472,506}

13 Suggested Areas of Research

This latest CPG on hypertension has incorporated key references from research done in Malaysia. It is also heartening to know that several multicentre international trials cited in this CPG also had Malaysia as one of the centers involved. There are however a few unanswered question unique to Malaysia which needed to be addressed in future CPGs. It is suggested that Malaysian researchers should focus on these research areas and funding authorities should give due importance to these areas of research.

Epidemiology

- Burden of disease in lower income group and association with BP
- Prevalence of hypertension in children and adolescents
- Cost effectiveness of hypertension treatment
- Health system research in hypertension

Drugs

- Differences in antihypertensive drug response among different ethnic groups

Monitoring

- Blood pressure goal for patients with hypertension with different co morbidities
- Intervention thresholds for people aged under 40 with hypertension
- Methods of assessing cardiovascular risk in people aged under 40 years with hypertension
- Barriers to good BP control in community
- Acceptability of ambulatory and home blood pressure monitoring
- Psychological impact of home blood pressure monitoring

Treatment

- Precision medicine and hypertension
- Should white coat hypertension be treated?
- New drug for resistant hypertension
- What are the reasons for treatment inertia among doctors?
- Randomised Control Trial comparing reduced sodium intake with usual diet on efficacy of blood pressure lowering
- Patient empowerment in blood pressure management

Complications

- Sleep apnea, obesity and hypertension
- Reasons for higher rate of haemorrhagic stroke in Asian compared to Caucasian

Risk Factors

- Hypertension and Dementia – link and prevention
- Use of Information and Communication Technology and blood pressure
- Risk factors of hypertension in younger age group
- Effects of environmental pollution on blood pressure

Pregnancy

- Screening and management of hypertension in early pregnancy
- Prevalence of preeclampsia and its outcome
- Risk of mortality in preeclampsia
- What level of proteinuria is considered significant in women with hypertension and its correlation with outcome?
- Is there a difference in maternal and fetal outcomes in using mercury sphygmomanometer vs automated device for BP measurement?
- Home Blood Pressure Monitoring in pregnancy

Appendices

APPENDIX 1 Estimated BP Values After 2 Weeks of Age in Infants from 26 to 44 Weeks Postconceptual Age

Postconceptual age	50th percentile	95th percentile	99th percentile
44 Weeks			
SBP	88	105	110
DBP	50	68	73
MAP	63	80	85
42 Weeks			
SBP	85	98	102
DBP	50	65	70
MAP	62	76	81
40 Weeks			
SBP	80	95	100
DBP	50	65	70
MAP	60	75	80
38 Weeks			
SBP	77	92	97
DBP	50	65	70
MAP	59	74	79
36 weeks			
SBP	72	87	92
DBP	50	65	70
MAP	57	72	71
34 Weeks			
SBP	70	85	90
DBP	40	55	60
MAP	50	65	70
32 Weeks			
SBP	68	83	88
DBP	40	55	60
MAP	48	62	69
30 Weeks			
SBP	65	80	85
DBP	40	55	60
MAP	48	65	68
28 Weeks			
SBP	60	75	80
DBP	38	50	54
MAP	45	58	63
26 Weeks			
SBP	55	72	77
DBP	30	50	56
MAP	38	57	63

APPENDIX 2 Blood Pressure Levels for Boys by Age and Height Percentile

Age year	BP Percentile	Systolic BP (mmHg)							Diastolic BP (mmHg)						
		Height Percentile							Height Percentile						
		5 th	10 th	25 th	50 th	75 th	90 th	95 th	5 th	10 th	25 th	50 th	75 th	90 th	95 th
1	50 th	85	85	86	86	87	88	88	40	40	40	41	41	42	42
	90 th	98	99	99	100	100	101	101	52	52	53	53	54	54	54
	95 th	102	102	103	103	104	105	105	54	54	55	55	56	57	57
2	50 th	87	87	88	89	89	90	91	43	43	44	44	45	46	46
	90 th	100	100	101	102	103	103	104	55	55	56	56	57	58	58
	95 th	104	105	105	106	107	107	108	57	58	58	59	60	61	61
3	50 th	88	89	89	90	91	92	92	45	46	46	47	48	49	49
	90 th	101	102	102	103	104	105	105	58	58	59	59	60	61	61
	95 th	106	106	107	107	108	109	109	60	61	61	62	63	64	64
4	50 th	90	90	91	92	93	94	94	48	49	49	50	51	52	52
	90 th	102	103	104	105	105	106	107	60	61	62	62	63	64	64
	95 th	107	107	108	108	109	110	110	63	64	65	66	67	67	68
5	50 th	91	92	93	94	95	96	96	51	51	52	53	54	55	55
	90 th	103	104	105	106	107	108	108	63	64	65	65	66	67	67
	95 th	107	108	109	109	110	111	112	66	67	68	69	70	70	71
6	50 th	93	93	94	95	96	97	98	54	54	55	56	57	57	58
	90 th	105	105	106	107	109	110	110	66	66	67	68	68	69	69
	95 th	108	109	110	111	112	113	114	69	70	70	71	72	72	73
7	50 th	94	94	95	97	98	98	99	56	56	57	58	58	59	59
	90 th	106	107	108	109	110	111	111	68	68	69	70	70	71	71
	95 th	110	110	111	112	114	115	116	71	71	72	73	73	74	74
8	50 th	95	96	97	98	99	99	100	57	57	58	59	59	60	60
	90 th	107	112	114	116	118	119	120	69	70	70	71	72	72	73
	95 th	111	112	112	114	115	116	117	72	73	73	74	75	75	75
9	50 th	96	97	98	99	100	101	101	57	58	59	60	61	62	62
	90 th	107	108	109	110	112	113	114	70	71	72	73	74	74	74
	95 th	112	112	113	115	116	118	119	74	74	75	76	76	77	77
10	50 th	97	98	99	100	101	102	103	59	60	61	62	63	63	64
	90 th	108	109	111	112	113	115	116	72	73	74	74	75	75	76
	95 th	112	113	114	116	118	120	121	76	76	77	77	78	78	78
11	50 th	99	99	101	102	103	104	106	61	61	62	63	63	63	63
	90 th	110	111	112	114	116	117	118	74	74	75	75	75	76	76
	95 th	114	114	116	118	120	123	124	77	78	78	78	78	78	78
12	50 th	101	101	102	104	106	108	109	61	62	62	62	63	63	63
	90 th	113	114	115	117	119	121	122	75	75	75	75	75	76	76
	95 th	116	117	118	121	124	126	128	78	78	78	78	78	79	79

Clinical Practice Guideline for Screening and Management of High Blood Pressure in Children and Adolescents. Pediatrics 2017 Sep;140(3) pii:e20171904.³⁹⁸

APPENDIX 3 Blood Pressure Levels for Girls by Age and Height Percentile

Age year	BP Percentile	Systolic BP (mmHg)							Diastolic BP (mmHg)						
		Height Percentile							Height Percentile						
		5 th	10 th	25 th	50 th	75 th	90 th	95 th	5 th	10 th	25 th	50 th	75 th	90 th	95 th
1	50th	84	85	86	86	87	88	88	41	42	42	43	44	45	46
	90th	98	99	99	100	101	102	102	54	55	56	56	57	58	58
	95th	101	102	102	103	104	105	105	59	59	60	60	61	62	62
2	50th	87	87	88	89	90	91	91	45	46	47	48	49	50	51
	90th	101	101	102	103	104	105	106	58	58	59	60	61	62	62
	95th	104	105	106	106	107	108	109	62	63	63	64	65	66	66
3	50th	88	89	89	90	91	92	93	48	48	49	50	51	53	53
	90th	102	103	104	104	105	106	107	60	61	61	62	63	64	65
	95th	106	106	107	108	109	110	110	64	65	65	66	67	68	69
4	50th	89	90	91	92	93	94	94	50	51	51	53	54	55	55
	90th	103	104	105	106	107	108	108	62	63	64	65	66	67	67
	95th	107	108	109	109	110	111	112	66	67	68	69	70	70	71
5	50th	90	91	92	93	94	95	96	52	52	53	55	56	57	57
	90th	104	105	106	107	108	109	110	64	65	66	67	68	69	70
	95th	108	109	109	110	111	112	113	68	69	70	71	72	73	73
6	50th	92	92	93	94	96	97	97	54	54	55	56	57	58	59
	90th	105	106	107	108	109	110	111	67	67	68	69	70	71	71
	95th	109	109	110	111	112	113	114	70	71	72	72	73	74	74
7	50th	92	93	94	95	97	98	99	55	56	56	57	58	59	60
	90th	106	107	108	109	110	111	112	68	68	69	70	71	72	72
	95th	109	110	111	112	113	114	115	72	72	73	73	74	74	75
8	50th	93	94	95	97	98	99	100	56	56	57	59	60	61	61
	90th	107	107	108	110	111	112	113	69	70	71	72	72	73	73
	95th	110	111	112	113	115	116	117	72	73	74	74	75	75	75
9	50th	96	95	97	98	99	100	101	57	58	59	60	60	61	61
	90th	108	108	109	111	112	113	114	71	71	72	73	73	73	73
	95th	112	112	113	114	116	117	118	74	74	75	75	75	75	75
10	50th	96	97	98	99	101	102	103	58	59	59	60	61	61	62
	90th	109	110	111	112	113	115	116	72	73	73	73	73	73	73
	95th	125	126	126	128	129	131	132	87	87	88	88	88	88	88
11	50th	98	99	101	102	104	105	106	60	60	60	61	62	63	64
	90th	111	112	113	113	116	118	120	74	74	74	74	74	75	75
	95th	115	116	117	118	120	123	124	76	77	77	77	77	77	77
12	50th	102	102	104	105	107	108	108	61	61	61	62	63	64	65
	90th	114	115	116	118	120	122	122	75	75	75	75	76	76	76
	95th	118	119	120	122	124	125	126	78	78	78	78	79	79	79

Clinical Practice Guideline for Screening and Management of High Blood Pressure in Children and Adolescents. Pediatrics 2017;140(3) pii:e20171904.³⁹⁸

APPENDIX 4 Dosing Recommendation for the Initial Prescription of Antihypertensive Drugs for Outpatient Management of Chronic Hypertension in Children and Neonates^{398,405}

Drugs	Doses	Frequency
Angiotensin-Converting Enzyme Inhibitors *		
Captopril **	0.3 mg/kg/dose (Max 6 mg/kg/day or 50 mg/day)	BD or TDS
Enalapril **	20-50 kg: Initially 2.5 mg/day (Max 20 mg/day) ≥50 kg: Initially 5 mg/day (Max 40 mg/day)	Once daily or BD
Angiotensin-Receptor Blockers *		
Irbesartan #	6-12 years: 75 – 150 mg/day >13 years: 150 – 300 mg/day	Once daily
Losartan **	≥6 years; 20-50 kg: Initially 0.7 mg/kg/day (Max 50 mg/day) >50kg: Initially 1.4 mg/kg/day (Max 100 mg/day)	Once daily
Valsartan **	≥6 years; <35kg: Initially 40 mg/day (Max 80 mg/day) 35-80kg: Initially 80 mg/day (Max 160 mg/day)	Once daily
Calcium Channel Blockers		
Amlodipine **	6-17 years: Initially 2.5 mg/day (Max 5 mg/day)	Once daily
Nifedipine (Immediate release) ***	1 month - 11 years: 0.2-0.3 mg/kg/day (Max 3 mg/kg/day or 60 mg/day) 12-17 years: 5-20 mg TDS (Max 60 mg/day)	TDS or QID
Felodipine #	≥6 years: 2.5 -10 mg	Once daily
Diuretics		
Chlorothiazide **	6 months - 12 years: 10-20 mg/kg/day <2 years: Max 375 mg/day 2-12 years: Max 1,000 mg/day	Once daily or BD
Hydrochlorothiazide **	6 months - 2 years: 1-2 mg/kg/day (Max 37.5 mg/day) >2-12 years: 1-2 mg/kg/day (Max 100 mg/day)	Once daily or BD
Furosemide **	1-3 mg/kg/day (Max 80 mg/day)	Once daily or BD
Spironolactone #	1-3 mg/kg/day (Max 100 mg/day)	Once daily or BD
Beta Adrenergic Blockers		
Metoprolol ***	1 month - 11 years: Initially 1 mg/kg/dose (Max 8 mg/kg/day or 200 mg/day) 12-17 years: Initially 50-100 mg/day (Max 200 mg/day)	BD
Propranolol **	Initially 1 mg/kg/day Maintenance 2-4 mg/kg/day (Max 4 mg/kg/day)	BD or TDS
Atenolol **	1 month - 11 years: 0.5-2 mg/day (Max 50 mg/day) Child 12-17 years: 25-50 mg/day (Max 50 mg/day)	Once daily or BD

* ARB and ACEI are contraindicated in pregnant adolescent and neonates less than 44 weeks (Perindopril is not indicated in for the management of chronic hypertension in children and adolescents).

** Referenced from 153rd Edition, MIMS, 2018.

*** British National Formulary for Children (BNFC) 2018-2019.

American Academy of Pediatrics (AAP) 2017.

APPENDIX 5 Clinical Questions

1. What is the prevalence of hypertension in adults?
2. What are the causes of hypertension in adults?
3. What are the diagnostic criteria of hypertension in adults, pregnant women and neonates/children/adolescents?
4. What is the role of home blood pressure monitoring and how it should be measured?
5. What is the role of ambulatory blood pressure monitoring?
6. What are the secondary causes of hypertension in adults, pregnant women and neonates/children/adolescents?
7. How should patients with hypertension be assessed clinically?
8. Which investigation should be done in newly diagnosed hypertension?
9. How should patients be stratified according to global cardiovascular risk?
10. What non-pharmacological intervention is recommended and beneficial?
11. What and how should pharmacological management be started?
12. What target blood pressure should be aimed for in general hypertensive population and in specific sub-groups?
13. When should target blood pressure be achieved?
14. When should combination therapy be used?
15. How to recognise, evaluate and manage resistant hypertension?
16. How should severe hypertension be assessed and managed?
17. How should specific sub-groups with hypertension be managed?
 - Diabetes
 - Renal disease
 - Heart disease
 - Stroke
 - Older adults
 - Women
 - Neonates, children and adolescents
18. What are the current available pharmacological treatment for hypertension?
19. How cost effective is treating hypertensive?
20. How should resistant and refractory hypertension be diagnosed, assessed and managed?
21. Should aspirin be prescribed to patients with hypertension?
22. What is the role of device based therapy in hypertension?
23. What key research areas should be focused on to address unanswered clinical questions?

References

1. 2017 ACC/AHA/AAPA/ABC/ACPM/AGS/APhA/ASH/ASPC/NMA/PCNA Guideline for the Prevention, Detection, Evaluation, and Management of High Blood Pressure in Adults: A Report of the American College of Cardiology/American Heart Association Task Force on Clinical Practice Guidelines. *J Am Coll Cardiol* 2017;DOI: 10.1016/j.jacc.2017.11.005
2. Omar MA, for Institute for Public Health (IPH) 2012. Second Malaysia Burden of Disease and Injury Study. NMRR-10-758-6818. (<http://iku.moh.gov.my/index.php/research-eng/list-of-research-eng/iku-eng/bod-eng>).
3. Nur Liana AM, Mohd AO, Yi Yi K, et al. Prevalence, awareness, treatment and control of hypertension in the Malaysian population: findings from the National Health and Morbidity Survey 2006-2015. *J Human Hypertens*. 2018. (<https://doi.org/10.1038/s41371-018-0082-x>)
4. National Institute for Health and Excellence Clinical Guideline 127: Hypertension. August 2011 (available at: <http://publications.nice.org.uk/hypertension-cg127/guidance>. Accessed 6 December 2017).
5. Kostis JB, Davis BR, Cutler J, et al. Prevention of heart failure by antihypertensive drug treatment in older persons with isolated systolic hypertension. SHEP Cooperative Research Group. *JAMA* 1997;278:212-6.
6. SHEP Cooperative Research Group. Prevention of stroke by antihypertensive drug treatment in older persons with isolated systolic hypertension. Final results of the Systolic Hypertension in the Elderly Program (SHEP). *JAMA* 1991;265:3255-64.
7. Staessen JA, Thijs L, Fagard R, et al. Predicting cardiovascular risk using conventional vs ambulatory blood pressure in older patients with systolic hypertension. Systolic Hypertension in Europe Trial Investigators. *JAMA* 1999;282:539-46.
8. Burt VL, Whelton P, Roccella EJ, et al. Prevalence of hypertension in the US adult population. Results from the Third National Health and Nutrition Examination Survey, 1988-1991. *Hypertension* 1995;25:305-13.
9. Franklin SS, Gustin W, Wong ND, et al. Hemodynamic patterns of age-related changes in blood pressure. The Framingham Heart Study. *Circulation* 1997;96:308-15.
10. Franklin SS, Larson MG, Khan SA, et al. Does the relation of blood pressure to coronary heart disease risk change with aging? The Framingham Heart Study. *Circulation* 2001;103:1245-9.
11. Briasoulis A, Androulakis E, Palla M, Papageorgiou N, Tousoulis D. White-coat hypertension and cardiovascular events: a meta-analysis. *J Hypertens* 2016;34(4):593-599.
12. Franklin SS, O'Brien E, Staessen JA. Masked hypertension: understanding its complexity. *Eur Heart J* 2017;38(15):1112-1118.
13. Quinn RR, Hemmelgarn BR, Padwal RS, et al. The 2010 Canadian Hypertension Education Program recommendations for the management of hypertension: part I - blood pressure measurement, diagnosis and assessment of risk. *Can J Cardiol* 2010;26(5):241-8.
14. Ministry of Health, Malaysia circular KKM 600-29/1/10, 2016.

15. Niiranen TJ, Hanninen MR, Johansson J, et al. Home-measured blood pressure is a stronger predictor of cardiovascular risk than office blood pressure: the Finn-Home study. *Hypertension* 2010;55(6):1346-52.
16. Asayama K, Ohkubo T, Hara A, et al. Repeated evening home blood pressure measurement improves prognostic significance for stroke: a 12-year follow-up of the Ohasama study. *Blood Press Monit.* 2009;14(3):93-8.
17. Stergiou GS, Bliziotis IA. Home blood pressure monitoring in the diagnosis and treatment of hypertension: a systematic review. *Am J Hypertens* 2011;24(2):123-34.
18. Jones MI, Greenfield SM, Bray EP, et al. Patient self-monitoring of blood pressure and self-titration of medication in primary care: the TASMINH2 trial qualitative study. *Br J Gen Pract* 2012;62(595):e135-42.
19. Bray EP, Holder R, Mant J, et al. Does self-monitoring reduce blood pressure? Meta-analysis with meta-regression of randomized controlled trials. *Ann Med* 2010;42:371-386.
20. Agarwal R, Bills JE, Hecht TJ, et al. Light role of home blood pressure monitoring in overcoming therapeutic inertia and improving hypertension control: a systematic review and meta-analysis. *Hypertension* 2011;57:29-38.
21. Eguchi K, Kuruwilla S, Ishikawa J, et al. Correlations between different measures of clinic, home, and ambulatory blood pressure in hypertensive patients. *Blood Press Monit* 2011;16(3):142-8.
22. Stergiou GS, Tzamouranis D, Nasothimiou EG, et al. Are there really differences between home and daytime ambulatory blood pressure? Comparison using a novel dual-mode ambulatory and home monitor. *J Hum Hypertens* 2010;24(3):207-12.
23. Parati G, Stergiou GS, Asmar R, et al. European Society of Hypertension practice guidelines for home blood pressure monitoring. *J Hum Hypertens* 2010;24(12):779-85.
24. International Society for Chronobiology. 2013 ambulatory blood pressure monitoring recommendations for the diagnosis of adult hypertension, assessment of cardiovascular and other hypertension-associated risk, and attainment of therapeutic goals. *Chronobiol Int* 2013; 30(3):355-410.
25. O'Brien E, Parati G, Stergiou G, et al. on behalf of the European Society of Hypertension Working Group on Blood Pressure Monitoring. European Society of Hypertension position paper on ambulatory blood pressure monitoring. *J Hypertens* 2013;31(9):1731-68.
26. 2018 ESH/ESC Guidelines for the management of arterial hypertension: The Task Force for the management of arterial hypertension of the European Society of Hypertension (ESH) and the European Society of Cardiology (ESC). *Eur Heart J* 2018;00:1-98 (doi:10.1093/eurheartj/ehy339)
27. Cheong AT, Tong SF, Sazlina SG, et al. Blood pressure control among hypertensive patients with and without diabetes mellitus in six public primary care clinics in Malaysia. *Asia Pac J Public Health* 2015;27(2):NP580-9.
28. Chia YC, Srinivas P. Cardiovascular disease risk in a semirural community in Malaysia. *Asia Pac J Public Health* 2009;21(4):410-20.
29. Chia YC. Review of tools of cardiovascular disease risk stratification: interpretation, customisation and application in clinical practice. *Singapore Med J* 2011;52(2):116

30. Chia YC, et al. Validation of the Framingham general cardiovascular risk score in a multiethnic Asian population: a retrospective cohort study. *BMJ Open* 2015;5:e007324.
31. Selvarajah S, Kaur G, Haniff J, Kee CC, Tee GH, van de Graaf Y, Nots M. Comparison of the Framingham Risk Score, SCORE and WHO/ISH cardiovascular risk prediction models in an Asian population. *Int J Cardiol.* 2014; 176: 211-218.
32. National Institutes of Health. National Heart, Lung, and Blood Institute. The seventh report of the joint national committee on prevention, detection, evaluation, and treatment of high blood pressure (JNC 7). 2003. *NIH publication* no. 03-5233.
33. Silaste ML, Junes R, Rantala AO, et al. Dietary and other non-pharmacological treatments in patients with drug-treated hypertension and control subjects. *J Intern Med* 2000;247:318-324.
34. Eriksson MK, Franks PW, Eliasson M. A 3-year randomized trial of lifestyle intervention for cardiovascular risk reduction in the primary care setting: the Swedish Björknäs study. *PLoS ONE* 2009;4(4):e5195.
35. Dickinson HO, Mason JM, Nicolson DJ, et al. Lifestyle interventions to reduce raised blood pressure : a systematic review of randomised controlled trials. *J Hypertens* 2006 Feb; 24(2):215-33.
36. Semlitsch T, Jeitler K, Berghold A, et al. Long-term effects of weight-reducing diets in people with hypertension. *Cochrane Database Sys Rev* 2016; Issue 3. Art. No.: CD008274.
37. Aucott L, Rothnie H, McIntyre L, et al. Long-term weight loss from lifestyle intervention benefits blood pressure? A systematic review. *Hypertension* 2009;54(4):756-62.
38. Taylor RS, Ashton KE, Moxham T, et al. Reduced dietary salt for the prevention of cardiovascular disease. *Cochrane Database Sys Rev* 2011;(7):CD009217. doi:10.1002/14651858.CD009217.
39. Aburto NJ, Ziolkovska A, Hooper L, et al. Effect of lower sodium intake on health: systematic review and meta-analyses. *BMJ* 2013; 346:f1326.
40. Graudal NA, Hubeck-Graudal T, Jurgens G. Effects of low sodium diet versus high sodium diet on blood pressure, renin, aldosterone, catecholamines, cholesterol, and triglyceride. *Cochrane Database Sys Rev* 2017; Issue 4. Art. No.: CD004022.
41. He FJ, Li J, MacGregor GA. Effect of longer term modest salt reduction on blood pressure: Cochrane systematic review and meta-analysis of randomised trials. *BMJ* 2013; 346:f1325.
42. World Health Organization. Reducing sodium intake to reduce blood pressure and risk of cardiovascular diseases in adults. Geneva, Switzerland: WHO 2017. Available at http://www.who.int/elena/titles/sodium_cvd_adults/en/ (accessed 6 Jan 2018).
43. Aburto NJ, Ziolkovska A. Effect of reduced sodium intake on cardiovascular disease, coronary heart disease and stroke. Geneva, Switzerland: World Health Organization 2012. http://apps.who.int/iris/bitstream/10665/79322/1/9789241504904_eng.pdf.
44. World Health Organization. Guideline: Sodium intake for adults and children. Geneva, Switzerland: WHO 2012. Available at http://apps.who.int/iris/bitstream/10665/77985/1/9789241504836_eng.pdf?ua=1&ua=1.
45. Adler AJ, Taylor F, Martin N, et al. Reduced dietary salt for the prevention of cardiovascular disease. *Cochrane Database Sys Rev* 2014; Issue 12. Art. No.: CD009217.

46. Rashidah A, Yeo PS, Noor Ani A, et al. Sodium intake among normotensive health staff assessed by 24 hour urinary excretion: a cross-sectional study. *Mal J Nutr* 2014;20:317-26.
47. Mente A, O'Donnell MJ, Rangarajan S, et al. Association of urinary sodium and potassium excretion with blood pressure. *N Engl J Med* 2014;371:601-11.
48. Norimah AK Jr, Safiah M, Jamal K, et al. Food consumption patterns: Findings from the Malaysian Adult Nutrition Survey (MANS). *Malays J Nutr* 2008;14(1):25-39.
49. Xin X, He J, Frontini MG, et al. Effects of alcohol reduction on blood pressure: a meta-analysis of randomized controlled trials. *Hypertension* 2001; 38(5):1112-7.
50. Roerecke M, Kaczorowski J, Tobe SW, et al. The effect of a reduction in alcohol consumption on blood pressure: a systematic review and meta-analysis. *Lancet Public Health* 2017;2: e108–20.
51. Semlitsch T, Jeitler K, Hemkens LG, et al. Increasing physical activity for the treatment of hypertension: a systematic review and meta-analysis. *Sports Med* 2013 Oct; 43(10): 1009–1023.
52. Cornelissen VA, Smart NA. Exercise training for blood pressure: a systematic review and meta-analysis. *J Am Heart Assoc* 2013;2:e004473
53. Huang G, Shi X, Gibson CA, et al. Controlled aerobic exercise training reduces resting blood pressure in sedentary older adults. *Blood Press* 2013;22(6):386-94.
54. Cornelissen VA, Fgard RH, Coeckelberghs E. Impact of resistance training on blood pressure and other cardiovascular risk factors: a meta-analysis of randomized, controlled trials. *Hypertension* 2011;58(5):950-8.
55. Owen A, Wiles J, Swaine I. Effect of isometric exercise on resting blood pressure: a meta analysis. *J Hum Hypertens* 2010;24(12):796-800.
56. Carlson DJ, Dieberg G, Hess NC, et al. Isometric exercise training for blood pressure management: a systematic review and meta-analysis. *Mayo Clin Proc* 2014;89(3):327-34.
57. Inder JD, Carlson DJ, Dieberg G, et al. Isometric exercise training for blood pressure management: a systematic review and meta-analysis to optimize benefit. *Hypertens Res* 2016;39(2):88-94.
58. Pedersen BK and Saltin B. Exercise as medicine – evidence for prescribing exercise as therapy in 26 different chronic diseases. *Scand J Med Sci Sports* 2015, 25: 1–72.
59. Rogers MW, Probst MM, Gruber JJ, et al. Differential effects of exercise training intensity on blood pressure and cardiovascular responses to stress in borderline hypertensive humans. *J Hypertens* 1996;14(11):1369-1375.
60. Ebrahim S, Smith GD. Lowering blood pressure: a systematic review of sustained effects of nonpharmacological interventions. *J Public Health Med* 1998;20:441-448.
61. Ndanuko RN, Tapsell LC, Charlton KE et al. Dietary patterns and blood pressure in adults: a systematic review and meta-analysis of randomized controlled trials. *Adv Nutri* 2016;7(1):76-89.
62. Miller V, Mente A, Dehghan M, et al. Fruit, vegetable, and legume intake, and cardiovascular disease and deaths in 18 countries (PURE): a prospective cohort study. *Lancet* 2017; 390: 2037–49.
63. Dickinson HO, Beyer FR, Ford GA, et al. Relaxation therapies for the management of primary hypertension in adults. *Cochrane Database Syst Rev* 2008, Issue 1. Art. No.: CD004935.

64. Hagins M, States R, Selfe T, et al. Effectiveness of yoga for hypertension: systematic review and meta-analysis. *Evidence-Based Complementary and Alternative Medicine* 2013;2013:-649836
65. Aburto NJ, Hanson S, Gutierrez H, et al. Effect of increased potassium intake on cardiovascular risk factors and disease: systematic review and meta-analyses. *BMJ* 2013;346:f1378.
66. Dickinson HO, Nicolson D, Cook JV, et al. Calcium supplementation for the management of primary hypertension in adults. *Cochrane Database Syst Rev* 2006, Issue 2. Art. No.: CD004639.
67. Beyer FR, Dickinson HO, Nicolson D, et al. Combined calcium, magnesium and potassium supplementation for the management of primary hypertension in adults. *Cochrane Database Syst Rev* 2006, Issue 3. Art. No.: CD004805.
68. Dickinson HO, Nicolson D, Campbell F, et al. Magnesium supplementation for the management of primary hypertension in adults. *Cochrane Database Syst Rev* 2006, Issue 3. Art. No.: CD004640.
69. Stabler SN, Tejani AM, Huynh F, et al. Garlic for the prevention of cardiovascular morbidity and mortality in hypertensive patients. *Cochrane Database Syst Rev* 2012; Issue 8. Art. No.: CD007653.
70. Steffen M, Kuhle C, Hensrud D, et al. The effect of coffee consumption on blood pressure and the development of hypertension: a systematic review and meta-analysis. *J Hypertens* 2012;30(12):2245-54.
71. Mesas AE, Leon-Muñoz LM, Rodríguez-Artalejo F, et al. The effect of coffee on blood pressure and cardiovascular disease in hypertensive individuals: a systematic review and meta-analysis. *Am J Clin Nutr* 2011;94:1113-26.
72. McCarty CA, Berg RL, Rottscheit CM, et al. The use of dietary supplements and their association with blood pressure in a large Midwestern cohort. *BMC Complementary Altern Med* 2013;13:339.
73. Cormick G, Ciapponi A, Cafferata ML, et al. Calcium supplementation for prevention of primary hypertension. *Cochrane Database Syst Rev* 2015, Issue 6. Art. No.: CD010037.
74. Greyling A, Ras RT, Zock PL, et al. The effect of black tea on blood pressure: a systematic review with meta-analysis of randomized controlled trials. *PLoS ONE* 2014;9(7): e103247.
75. Peng X, Zhou R, Wang B, et al. Effect of green tea consumption on blood pressure: A meta-analysis of 13 randomized controlled trials. *Sci Rep* 2014; 4:6251.
76. Blood Pressure Lowering Treatment Trialists' Collaboration. Effects of different blood pressure lowering regimens on major cardiovascular events: results of prospectively-designed overviews of randomised trials. *Lancet* 2003;362:1527-35.
77. Lonn EM, Bosch J, Lopez – Jaramillo P, et al. Blood pressure lowering in intermediate –risk persons without cardiovascular disease. *N Eng J Med* 2016;373:2103-2116.
78. Thompson AM, Hu T, Eshelbrenner CL, et al. antihypertensive treatment for primary prevention of cardiovascular events in mild hypertension. *JAMA* 2011; 305(9): 913-922.
79. PROGRESS Collaborative Group. Randomised trial of a perindopril-based blood-pressure-lowering regimen among 6,105 individuals with previous stroke or transient ischaemic attack. *Lancet* 2001;358(9287):1033-41.

80. Pepine CJ, Handberg EM, Coper-DeHoff RM, et al. A calcium antagonist vs a non-calcium antagonist hypertension treatment strategy for patients with coronary artery disease. The international verapamil – trandolapril study (INVEST): Randomized controlled trial. *JAMA* 2003;290(21): 2805-2816.
81. European trial on reduction of cardiac events with perindopril in stable coronary artery disease investigators. Efficacy of perindopril in reduction of cardiovascular events among patients with stable coronary artery disease: randomised, double-blind, placebo-controlled, multicentre trial (the EUROPA study). *Lancet* 2003;362(9386):782-8.
82. Stevo Julius, Sverre E Kjeldsen, Michael Weber, et al. Outcome in hypertensives patients at high cardiovascular risk treated with regimens based on valsartan or amlodipine: the VALUE randomized trial. *Lancet* 2004;363(9426):2022-2031.
83. Patel A. ADVANCE Collaborative Group. Effects of a fixed combination of perindopril and indapamide on macrovascular and microvascular outcomes in patients with type 2 diabetes mellitus (the ADVANCE trial): a randomised controlled trial. *Lancet* 2007;370(9590):829-40.
84. Jamerson K, Weber MA, Bakris GL, et al. for ACCOMPLISH Trial Investigators. Benazepril plus amlodipine or hydrochlorothiazide for hypertension in high-risk patients. *N Engl J Med*. 2008; 359:2417-2428.
85. Hansson L, Lindholm LH, Ekblom T, et al. Randomised trial of old and new antihypertensive drugs in elderly patients: cardiovascular mortality and morbidity the Swedish trial in old patients with hypertension-2 study. *Lancet* 1999;354:1751-6.
86. Brown MJ, Palmer CR, Castaigne A, et al. Morbidity and mortality in patients randomised to double-blind treatment with a long-acting calcium-channel blocker or diuretic in the International Nifedipine GITS study: Intervention as a Goal in Hypertension Treatment (INSIGHT). *Lancet* 2000;356:366-72.
87. Hansson L, Lindholm LH, Nishanen L, et al. Effect of angiotension-converting-enzyme inhibition compared with conventional therapy on cardiovascular morbidity and mortality in hypertension: the Captopril Prevention Project (CAPPP) randomised trial. *Lancet* 1999;353:611-6.
88. Wing LMH, Reid CM, Ryan P, et al. A comparison of outcomes with angiotensin-converting-enzyme inhibitors and diuretics for hypertension in the elderly. *N Engl J Med* 2003;348:583-92.
89. Davis BR, Cutler JA, Gordon DJ, et al. Major outcomes in high-risk hypertensive patients randomized to angiotensin-converting enzyme inhibitor or calcium channel blocker vs. diuretic. The Antihypertensive and Lipid-Lowering Treatment to Prevent Heart Attack Trial (ALLHAT). *JAMA* 2002;288:2981-97. (Errata in 2003;289:178 and 2004; 291:2196.).
90. James PA, Oparil S, Carter BL, et al. 2014 Evidence-based guideline for the management of high blood pressure in adults report from the panel members appointed to the eighth joint national committee (JNC8). *JAMA* 2014;311(5):507-520.
91. Weber MA, Schiffrin EL, White WB, et al. Clinical practice guidelines for the management of hypertension in the community: a statement by the American Society of Hypertension and the International Society of Hypertension. *J Clin Hypertens*. 2014;16(1):14-26. doi:10.1111/jch.12237.
92. Lindholm LH, Carlberg B, Samuelsson O. Should β -blockers remain first choice in the treatment of primary hypertension? A meta-analysis. *Lancet* 2005;366:1545-53.

93. Khan N, McAlister FA. Re-examining the efficacy of beta-blockers for the treatment of hypertension: a meta-analysis. *CMAJ* 2006;174(12):1737-42.
94. Bradley HA, Wiysonge CS, Volmink JA, et al. How strong is the evidence for the use of beta-blockers as first-line therapy for hypertension? Systematic review and meta-analysis. *J Hypertens* 2006;24(11):2131-41.
95. Wiysonge CS, Bradley HA, Mayosi BM, et al. Beta-blockers for hypertension. *Cochrane Database Syst Rev* 2007;(1): CD002003.
96. Law MR, Morris JK, Wald NJ. Use of blood pressure lowering drugs in the prevention of cardiovascular disease: meta-analysis of 147 randomised trials in the context of expectations from prospective epidemiological studies. *BMJ* 2009;338:b1665.
97. Chrysant SG, Chrysant GS. Current status of beta blockers for the treatment of hypertension: an update. *Drugs Today* 2012;48(5):353-66.
98. Wright JM, Musini VM. First-line drugs for hypertension. *Cochrane Database Syst Rev* 2009;3:CD001841.
99. Wong GW, Boyda HN, Wright JM. Blood pressure lowering efficacy of beta-1 selective beta blockers for primary hypertension. *Cochrane Database Syst Rev*. 2016;3:CD007451.
100. Cruickshank JM. The role of beta-blockers in the treatment of hypertension. *Adv Exp Med Biol*. 2017;956:149-166.
101. Hackam DG, Quinn RR, Ravani P, et al. The 2013 Canadian Hypertension Education Program recommendations for blood pressure measurement, diagnosis, assessment of risk, prevention, and treatment of hypertension. *Can J Cardiol* 2013;29 (5):528-42.
102. Ogihara T, Kikuchi K, Matuoka H, et al. The Japanese Society of Hypertension Guidelines for the management of hypertension (JSH 2009). *Hypertens Res* 2009;32(1):3-107.
103. Steward S, Carrington MJ, Swemmer CH, et al. Determinants of achieving early blood pressure control with monotherapy in a primary care setting. *J Clin Hypertens* 2013;15;9:674-680.
104. Bangalore S, Kamalakkannan G, Parkar S, et al. Fixed-dose combinations improve medication compliance: a meta-analysis. *Am J Med* 2007;120:713-19.
105. Ajay K.G., Arshad S. and Neil R.P. Compliance, safety, and effectiveness of fixed-dose combinations of antihypertensive agents: a meta-analysis. *Hypertension* 2010;55:399-407.
106. Alan HG, Jan NB, Barry LC, et al. Position article combination therapy in hypertension. *J Am Soc Hypertens* 2010;4(2):90-98.
107. Feldman RD, Zou GY, Vandervoort MK, et al. A simplified approach to the treatment of uncomplicated hypertension: a cluster randomized, controlled trial. *Hypertension* 2009;53:646-53.
108. Brown MJ, McInnes GT, Papst CC, et al. Aliskiren and the calcium-channel blocker amlodipine combination as an initial treatment strategy for hypertension. *Lancet* 2011;377(9762):312-320.
109. MacDonald TM, Williams B, Caulfield M, et al. Monotherapy versus dual therapy for the initial treatment of hypertension (PATHWAY -1): a randomized double-blind controlled trial. *BMJ Open* 2015;5:e007645.
110. Dickson M, Plauschinat CA. Compliance with antihypertensive therapy in the elderly; a comparison of fixed-dose combination amlodipine/benazepril versus component-based free-combination therapy. *Am J Cardiovasc Drugs* 2008;8(1):45-50.

111. The Blood Pressure Lowering Treatment Trialists' Collaboration. Blood pressure – lowering treatment based on cardiovascular risk: a meta-analysis of individual patient data. *Lancet*. 2014;384(9943):591-598.
112. Dahlof B, Devereux RB, Kjeldsen SE, et al. Cardiovascular morbidity and mortality in the Losartan Intervention For Endpoint reduction in hypertension study (LIFE): a randomised trial against atenolol. *Lancet* 2002;359:995-1003.
113. Beckett NS, Peters R, Fletcher AE, et al. Treatment of hypertension in patients 80 years of age or older. *N Engl J Med* 2008;358(18):1887-98.
114. Matsuzaki M, Ogihara T, Umemoto S, et al. Prevention of cardiovascular events with calcium channel blocker-based combination therapies in patients with hypertension: a randomized controlled trial. *J Hypertens* 2011;29:1649-1659.
115. Dahlof B, Sever PS, Poulter NR, et al. Prevention of cardiovascular events with an antihypertensive regimen of amlodipine adding perindopril as required versus atenolol adding bendroflumethiazide as required, in the Anglo-Scandinavian Cardiac Outcomes Trial-Blood Pressure Lowering Arm (ASCOT-BPLA): a multicentre randomised controlled trial. *Lancet* 2005;366(9489):895-906.
116. Ettehad D, Emdin CA, Kiran A, et al. Blood pressure lowering for prevention of cardiovascular disease and death: a systematic review and meta-analysis. *Lancet* 2016;387(10022):957-967.
117. The SPRINT Research Group. A randomized trial of intensive versus standard blood-pressure control. *N Engl J Med* 2015; 373:2103-2116.
118. Moser M, Setaro JF. Resistant or difficult to control hypertension. *N Engl J Med* 2006;355:385-92.
119. van der Wardt V, Harrison JK, Welsh T, et al. Withdrawal of antihypertensive medication: a systematic review. *J Hypertens* 2017; 35(9):1742-1749.
120. Patel KK, Young L, Howell EH, et al. Characteristics and outcomes of patients presenting with hypertensive urgency in the office setting. *JAMA Intern Med* 2016;176(7):981-988.
121. Grassi D, O'Flaherty M, Pellizzari M, et al. on behalf of the Group of Investigators of the REHASE Program. Hypertensive Urgencies in the Emergency Department: Evaluating Blood Pressure Response to Rest and to Antihypertensive Drugs With Different Profiles. *J Clin Hypertens* 2008;10:662-667.
122. Park SK, Kim WJ, Lee DY, et al. Comparing the clinical efficacy of resting and antihypertensive medication in patients of hypertensive urgency: a randomized, control trial. *J Hypertens* 2017;35(1):1-7.
123. Pergolini M. The Management of hypertensive crises: a clinical review. *Clin Ter* 2009;160(2):151-7.
124. van den Born BJ, Beutler JJ, Gaillard CA, et al. Dutch guideline for the management of hypertensive crisis. 2010 revision. *Neth J Med* 2011;69(5):248-55.
125. Suneja M, Sanders ML. Hypertensive Emergency. *Med Clin North Am* 2017;101(3):465-478.
126. Tisdale JE, Huang MB, Borzak S. Risk factors for hypertensive crisis: importance of out-patient blood pressure control. *Fam Pract* 2004;21(4):420-4.
127. Mancia G, Fagard R, Krzysztof Narkiewicz JR, et al. 2013 Practice guidelines for the management of arterial hypertension of the European Society of Hypertension (ESH) and the European Society of Cardiology (ESC) ESH/ESC Task Force for the Management of Arterial Hypertension. *J Hypertens* 2013;31:1925-1938.

128. Chobanian AV, Bakris GL, Black HR, et al. Seventh Report of the Joint National Committee on prevention, detection, evaluation, and treatment of high blood pressure. *Hypertension* 2003;42:1206–1252.
129. Marik PE, Rivera R. Hypertensive emergencies: an update. *Curr Opin Crit Care* 2011;17(6):569-80.
130. González Pacheco H, Morales Victorino N, Núñez Urquiza JP, et al. Patients with hypertensive crises who are admitted to a coronary care unit: clinical characteristics and outcomes. *J Clin Hypertens* 2013;15(3):210-4.
131. Muiesan ML, Salvetti M, Amadoro V, et al. on behalf of the Working Group on Hypertension, Prevention, Rehabilitation of the Italian Society of Cardiology, the Società Italiana dell'Ipertensione Arteriosa (SIIA). An update on hypertensive emergencies and urgencies. *J Cardiovasc Med* 2015; 16:372–382.
132. Varon J, Soto-Ruiz KM, Baumann BM, et al. The management of acute hypertension in patients with renal dysfunction: labetalol or nicardipine? *Postgrad Med* 2014;126:124-30.
133. Peacock IV WF, Hilleman DE, Levy PD, et al. A systematic review of nicardipine vs labetalol for the management of hypertensive crises. *Am J Emerg Med* 2012;30(6):981-93
134. Peacock WF, Varon J, Baumann BM, et al. CLUE: randomized comparative effectiveness trial of IV nicardipine versus labetalol use in the emergency department. *Crit Care* 2011;15:R157.
135. Elliot WJ. Hypertensives emergencies. *Crit Care Clin* 2001;17:435-451.
136. Editorial. Thought for autoregulation in the hypertensive patient. *Lancet* 1979;314(8141):510.
137. Messerli FH, Kowey P, Grodzicki T. Sublingual nifedipine for hypertensive emergencies. *Lancet* 1991;338: 881.
138. Ministry of Health, Malaysia. National Diabetes Registry report 2017. (Pending publication with permission from Disease Control Division, MOH).
139. ADA Standards of Medical Care in Diabetes. *Diabetes Care* 2017;40(Suppl. 1):S1–S2
140. Lewis EJ, Hunsicker LG, Bain RP, et al. The effect of angiotensin-converting-enzyme inhibition on diabetic nephropathy. *N Engl J Med* 1993;329:1456-1462.
141. Ravid M, Lang R, Rachmani R, et al. Long-term renoprotective effect of angiotensin-converting enzyme inhibition in non-insulin-dependent diabetes mellitus. A 7-year follow-up study. *Arch Intern Med* 1996;156:286-289.
142. Kasiske BL, Kalil RS, Ma JZ, et al. Effect of antihypertensive therapy on the kidney in patients with diabetes: a meta-regression analysis. *Ann Int Med* 1993;118:129-138.
143. Heeg JE, de Jong PE, Van de Hem GK, de Zeeuw D. Efficacy and variability of the antiproteinuric effect of ACE Inhibition by lisinopril. *Kidney Int* 1989;36(2):272-9.
144. Parving HH, Lehnert H, Brochner-Mortensen J, et al. The effect of irbesartan on the development of diabetic nephropathy in patients with type 2 diabetes. *N Engl J Med* 2001;345:870-878.
145. Brenner BM, Cooper ME, de Zeeuw D, et al. Effects of losartan on renal and cardiovascular outcomes in patients with type 2 diabetes and nephropathy. *N Engl J Med* 2001;345:861-869.
146. Lewis EJ, Hunsicker LG, Clarke WR, et al. Renoprotective effect of the angiotensin-receptor antagonist irbesartan in patients with nephropathy due to type 2 diabetes. *N Engl J Med* 2001;345:851-60.

147. Schrier RW, Estacio RO, Mehler, PS, Hiatt WR. Appropriate blood pressure control in hypertensive and normotensive type 2 diabetes mellitus: a summary of the ABCD trial. *Nat Clin Pract Nephrol* 2007;3:428–438.
148. Gaede P, Tarnow L, Vedel P, et al. Remission to normoalbuminuria during multifactorial treatment preserves kidney function in patients with type 2 diabetes and microalbuminuria. *Nephrol Dial Transplant* 2004;19:2784–8.
149. Bakris GL, Williams M, Dworkin L, et al. Preserving renal function in adults with hypertension and diabetes: a consensus approach. *Am J Kidney Dis* 2000;36:646–661.
150. UK Prospective Diabetes Study Group. Tight blood pressure control and risk of macrovascular and microvascular complications in type 2 diabetes: UKPDS 38. *BMJ* 1998;317:703–713.
151. The ACCORD Study Group. Effects of Intensive Blood-Pressure Control in Type 2 Diabetes Mellitus. *N Engl J Med* 2010;362(17):1575–1582.
152. Eryd SA, Gudbjörnsdóttir S, Manhem K, et al. Blood pressure and complications in individuals with type 2 diabetes and no previous cardiovascular disease: national population based cohort study. *BMJ* 2016;354:i4070.
153. Emdin CA, Rahimi K, Neal B, et al. Blood pressure lowering in type 2 diabetes: a systematic review and meta-analysis. *JAMA* 2015;313(6):603–615.
154. Williams B. Treating Hypertension in patients with diabetes. When to start and how low to go? *JAMA* 2015;313(6):573–574.
155. Heart Outcomes Prevention Evaluation (HOPE) Study Investigators. Effects of ramipril on cardiovascular and microvascular outcomes in people with diabetes mellitus: results of the HOPE study and MICRO- HOPE substudy. *Lancet* 2000;355:253–9.
156. Howard BV, Roman RB, MJ, Devereux, et al. Effect of lower targets for blood pressure and LDL cholesterol on atherosclerosis in diabetes; The SANDS randomized trial. *JAMA* 2008;299:1678–1689.
157. Perkovic V, de Galan BE, Ninomiya T, et al. Lowering Blood Pressure Reduces Renal Events in Type 2 Diabetes. *J Am Soc Nephrol* 2009;20(4):883–892.
158. Barnett AH, Bain SC, Bouter P, et al. Angiotensin-receptor blockade versus converting enzyme inhibition in type 2 diabetes and nephropathy. *N Engl J Med* 2004;351:1952–61.
159. Ruggenenti P, Fassi A, Ilieva AP, et al. Preventing microalbuminuria in type 2 diabetes. *N Engl J Med* 2004;351:1941–51.
160. Vasan RS, Larson MG, Leip EP, et al. Assessment of frequency of progression to hypertension in non-hypertensive participants in the Framingham Heart Study: a cohort study. *Lancet* 2001;358:1682–86.
161. Mazidi M, Rezaie P, Gao HK, et al. Effect of Sodium-glucose cotransport-2 inhibitors on blood pressure in people with type 2 diabetes mellitus: a systematic review and meta-analysis of 43 randomized control trials with 22,528 patients. *J Am Heart Assoc* 2017;6(6).pii:e004007.
162. Zinman B, Wanner C, Lachin JM, et al. Empagliflozin, cardiovascular outcomes, and mortality in type 2 diabetes. *N Engl J Med* 2015;373:2117–28.
163. Neal B, Perkovic V, Kenneth Mahaffey KW, et al. Canagliflozin and cardiovascular and renal events in type 2 diabetes. *N Engl J Med* 2017; 377:644–657.
164. Klag MJ, Whelton PK, Randall BL, et al. End-stage renal disease in African-American and white men.16 year MRFIT findings. *JAMA* 1997; 277:1293–1298.

165. Madhavan S, Stockwell D, Cohen H, et al. Renal function during antihypertensive treatment. *Lancet* 1995; 345:749-751.
166. Chronic Kidney Disease Work Group. K/DOQI clinical practice guidelines for chronic kidney disease: Evaluation, classification, and stratification. Kidney Disease Outcome Quality Initiative. *Am J Kidney* 2002;39:S1-S266.
167. Blood Pressure Lowering Treatment Trialists' Collaboration, Ninomiya T, Perkovic V, Turnbull F et al. Blood pressure lowering and major cardiovascular events in people with and without chronic kidney disease: meta-analysis of randomised controlled trials. *BMJ* 2013;347: f5680).
168. Heerspink HJ, Ninomiya T, Zoungas S et al Effects of lowering blood pressure on cardiovascular events and mortality in patients on dialysis: a systematic review and meta-analysis of randomised controlled trials. *Lancet* 2009;379(9668) :1009-15.
169. Agarwal R, Sinha AD. Cardiovascular protection with antihypertensive drugs in dialysis patients: systematic review and meta-analysis. *Hypertension* 2009;53(5):860-66.
170. Klahr S, Levey AS, Beck GJ et al. The effects of dietary protein restriction and blood pressure control on the progression of chronic kidney disease. *N Eng J Med* 1994;330:877-884.
171. Appel LJ, Wright JT Jr, Greene T et al. Intensive blood-pressure control in hypertensive chronic kidney disease. *N Engl J Med*. 2010;363:918-29.
172. Ruggenti P, Perna A, Loriga G et al. Blood-pressure control for renoprotection in patients with non- diabetic chronic renal disease (REIN-2): multicentre, randomised controlled trial. *Lancet* 2005 Mar 12;365(9463):939-46.
173. Jafar TH, Stark PC, Schmid CH et al. Progression of chronic kidney disease: the role of blood pressure control, proteinuria, and angiotensin-converting enzyme inhibition. A patient level meta-analysis. *Ann Intern Med* 2003;139:244-252.
174. Leung AA, Daskalopoulou SS, Dasgupta K et al. Hypertension Canada's 2017 Guidelines for Diagnosis, Risk Assessment, Prevention, and Treatment of Hypertension in Adults. *Can J Cardiol* 2017;33(5):557-576.
175. National Heart Foundation of Australia. Guideline for the diagnosis and management of hypertension in adults 2016, Melbourne: National Heart Foundation of Australia, 2016.
176. Gansevoort RT, Sluiter WJ, Hemmelder MH, et al. Antiproteinuric effect of blood pressure lowering agents: meta-analysis of comparative trials. *Nephrol Dial Transplant* 1995; 10:1963-1974.
177. Giatras I, Lau J, Levey AS, et al. Effect of angiotensin-converting enzyme inhibitors on the progression of nondiabetic renal disease: a meta-analysis of randomized trials. *Ann Intern Med* 1997; 127:337-345.
178. GISEN Group. Randomised placebo-controlled trial of effect of ramipril on decline of glomerular filtration rate and risk of terminal renal failure in proteinuric, non-diabetic nephropathy. *Lancet* 1997; 349:1857- 1863.
179. Kunz R, Friedrich C, Wolpers M and Mann JE. Meta-analysis: Effect of monotherapy and combination therapy with inhibitors of the renin-angiotensin system on proteinuria in renal disease. *Ann Intern Med* 2008;148:30-48.
180. Makani H, Bangalore S, Desouza KA, Shah A, Messerli FH. Efficacy and safety of dual blockade of the renin-angiotensin system: meta-analysis of randomized trials 2013;346: f360 doi: 10.1136/bmj.f360.

181. Bakris GL, Weir MR. Angiotensin-converting enzyme inhibitor-associated elevations in serum creatinine; is this a cause for concern? *Arch intern Med* 2000;160(5):685-93.
182. National Kidney Foundation K/DOQI. Clinical practice guidelines on hypertension and antihypertensive agents in chronic kidney disease. *Am J Kidney Dis* 2004;43(suppl 1):S1-S290.
183. Taal MW, Brenner BM. Evolving strategies for renoprotection: non-diabetic chronic renal disease. *Curr Opin Nephrol Hypertens* 2001;10:523-531.
184. Textor S C, Lerman L. State of the Art: Renovascular hypertension and ischaemic nephropathy. *Am J Hypertens* 2010;23(11):1159-1169.
185. Babool K, Evans C, Moore RH. Incidence of end-stage renal disease in medically treated patients with severe bilateral atherosclerotic renovascular disease. *Am J Kidney* 1998;31:971-977.
186. Mailloux LU, Napolitano B, Belluci AG, et al. Renal vascular disease causing end-stage renal disease, incidence, clinical correlates, and outcomes: a 20-year clinical experience. *Am J Kidney* 1994;24:622- 629.
187. Robert DS, Stephen CT. Renal artery stenosis. *N Engl J Med* 2001;344:431-442.
188. Textor S. Treatment of bilateral atherosclerotic renal artery stenosis or stenosis to a solitary functioning kidney. In UpToDate www.uptodate.com, accessed 13/10/17.
189. Bax L, Woittiez AJ, Kouwenberg HJ, et al. Stent placement in patients with atherosclerotic renal artery stenosis and impaired renal function: a randomized trial. *Ann Intern Med*. 2009;150(12):840-8.
190. The ASTRAL Investigators. Revascularization versus medical therapy for renal-artery stenosis. *N Engl J Med*. 2009;361:1953-1962.
191. Cooper CJ, Murphy TP, Cutlip DE, et al. Stenting and medical therapy for atherosclerotic renal-artery stenosis. *N Engl J Med* 2014;370:13-22.
192. Jaff MR, White CJ. Vascular Disease. Diagnostic and therapeutic approaches. *Cardiotext Publishing* 2011.
193. Slovut DP, Olin JW. Current concepts: Fibromuscular dysplasia. *N Engl J Med*. 2004; 350:1862-1871.
194. Sundstrom J, Arima H, Woodward M, et al. Blood pressure-lowering treatment based on cardiovascular risk: a meta-analysis of individual patient data. *Lancet* 2014;384:591-598.
195. Yap YG, Duong T, Bland JM, et al. Prognostic value of blood pressure measured during hospitalization after acute myocardial infarction: an insight from survival trials. *J Hypertens* 2007;25:307-313.
196. Staessen JA, Wang JG, Thijs L. Cardiovascular prevention and blood pressure reduction: a quantitative overview updated until 1 March 2003. *J Hypertens* 2003;21(6):1055-76.
197. Thomopoulos C, Parati G, Zanchetti A. Effects of blood pressure lowering on outcome incidence in hypertension: 4 Effects of various classes of antihypertensive drugs--overview and meta-analyses. *J Hypertens* 2015;33(2):195-211.
198. Houghton T, Freemantle N, Cleland JG. Are beta-blockers effective in patients who develop heart failure soon after myocardial infarction? A meta-regression analysis of randomized trials. *Eur J Heart Failure* 2000; 2(3):333-40.

199. Flather MD, Yusuf S, Kober L et al. Long – term ACE-inhibitor therapy in patients with heart failure or left ventricular dysfunction: a sytematic overview of data from individual patient. ACE-inhibitor Myocardial infarction Collaborative Group. *Lancet* 2000;355(9215):1575-81.
200. Werner C, Baumhakel M, Teo KK et al. RAS blockade with ARB and ACE inhibitor: current perspective on rationale and patient selection. *Clin Res Cardiol* 2008; 97(7):418-31.
201. Bangalore S, Makani H, Radford M, et al. Clinical outcomes with β -blockers for myocardial infarction: a meta-analysis of randomized trials. *Am J Med* 2014 Oct;127(10):939-53.
202. The Miami Trial Research Group. Metoprolol in acute myocardial infarction (MIAMI). A randomised placebo-controlled international trial. *EHJ* 1985;6(3):199-226.
203. Chen ZM, Pan HC, Chen YP, et al. Early intravenous then oral metoprolol in 45,852 patients with acute myocardial infarction: randomised placebo-controlled trial. *Lancet* 2005;366(9497):1622-32.
204. ISIS-1 (First International Study Of Infarct Survival) collaborative group. Randomised trial of intravenous atenolol among 16 027 cases of suspected acute myocardial infarction: ISIS-1. *Lancet* 1986;2(8498):57-66.
205. Veterans Administration Cooperative Study Group on antihypertensive agents. Effects of treatment on morbidity in hypertension. II. Results in patients with diastolic blood pressure averaging 90 through 114 mmHg. *JAMA* 1970;213(7):1143-52.
206. McAlister FA, Wiebe N, Ezekowitz JA, et al. Meta-analysis: beta-blocker dose, heart rate reduction and death in patients with heart failure. *Ann Intern Med* 2009;150(11):784-94.
207. Fauchier L, Pierra B, de Labriollet A, et al. Comparison of the beneficial effect of beta-blockers on mortality in patients with ischaemic and non-ischaemic systolic heart failure: a meta-analysis of randomized controlled trials. *Eur J Heart Failure* 2007;9(11):1136-9.
208. Chatterjee S, Biondi-Zoccai G, Abbate A, et al. Benefits of β -blockers in patients with heart failure and reduced ejection fraction: network meta-analysis. *BMJ* 2013;346:f55.
209. Shekelle PG, Rich MW, Morton SC, et al. Efficacy of angiotensin-converting enzyme inhibitors and beta blockers in the management of left ventricular systolic dysfunction according to race, gender and diabetes status: a meta analysis of major clinical trials. *J Am Coll Cardiol* 2003;41(9):1529-38.
210. Ezekowitz JA, McAlister FA. Aldosterone blockade and left ventricular dysfunction: a systematic review of randomized clinical trials. *Eur Heart J* 2009;30(4):409-72.
211. Heran BS, Musini VM, Bassett K, et al. Angiotensin receptor blockers for heart failure. *Cochrane Database Syst Rev* 2012;4:CD003040.
212. Demers C, Mody A, Teo KK. ACE inhibitors in heart failure: what more do we need to know? *Am J Cardiovasc Drugs* 2005;5(6):351-9.
213. Yusuf S, Pfeffer MA, Swedberg K, et al. CHARM Investigators and Committees. Effects of candesartan in patients with chronic heart failure and preserved left-ventricular ejection fraction: the CHARM-Preserved Trial. *Lancet* 2003;362:777-781.
214. Cleland JG, Tendera M, Adamus J, et al. PEP-CHF Investigators. The perindopril in elderly people with chronic heart failure (PEP-CHF) study. *Eur Heart J* 2006; 27: 2338–2345.
215. Massie BM, Carson PE, McMurray JJ, et al. I-PRESERVE Investigators. Irbesartan in patients with heart failure and preserved ejection fraction. *N Engl J Med* 2008;359: 2456–2467.

216. Komajda M, Lutiger B, Madeira H, et al. Tolerability of carvedilol and ACE-Inhibition in mild heart failure. Results of CARMEN (Carvedilol ACE-Inhibitor Remodelling Mild CHF EvaluationN). *Eur J Heart Fail* 2004;6(4):467-75.
217. Willenheimer R, van Veldhuisen DJ, Ponikowski P, et al. Beta-blocker treatment before angiotensin-converting enzyme inhibitor therapy in newly diagnosed heart failure. *J Am Coll Cardiol* 2005;46(1):182.
218. Yip GW, Wang M, Wang T, et al. The Hong Kong diastolic heart failure study: a randomised controlled trial of diuretics, irbesartan and ramipril on quality of life, exercise capacity, left ventricular global and regional function in heart failure with a normal ejection fraction. *Heart* 2008;94(5):573-80.
219. Pitt B, Pfeffer MA, Assmann SF, et al. Spironolactone for heart failure with preserved ejection fraction. *N Engl J Med* 2014;370:1383-1392.
220. Al-Mallah MH, Tleyjeh IM, Abd-Latif AA, et al. Angiotensin-converting enzyme inhibitors in coronary artery disease and preserved left ventricular systolic function: a systematic review and meta-analysis of randomized controlled trials. *J Am Coll Cardiol* 2006;47(8):1575-83.
221. Lund LH, Benson L, Dahlström U, et al. Association between use of renin-angiotensin system antagonists and mortality in patients with heart failure and preserved ejection fraction. *JAMA* 2012;308:2108-2117.
222. Kannel WB, Wolf PA, Benjamin EJ, Levy D. Prevalence, incidence, prognosis, and predisposing conditions for atrial fibrillation: population-based estimates. *Am J Cardiol* 1998;82:2N-9N.
223. Conen D, Tedrow UB, Koplan BA, et al. Influence of systolic and diastolic blood pressure on the risk of incident atrial fibrillation in women. *Circulation* 2009;119(16):2146-52.
224. Lim CW, Kasim S, Ismail JR, et al. Prevalence of atrial fibrillation in the Malaysian communities. *Heart Asia* 2016;8(2):62-66.
225. Krahn AD, Manfreda J, Tate RB, et al. The natural history of atrial fibrillation: incidence, risk factors, and prognosis in the Manitoba Follow-Up Study. *Am J Med* 1995;98(5):476-84.
226. Stewart S, Hart CL, Hole DJ, McMurray JJ. A population-based study of the long-term risks associated with atrial fibrillation: 20-year follow-up of the Renfrew/Paisley study. *Am J Med* 2002;113(5):359-64.
227. Okin PM, Wachtell K, Devereux RB, et al. Regression of electrocardiographic left ventricular hypertrophy and decreased incidence of new-onset atrial fibrillation in patients with hypertension. *JAMA* 2006;296(10):1242-8.
228. Rienstra M, Van Veldhuisen DJ, Harry Crijns JGM, et al. for the RACE investigators. Enhanced cardiovascular morbidity and mortality during rhythm control treatment in persistent atrial fibrillation in hypertensives: data of the RACE study. *Eur Heart J* 2007;28(6):741-751.
229. Lip GY, Frison L, Grind M; SPORTIF Investigators. Effect of hypertension on anticoagulated patients with atrial fibrillation. *Eur Heart J* 2007;28(6):752-9.
230. Albertsen IA, Rasmussen LH, Overvad TF, et al. Risk of stroke or systemic embolism in atrial fibrillation patients treated with warfarin. A systematic review and meta-analysis. *Stroke* 2013;44:1329-1336.

231. López-López JA, Sterne JAC, Thom HHZ, et al. Oral anticoagulants for prevention of stroke in atrial fibrillation: systematic review, network meta-analysis, and cost effectiveness analysis. *BMJ* 2017;359:j5058.
232. Fogari R, Mugellini A, Destro M, et al. Losartan and prevention of atrial recurrence in hypertensive patients. *J Cardiovasc Pharmacol* 2006;47:46-50.
233. Madrid AH, Bueno MG, Rebollo JMG, et al. Use of irbesartan to maintain sinus rhythm in patients with long-lasting persistent atrial fibrillation. *Circulation* 2002;106:331-6.
234. Ducharme A, Swedberg K, Pfeffer MA, et al. Prevention of atrial fibrillation in patients with symptomatic chronic heart failure by candesartan in the Candesartan in heart failure: Assessment of Reduction in Mortality and morbidity (CHARM) program. *Am Heart J* 2006;151:985-91.
235. Maggioni AP, Latini R, Carson PE, et al. Valsartan reduces the incidence of atrial fibrillation in patients with heart failure: results from the Valsartan Heart Failure Trial (Val-HeFT). *Am Heart J* 2005;149:548-57.
236. Healey JS, Baranchuk A, Crystal E, et al. Prevention of atrial fibrillation with angiotensin-converting enzyme inhibitors and angiotensin receptor blockers: a meta-analysis. *J Am Coll Cardiol* 2005;45(11):1832-9.
237. Lip GY, Frison L, Grind M. Angiotensin converting enzyme inhibitor and angiotensin receptor blockade use in relation to outcomes in anticoagulated patients with atrial fibrillation. *J Intern Med* 2007; 261 (6):577-86.
238. de Denus S, Sanoski CA, Carlsson J, et al. Rate vs rhythm control in patients with atrial fibrillation: a meta-analysis. *Arch Intern Med* 2005;165(3):258-62.
239. Badheka AO, Patel NJ, Grover PM, et al. Optimal blood pressure in patients with atrial fibrillation (from the AFFIRM Trial). *Am J Cardiol* 2014;114(5):727-36.
240. Selvin E, Erlinger T. Prevalence of risk factors for peripheral arterial disease in the United States. Result from the National Health and Nutrition Examination Survey 1999-2000. *Circulation* 2004;110:738-743.
241. Lane DA, Lip GY. Treatment of hypertension in peripheral arterial disease. *Cochrane Database Syst Rev* 2009;(4):CD003075.
242. Fowkes FG, Rudan D, Rudan I, et al. Comparison of global estimates of prevalence and risk factors for peripheral artery disease in 2000 and 2010: a systematic review and analysis. *Lancet* 2013;382(9901):1329-40.
243. Selvin E, Hirsch AT. Contemporary Risk factor control and walking dysfunction in individuals with peripheral artery disease: NHANES 1999-2004. *Atherosclerosis* 2008;201(2):425-433.
244. Clement DL. Treatment of hypertension in patients with peripheral arterial disease: an update. *Curr Hypertens Rep* 2009 Aug;11(4):271-6.
245. The Heart Outcomes Prevention Evaluation Study Investigators. Effects of an angiotensin-converting enzyme inhibitor, ramipril, on cardiovascular events in high risk patients. *N Engl J Med* 2000;342:145-153.
246. Travis M Falconer, John W Eikboom, Graeme J Hankey et al. Management of peripheral arterial disease in the elderly: focus on cilostazol. *Clin Interv Aging* 2008; 3(1):17-23.
247. Singer DR, Kite A. Management of hypertension in peripheral arterial disease: does the choice of drugs matter? *Eur J Vasc Endovasc Surg* 2008;35(6):701-8.

248. Diez J, Gonzalez A, Lopez B, et al. Effects of antihypertensive agents on the left ventricle: clinical implication. *Am J Cardiovasc Drugs* 2001;1(4):263-73.
249. Fagard RH, Celis H, Wouters S. Regression of left ventricular mass by antihypertensive treatment: a meta analysis of randomized comparative studies. *Hypertension* 2009;54(5):1084-91.
250. Levy D, Garrison RJ, Savage DD, et al. Prognostic implications of echocardiographically determined left ventricular mass in the Framingham Heart Study. *N Engl J Med* 1990;322(22):1561-6.
251. Verdecchia P, Schillaci G, Borgioni C, et al. Prognostic significance of serial changes in left ventricular mass in essential hypertension. *Circulation* 1998;97(1):48-54.
252. Klingbeil AU, Schneider M, Martus P, et al. A meta-analysis of the effects of treatment on left ventricular mass in essential hypertension. *Am J Med* 2003;115(1):41-6.
253. Verdecchia P, Staessen JA, Angeli F, et al. Usual versus tight control of systolic blood pressure in non-diabetic patients with hypertension (Cardio-Sis): an open-label randomised trial. *Lancet* 2009;374(9689):525-33.
254. Mackay J, Mensah G. The atlas of heart disease and stroke. Geneva, Switzerland: World Health Organization 2004. Available at http://www.who.int/cardiovascular_diseases/resources/atlas/en/
255. Lewington S, Clarke R, Qizilbash N, et al. Age-specific relevance of usual blood pressure to vascular mortality: a meta-analysis of individual data for one million adults in 61 prospective studies. *Lancet* 2002;360:1903-1913.
256. Lawes CM, Rodgers A, Bennett DA, et al. Asia Pacific Cohort Studies Collaboration: Blood pressure and cardiovascular diseases in the Asia Pacific region. *J Hypertens* 2003;21:707-716.
257. Asia Pacific Cohort Studies Collaboration. Blood pressure indices and cardiovascular disease in the Asia Pacific region. A pooled analysis. *Hypertension* 2003;42:69-75.
258. Nazifah SN, Azmi IK, Hamidon BB et al. National Stroke Registry: Terengganu and Seberang Jaya experience. *Med J Malaysia* 2012;67(3): 302-304.
259. Venketasubramanian N. The epidemiology of stroke in ASEAN countries – A review. *Neurol J SEA* 1998;3:9-14.
260. Chalmers J, Todd A, Chapman N, et al: International Society of Hypertension Writing Group. International Society of Hypertension Writing Group. International Society of Hypertension (ISH): Statement on Blood Pressure Lowering and Stroke Prevention. *J Hypertens* 2003;21:651–663.
261. Liu L, Wang JG, Gong L, et al. Comparison of active treatment and placebo in older Chinese patients with isolated systolic hypertension. Systolic Hypertension in China (Sys-China) Collaborative Group. *J Hypertens* 1998;16:1823-1829.
262. Staessen JA, Fagard R, Thijs L, et al. Randomised double-blind comparison of placebo and active treatment for older patients with isolated systolic hypertension. The Systolic Hypertension in Europe (Syst-Eur) Trial Investigators. *Lancet* 1997;350:757-764.
263. Liu L, Zhao Y, Liu G, et al. FEVER Study Group. The Felodipine Event Reduction (FEVER) Study: a randomized long-term placebo controlled trial in Chinese hypertensive patients. *J Hypertens* 2005;23:2157-2172.
264. MRC Working Group. Medical Research Council trial on treatment of hypertension in older adults: principal results. *BMJ* 1992, 304 (6824) : 405-412.

265. Spengos K, Tsivgoulis G, Zakopoulos N. Blood pressure management in acute stroke: a long-standing debate. *Eur Neurol* 2006;55:123–135.
266. CAST (Chinese Acute Stroke Trial) Collaborative Group: CAST: randomized placebo controlled trial of early aspirin use in 20,000 patients with acute ischaemic stroke. *Lancet* 1997;349:1641–1649.
267. International Stroke Trial Collaborative Group: The International Stroke Trial (IST): a randomised trial of aspirin, subcutaneous heparin, both, or neither among 19,435 patients with acute ischaemic stroke. *Lancet* 1997;349:1569–1581.
268. Kidwell CS, Saver JL, Mattiello J, et al. Thrombolytic reversal of acute human cerebral Ischaemic injury shown by diffusion/perfusion magnetic resonance imaging. *Ann Neurol* 2000;47:621–469.
269. Phillips SJ. Pathophysiology and management of hypertension in acute Ischaemic stroke. *Hypertension* 1994;23:131–136.
270. Adams HP Jr, Adams RJ, Brott T, et al. Stroke Council of the American Stroke Association. Guidelines for the early management of patients with Ischaemic stroke: A scientific statement from the Stroke Council of the American Stroke Association. *Stroke* 2003;34:1056–1083.
271. Broderick JP, Adams HP Jr, Barsan W, et al. Guidelines for the management of spontaneous intracerebral hemorrhage: a statement for healthcare professionals from a special writing group of the Stroke Council, American Heart Association. *Stroke* 1999;30: 905–915.
272. Anderson CS, Huang Y, Wang JG, et al. Intensive blood pressure reduction in acute cerebral haemorrhage trial (INTERACT): a randomised pilot trial. *Lancet Neurol*. 2008;7(5):391-9.
273. Anderson CS, Heeley E, Huang Y, et al. Rapid blood-pressure lowering in patients with acute intracerebral hemorrhage. *N Engl J Med* 2013; 368:2355–2365.
274. Sakamoto Y, Koga M, Yamagami H, et al. Systolic blood pressure after intravenous antihypertensive treatment and clinical outcomes in hyperacute intracerebral hemorrhage: the stroke acute management with urgent risk-factor assessment and improvement intracerebral hemorrhagic study. *Stroke* 2013 July;44(70):1846-51.
275. Anderson CS, Heeley E, Huang Y, et al. Rapid blood-pressure lowering in patients with acute intracerebral haemorrhage. *N Engl J Med* 2013 Jun 20;368(25): 2355-65.
276. Adams H, Adams R, Del Zoppo G, Goldstein LB. Stroke Council of the American Heart Association; American Stroke Association (2005). Guidelines for the early management of patients with Ischaemic stroke: 2005 guidelines update a scientific statement from the Stroke Council of the American Heart Association / American Stroke Association. *Stroke* 2005;36:916–923.
277. Warlow C, Sudlow C, Dennis M, et al. Stroke. *Lancet* 2003;362:1211-1224.
278. Kein R, Steinke W, Daffertshofer M, et al. Stroke recurrence in patients with symptomatic vs asymptomatic middle cerebral artery disease. *Neurology* 2005;65(6):859-64.
279. Wolfe CD, Crichton SL, Heuschmann PU, et al. Estimates of outcomes up to ten years after stroke: analysis from prospective South London Stroke Register. *PLoS Med* 2011;8(5):E1001033.
280. Reboli G, Angeli F, de Simone G, et al. Tight versus standard blood pressure control in patients with hypertension with and without cardiovascular disease. *Hypertension* 2014;63:475-482.

281. Staessen JA, Li Y, Thijs L, Wang JG. Blood pressure reduction and cardiovascular prevention: an update including the 2003–2004 secondary prevention trials. *Hypertens Res* 2005; 28:385–407.
282. Lawes CM, Bennett DA, Feigin VL, Rodgers A. Blood pressure and stroke: an overview of published reviews. *Stroke* 2004;35:776–785.
283. Schrader J, Luders S, Kulschewski A, et al. Morbidity and mortality after stroke, Eprosartan compared with Nitrendipine for secondary prevention. Principal results of a prospective randomized controlled study (MOSES). *Stroke* 2005;36:1218–1226.
284. Yusuf S, Diener HC, Sacco RL, et al. ProFESS Study Group. Telmisartan to Prevent Recurrent Stroke and Cardiovascular Events. *N Engl J Med* 2008; 359:1225–1237.
285. Sandset EC, Bath MWP, Boysen G, Jatuzis D. SCAST Study Group. The angiotensin-receptor blocker candesartan for treatment of acute stroke (SCAST): a randomised, placebo-controlled, double-blind trial. *Lancet* 2011;377(9767):741–750.
286. Reboli G, Angeli F, Cavallini C, et al. Comparison between angiotensin-converting enzyme inhibitors and angiotensin receptor blockers on the risk of myocardial infarction, stroke and death: a meta-analysis. *J Hypertens* 2008;26(7):1282–1289.
287. SPS3 Study Group. Benavente OR, Coffy CS, Conwif R, et al. Blood-pressure targets in patients with recent lacunar stroke: the SPS3 randomised trial. *Lancet* 2013;382(9891):507–15.
288. Ministry of Health Malaysia. National Health and Morbidity Survey 2015, Volume 2. Non-communicable diseases & other health problems. (<http://www.iku.gov.my/index.php/research-eng/list-of-research-eng/iku-eng/nhms-eng/nhms-2015>).
289. Department of Statistics Official Portal, <https://www.dosm.gov.my>, Population Projection (Revised) 2010–2040.
290. Vokonas PS, Kannel WB, Cupples LA. Epidemiology and risk of hypertension in the elderly: the Framingham Study. *J Hypertens Suppl* 1988;6(1):S3–9.
291. Amery A, Birkenhäger W, Brixko P, et al. Mortality and morbidity results from the European Working Party on High Blood Pressure in the Elderly trial. *Lancet* 1985;1(8442):1349–54.
292. Dahlöf B, Lindholm LH, Hansson L, et al. Morbidity and mortality in the Swedish Trial in Old Patients with Hypertension (STOP-Hypertension). *Lancet* 1991;338(8778):1281–5.
293. Tuomilehto J. Medical Research Council trial of treatment of hypertension in older adults: principal results. MRC Working Party. *BMJ* 1992;304:405–412.
294. Hansson L, Julius S, Carruthers SG, et al. Effects of intensive blood pressure lowering and low dose aspirin in patients with hypertension: principal results of the Hypertension Optimal Treatment (HOT) randomised trial. *Lancet* 1998;351:1755–62.
295. Lithell H, Hansson L, Skoog I, et al. SCOPE Study Group. The Study on Cognition and Prognosis in the Elderly (SCOPE): principal results of a randomized double-blind intervention trial. *J Hypertens* 2003;21(5):875–86.
296. Qaseem A, Wilt TJ, Rich R, et al. Clinical Guidelines Committee of the American College of Physicians and the Commission on Health of the Public and Science of the American Academy of Family Physicians. Pharmacologic treatment of hypertension in adults aged 60 years or older to higher versus lower blood pressure targets: a clinical practice guideline from the American College of Physicians and the American Academy of Family Physicians. *Ann Intern Med* 2017;166(6):430–437.

297. Weiss J, Freeman M, Low A, et al. Benefits and harms of intensive blood pressure treatment in adults aged 60 years or older: a systematic review and meta-analysis. *Ann Intern Med* 2017; 166(6):419-429.
298. Barnett K, Mercer SW, Norbury M, et al. Epidemiology of multimorbidity and implications for health care, research, and medical education: a cross-sectional study. *Lancet* 2012; 380(9836):37-43.
299. Fortin M, Bravo G, Hudon C, et al. Prevalence of multimorbidity among adults seen in family practice. *Ann Fam Med* 2005;3(3):223-8.
300. Onder G, Landi F, Fusco D, et al. Recommendations to prescribe in complex older adults: results of the CRITERIA to assess appropriate Medication use among Elderly complex patients (CRIME) project. *Drugs Aging* 2014;31(1):33-45.
301. Odden MC, Peralta CA, Haan MN, Covinsky KE. Rethinking the association of high blood pressure with mortality in elderly adults: the impact of frailty. *Arch Intern Med* 2012;172(15):1162-8.
302. Sabayan B, van Vliet P, de Ruijter W, et al. High blood pressure, physical and cognitive function, and risk of stroke in the oldest old: the Leiden 85-plus Study. *Stroke* 2013;44(1):15-20.
303. Ogliari G, Westendorp RG, Muller M, et al. Blood pressure and 10-year mortality risk in the Milan Geriatrics 75+ Cohort Study: role of functional and cognitive status. *Age Ageing* 2015;44(6):932-7.
304. Windham BG, Griswold ME, Lrette S, et al. Effects of age and functional status on the relationship of systolic blood pressure with mortality in mid and late life: the ARIC study. *J Gerontol A Biol Sci Med Sci* 2017;72(1):89-94.
305. Weidung B, Littbrand H, Nordström P, et al. The association between SBP and mortality risk differs with level of cognitive function in very old individuals. *J Hypertens* 2016; 34(4): 745-752.
306. Stessman J, Bursztyrn M, Gershinsky Y, et al. Hypertension and its treatment at age 90 years: Is there an association with 5-year mortality? *J Am Med Dir Assoc* 2017;18(3):277.e13-277.e19.
307. Wu C, Smit E, Peralta CA, et al. Functional status modifies the association of blood pressure with death in elders: health and retirement study. *J Am Geriatr Soc* 2017;65(7):1482-1489.
308. Messerli FH, Mancia G, Conti CR, et al. Dogma disputed: can aggressively lowering blood pressure in hypertensive patients with coronary artery disease be dangerous? *Ann Intern Med* 2006;144(12):884-93.
309. Banach M, Aronow WS. Blood pressure j-curve: current concepts. *Curr Hypertens Rep* 2012;14(6): 556-566.
310. Okin PM, Hille DA, Kjeldsen SE, et al. Impact of lower achieved blood pressure on outcomes in hypertensive patients. *J Hypertens* 2012;30(4):802-10.
311. Mancia G, Kjeldsen SE, Zappe DH, et al. Cardiovascular outcomes at different on-treatment blood pressures in the hypertensive patients of the VALUE trial. *Eur Heart J* 2016;37(12):955-64.
312. Kang YY, Wang JG. The j-curve phenomenon in hypertension. *Pulse (Basel)* 2016;4(1):49-60.

313. Yano Y, Rakugi H, Bakris GL, et al. On-treatment blood pressure and cardiovascular outcomes in older adults with isolated systolic hypertension. *Hypertension* 2017;69(2):220-227.
314. Böhm M, Schumacher H, Teo KK, et al. Achieved blood pressure and cardiovascular outcomes in high-risk patients: results from ONTARGET and TRANSCEND trials. *Lancet* 2017;389(10085):2226-2237.
315. Mukete BN, Ferdinand KC. Polypharmacy in older adults with hypertension: a comprehensive review. *J Clin Hypertens (Greenwich)* 2016;18(1):10-8.
316. Todd A, Husband A, Andrew I, et al. Inappropriate prescribing of preventative medication in patients with life-limiting illness: a systematic review. *BMJ Support Palliat Care*. 2017;7(2):113-121.
317. Freeman R, Wieling W, Axelrod FB, et al. Consensus statement on the definition of orthostatic hypotension, neurally mediated syncope and the postural tachycardia syndrome. *Clin Auton Res* 2011;21(2):69-72.
318. Rutan GH, Hermanson B, Bild DE, et al. Orthostatic hypotension in older adults. The Cardiovascular Health Study. CHS Collaborative Research Group. *Hypertension* 1992;19(6 Pt 1):508-19.
319. Kamaruzzaman S, Watt H, Carson C, Ebrahim S. The association between orthostatic hypotension and medication use in the British Women's Heart and Health Study. *Age Ageing* 2010;39(1):51-6.
320. Gangavati A, Hajjar I, Quach L, Jones RN, et al. Hypertension, orthostatic hypotension, and the risk of falls in a community-dwelling elderly population: the maintenance of balance, independent living, intellect, and zest in the elderly of Boston study. *J Am Geriatr Soc* 2011;59(3):383-9.
321. Valbusa F, Labat C, Salvi P, et al. PARTAGE investigators. Orthostatic hypotension in very old individuals living in nursing homes: the PARTAGE study. *J Hypertens* 2012;30(1):53-60.
322. Zia A, Kamaruzzaman SB, Tan MP. Blood pressure lowering therapy in older people: Does it really cause postural hypotension or falls? *Postgrad Med* 2015;127(2):186-93.
323. Heitterachi E, Lord SR, Meyerkort P, et al. Blood pressure changes on upright tilting predict falls in older people. *Age Ageing* 2002;31(3):181-6.
324. Ooi WL, Hossain M, Lipsitz LA. The association between orthostatic hypotension and recurrent falls in nursing home residents. *Am J Med* 2000;108(2):106-11.
325. Kenelly SP, Lawlor BA, Kenny RA. Blood Pressure and Dementia – a Comprehensive Review. *Ther Adv Neurol Disord* 2009; 2(4): 241–260.
326. Iadecola C, Yaffe K, Biller J, et al. American Heart Association Council on hypertension; Council on clinical cardiology; Council on cardiovascular disease in the young; Council on cardiovascular and stroke nursing; Council on quality of care and outcomes research; and stroke council. Impact of hypertension on cognitive function: a scientific statement from the American Heart Association. *Hypertension* 2016;68(6):e67-e94.
327. Prince MJ, Bird AS, Blizard RA, Mann AH. Is the cognitive function of older patients affected by antihypertensive treatment? Results from 54 months of the Medical Research Council's trial of hypertension in older adults. *BMJ* 1996;312(7034):801-5.
328. Forette F, Seux ML, Staessen JA, Thijs L, et al. Prevention of dementia in randomised double-blind placebo-controlled Systolic Hypertension in Europe (Syst-Eur) trial. *Lancet* 1998;352(9137):1347-51.

329. Peters R, Beckett N, Forette F, et al. HYVET investigators. Incident dementia and blood pressure lowering in the Hypertension in the Very Elderly Trial cognitive function assessment (HYVET-COG): a double-blind, placebo controlled trial. *Lancet Neurol* 2008;7(8):683-9.
330. Mossello E, Pieraccioni M, Nesti N, et al. Effects of low blood pressure in cognitively impaired elderly patients treated with antihypertensive drugs. *JAMA Intern Med* 2015;175(4):578-85.
331. Berrut G, Andrieu S, et al. Promoting access to innovation for frail old persons. IAGG (International Association of Gerontology and Geriatrics), WHO (World Health Organization) and SFGG (Société Française de Gériatrie et de Gériatologie) Workshop-Athens January 20-21, 2012. *J Nutr Health Aging* 2013;17(8):688-93.
332. Poudel A, Hubbard RE, Nissen L, Mitchell C. Frailty: a key indicator to minimize inappropriate medication in older people. *QJM* 2013;106(10):969-75.
333. Charlesworth CJ, Peralta CA, Odden MC. Functional status and antihypertensive therapy in older adults: a new perspective on old data. *Am J Hypertens* 2016;29(6):690-5.
334. Fried LP, Tangen CM, Walston J, et al. Cardiovascular Health Study Collaborative Research Group. Frailty in older adults: evidence for a phenotype. *J Gerontol A Biol Sci Med Sci* 2001;56(3):M146-56.
335. Potter K, Flicker L, Page A, Etherton-Beer C. Deprescribing in Frail Older People: A Randomised Controlled Trial. *PLoS One* 2016;11(3):e0149984.
336. Whelton PK, Appel LJ, Espeland MA, et al. Sodium restriction and weight loss in the treatment of hypertension in older persons: A randomised controlled trial of non-pharmacologic interventions in the elderly (TONE). *JAMA* 1998;279(11):839-846.
337. Tranquilli AL, Dekker G, Magee L, et al. The classification, diagnosis and management of the hypertensive disorders of pregnancy: A revised statement from the ISSHP. *Pregnancy Hypertens* 2014;4(2):97-104.
338. Report of the National High Blood Pressure Education Program Working Group on high blood pressure in pregnancy. *Am J Obstet Gynecol* 2000;183:S1-22.
339. Brown MA, Roberts LA, Mackenzie C, et al. A prospective randomized study of automated versus mercury blood pressure recordings in hypertensive pregnancy (PRAM Study). *Hypertens Pregnancy* 2012;31:1:107-119.
340. World Health Organization. Hypertension control: Report of a WHO expert committee. *WHO Technical Report Series* 1996;862:1-83
341. Davey DA. Hypertensive disorders of pregnancy. In Dewhurst's Textbook of Obstetrics and Gynaecology for Postgraduates Whitfield CR (editor), 5th edition, 1995; 175-227.
342. Boubouchairopoulou N, Karpettas N, Athanasakis K, et al. Cost estimation of hypertension management based on home blood pressure monitoring alone or combined office and ambulatory blood pressure measurements. *J Am Soc Hypertens* 2014;8(10):732-738.
343. Brown MA, Mangos G, Davis G, et al. The natural history of white coat hypertension during pregnancy. *BJOG* 2005;112(5):601-606
344. Brown MA. Is there a role for ambulatory blood pressure monitoring in pregnancy? *Clin Exp Pharmacol Physiol* 2014;41(1):16-21.

345. Montan S. Drugs used in hypertensive diseases in pregnancy. *Curr Opin Obstet Gynecol* 2004;16:111-115.
346. The Royal College of Obstetricians and Gynaecologists & Royal College of Midwives. Hypertension in pregnancy: the management of hypertensive disorders during pregnancy. *NICE Clinical Guideline* 2011.
347. Bullo M, Tschumi S, Bucher BS, et al. Pregnancy outcome following exposure to angiotensin-converting enzyme inhibitors or angiotensin receptor antagonists: a systematic review. *Hypertension* 2012;60:444-450.
348. Magee LA, von Dadelszen P, Rey E, et al. Less tight versus tight control of hypertension in pregnancy. *N Engl J Med* 2015;372(5):407-417.
349. National Collaborating Centre for Women's and Children's Health (UK). Hypertension in pregnancy: The management of hypertensive disorders during pregnancy. RCOG Press: London, 2011;1194
350. Magee LA, Helewa M, Moutquin JM, et al. Diagnosis, evaluation and management of the hypertensive disorders of pregnancy. *J Obstet Gyneacol Canada* 2008; 30(3) Suppl.1:S1-48.
351. Hermida RC, Ayala DE, Fernandez JR, et al. Administration time-dependent effects of aspirin in women at differing risk for preeclampsia. *Hypertension* 1999;34:1016-1023.
352. Park F, Russo K, Williams P, et al. Prediction and prevention of early-onset pre-eclampsia: impact of aspirin after first-trimester screening. *Ultrasound Obstet Gynecol* 2015;46(4):419-423.
353. Leitch H, Egarter C, Husslein P, et al. A meta-analysis of low dose aspirin for the prevention of intrauterine growth retardation. *BJOG* 1997;104:450-459.
354. Hofmeyr GJ, Belizan JM, von Dadelszen P; Calcium and Pre-eclampsia (CAP) Study Group. Low-dose calcium supplementation for preventing pre-eclampsia: a systematic review and commentary. *BJOG* 2014;121(8): 951-957.
355. Purswani JM, Gala P, Dwarkanath P, et al. The role of vitamin D in preeclampsia: a systematic review. *BMC Pregnancy and Childbirth* 2017;17(1):231.
356. Makrides M, Duley L, Olsen SF. Marine oil, and other prostaglandin precursor, supplementation for pregnancy uncomplicated by pre-eclampsia or intrauterine growth restriction. *Cochrane Database Syst Rev* 2006;3:CD003402.
357. Meher S, Duley L. Garlic for preventing pre-eclampsia and its complications. *Cochrane Database Syst Rev* 2006;3:CD006065
358. Thaver D, Saeed MA, Bhutta ZA. Pyridoxine (vitamin B6) supplementation in pregnancy. *Cochrane Database Syst Rev* 2006;3:CD000179.
359. Poston L, Briley AL, Seed PT, et al. Vitamin C and vitamin E in pregnant women at risk for pre-eclampsia (VIP trial): randomised placebo-controlled trial. *Lancet* 2006;367(9517):1145-54.
360. Verloren S, Galindo A, Schlembach D, et al. An automated method for the determination of the sFlt-1/PIGF ratio in the assessment of preeclampsia. *Am J Obstet Gynecol* 2010;202(2):161.e1-.e11.
361. Bateman BT, Huybrechts KF, Fischer MA, et al. Chronic hypertension in pregnancy and the risk of congenital malformations: a cohort study. *Am J Obstet Gynecol* 2015;212(3):337 e1-14.

362. McCaw-Binns AM, Ashley DE, Knight LP, et al. Strategies to prevent eclampsia in a developing country: I Reorganization of maternity services. *Int J Obstet Gynecol* 2004;87:286-294.
363. MacGillivray I, McCaw-Binns AM, Ashley DE, et al. Strategies to prevent eclampsia in a developing country: II. Use of a maternal pictorial card. *Int J Obstet Gynecol* 2004; 87(3):295-300.
364. American college of Obstetricians and Gynecologists; Task force on Hypertension in pregnancy. Hypertension in pregnancy. Report of the American College of Obstetricians and Gynecologists' task force on hypertension in pregnancy. *Obstet Gynecol* 2013;122(5):1122-1131.
365. Magee LA, Cham C, Waterman EJ, et al. Hydralazine for treatment of severe hypertension in pregnancy: a meta-analysis. *BMJ* 2003;327(7421):955-960.
366. Brown MA, Buddle ML, Farrel T, et al. Efficacy and safety of nifedipine tablets for the acute treatment of severe hypertension in pregnancy. *Am J Obstet Gynecol* 2002;187:1046-1050.
367. The Eclampsia Trial Collaborative Group. Which anticonvulsant for women with eclampsia? Evidence from the Collaborative Eclampsia Trial. *Lancet* 1995;345:1455-1463.
368. The American College of Obstetricians and Gynecologists Committee on Obstetric Practice. Committee opinion No. 692: Emergent therapy for acute-onset, severe hypertension during pregnancy and the postpartum period. *Obstet Gynecol* 2017;129:e90-e95.
369. The Magpie Trial Collaborative Group. Do women with pre-eclampsia, and their babies, benefit from magnesium sulphate? The Magpie Trial: a randomised placebo-controlled trial. *Lancet* 2002;359:1877- 1890.
370. Doyle LW, Crowther CA, Middleton P, et al. Magnesium sulphate for women at risk of preterm birth for neuroprotection of the fetus. *Cochrane Database Syst Rev* 2009;1:CD004661
371. Ferrazanni S, De Carolis S, Pomini F, et al. The duration of hypertension in the puerperium of preeclamptic women: relationship with renal impairment and week of delivery. *Am J Obstet Gynecol* 1994;171(2): 506- 512.
372. Matthys LA, Coppage KH, Lambers DS, et al. Delayed postpartum preeclampsia: an experience of 151 cases. *Am J Obstet Gynecol* 2004;190:1464-1466.
373. Royal College of Obstetricians and Gynaecologists. The management of severe pre-eclampsia/eclampsia. RCOG Guideline No.10(A) March 2006.
374. Lykke JA, Langhoff-Roos J, Sibai BM, et al. Hypertensive pregnancy disorders and subsequent cardiovascular morbidity and type 2 diabetes mellitus in the mother. *Hypertension* 2009;53(6):944-951.
375. Goldenberg RL, McClure EM, MacGuire ER, et al. Lessons for low-income regions following the reduction in hypertension-related maternal mortality in high-income countries. *Int J Gynecol Obstet* 2011;113:91-95.
376. Frishman W, Schlocker SJ, Awad K, Tejani N. Pathophysiology and medical management of systemic hypertension in pregnancy. *Cardiol Rev* 2005;13:274-284.
377. Reproductive Health Supplies Coalition. Magnesium sulphate. Product Brief: Caucus on New and Underused Reproductive Health Technologies 2012.

378. Duley L, Henderson-Smart DJ, Walker GJ, et al. Magnesium sulphate versus diazepam infusion in eclampsia. *Cochrane Database Syst Rev* 2010 Dec;12:CD000127.
379. Boldo A, White WB. Blood pressure effects of the oral contraceptive and postmenopausal hormone therapies. *Endocrinol Metab Clin North Am* 2011;40:419-432.
380. Nichols M, Robinson G, Bounds W, et al. Effect of four combined oral contraceptives on blood pressure in the pill-free interval. *Contraception* 1993;47:367-376.
381. Williams B, Poulter NR, Brown MJ, et al. Guidelines for management of hypertension: report of the fourth working party of the British Hypertension Society, 2004-BHS IV. *J Hum Hypertens* 2004;18:139-85
382. World Health Organisation. Medical eligibility criteria for contraceptive use. 4th edition. Geneva: WHO 2009.
383. Oelkers W, Foidart JM, Dombrovicz N, et al. Effects of a new oral contraceptive containing an anti-mineralocorticoid progestogen, drospirenone, on the renin-aldosterone system, body weight, blood pressure, glucose tolerance, and lipid metabolism. *J Clin Endocrinol Metab* 1995;80:1816-1821.
384. Hulley S, Grady D, Bush T, et al. Randomised trial of estrogen plus progestin for secondary prevention of coronary heart disease in post menopausal women. Heart and Estrogen/progestin Replacement Study (HERS) Research Group. *JAMA* 1998;280(7):605-613.
385. Anderson GL, Limacher M, Assaf AR, et al. Effects of conjugated equine estrogen in postmenopausal women with hysterectomy: The Women's Health Initiative randomized controlled trial. *JAMA* 2004;291(14):1701-1712.
386. White WB, Hanes V, Chauhan V, Pitt B. Effects of a new hormone therapy, drospirenone and 17-beta-estradiol, in post-menopausal women with hypertension. *Hypertension* 2006;48: 246-253.
387. Seliem WA, Falk MC, Shadbolt B, et al. Antenatal and postnatal risk factors for neonatal hypertension and infant follow-up. *Pediatr Nephrol* 2007; 22(12):2081-7
388. Barrington KJ. Umbilical artery catheters in the newborn: effects of catheter materials. *Cochrance Database of Sys Rev* 1999;1:CD000949
389. Dionne JM, Abitbol CL, Flynn JT. Hypertension in infancy: diagnosis, management and outcome. *Pediatr Nephrol* 2012;27:17-32
390. Nwankwo MU, Lorenz JM, Gardiner JC. A standard protocol for blood pressure measurements in the newborn. *Pediatrics* 1997;99:E10
391. Tack ED, Perlman JM. Renal failure in sick hypertensive premature infants receiving captopril therapy. *J Pediatr* 1988;112:805-810
392. Guron G, Friberg P. An intact renin-angiotensin system is a prerequisite for normal renal development. *J Hypertens* 2000;18:123-137
393. Sreeramareddy CT, Chew WF, Poulsaeman V, et al. Blood pressure and its associated factors among primary school children in suburban Selangor, Malaysia: A cross-sectional survey. *J Family Community Med* 2013; 20(2):90-97
394. Sorof JM, Lai D, Turner J, et al. Overweight ethnicity and the prevalence of hypertension in school-aged children. *Pediatrics* 2004;113:475-82
395. Jago R, Harrell JS, McMurray RG, et al. Prevalence of abnormal lipid and blood pressure values among an ethnically diverse population of eighth grade adolescents and screening implications. *Pediatrics* 2006;117:2065-73

396. Institute for Public Health (IPH), National Institute of Health Malaysia 2016. National Health and Morbidity Survey (NHMS) 2016: Maternal and child health. Vol. II: Findings, 2016:272
397. Institute for Public Health (IPH) 2011. National Health and Morbidity Survey 2011. Vol.II Non-communicable disease; 2011:98
398. Flynn JT, Kaelber DC, Baker-Smith CM, et al. Clinical practice guideline for screening and management of high blood pressure in children and adolescents. *Pediatrics* 2017;140(3) pii:e20171904
399. Stergiou GS, Boubouchairopoulou N, Kollias A. Accuracy of automated blood pressure measurement in children: Evidence, issue, and perspectives. *Hypertension* 2017;69:1000-1006
400. Flynn JT, Alderman MH. Characteristics of children with primary hypertension seen at a referral center. *Pediatr Nephrol* 2005;20:961-966
401. Flynn JT, Daniels SR, Hayman LL, et al. Update: Ambulatory blood pressure monitoring in children and adolescents: A scientific statement from the American Heart Association. *Hypertension* 2014;63:1116-1135
402. Sun SS, Grave CD, Siervogel RM, et al. Systolic blood pressure in childhood predicts hypertension and metabolic syndrome later in life. *Pediatrics* 2007;119:237-246
403. Juhola J, Magnussen CG, Viikari JS, et al. Tracking of serum lipid levels, blood pressure, and body mass index from childhood to adulthood: The cardiovascular risk in Young Finns Study. *J Pediatr* 2011;159(4):584-590
404. Chen X, Wang Y. Tracking of blood pressure from childhood to adulthood: A systematic review and meta-regression analysis. *Circulation* 2008;117(25):3171-3180
405. Lexi-Comp, Inc. (Lexi-Drugs®). Lexi-Comp, Inc; Dec 9, 2017.
406. W.A. Wan Azman, K.H. Sim. (Eds). Annual Report of the NCVD-ACS Registry, Year 2011-2013. Kuala Lumpur, Malaysia: National Cardiovascular Disease Database, 2015.
407. Lam CS, Teng TK, Tay WT, et al. Regional and ethnic differences among patients with heart failure in Asia: The Asian sudden cardiac death in heart failure registry. *Eur Heart J* 2016;37(41): 3141-3153
408. 23rd Report of the Malaysian Dialysis and Transplant Registry 2015 (Edited by Wong HS and Goh BL) Accessed 7th January 2018.
409. Alefan Q, Ibrahim M, Razak TA, et al. Cost-effectiveness of antihypertensive treatment in Malaysia. *Malaysian Journal of Pharmaceutical Sciences*. 2009;7(2):137-152.
410. Azimatun Noor A, Amrizal MN, T Weng Kang et al. Cost analysis of Hypertension management in an urban primary medical centre Kuala Lumpur. *Malaysian Journal of Public Health Medicine* 2014;14(3):18-23.
411. Health IMS health-data assessed on September 2017. Subscribed data available from IMS Health Malaysia Sdn. Bhd.
412. Mustapha FI, Azmi S, Abdul Manaf MR, Zanariah Hussein Z, et al. What are the direct medical costs of managing type 2 diabetes mellitus in Malaysia? *Med J Malaysia* 2017;72:271-277.
413. National Renal Registry, Malaysia. 23rd Report of the Malaysian Dialysis and Transplant Registry 2015. (<https://www.msn.org.my/nrr/mdtr 2015.jsp>) Accessed Jan 2018.

414. Chanhyun Park, Guijing Wang, Jefferey M.Durthaler, Jing Fang. Cost-effectiveness analyses of antihypertensive Medicines: a systematic review. *Am J Prev Med* 2017;53(6S2):S131–S142.
415. Dehmer SP, Maciosek MV, La France AB, et al. Health benefits and cost – effectiveness of asymptomatic screening for hypertension, and high cholesterol and aspirin counselling for primary prevention. *Ann Fam Med* 2017;15:23-36.
416. Zhang D, Wang G, Joo H. A systematic review of economic evidence on community hypertension interventions. *Am J Prev Med* 2017;53(6S2):S121-130.
417. Pharmaceutical Services Division. Malaysian Statistics on Medicines 2009 & 2010.
418. World Health Organisation. A global brief on hypertension. Silent killer, global public health crisis. Geneva, Switzerland: WHO 2013. Available at http://www.int/cardiocvascular_disease/publications/global_brief_hypertension/en/
419. Rahman M, Pressel S, David BR, et al. Cardiovascular outcomes in high-risk hypertensive patients stratified by baseline glomerular filtration rate. *Ann Intern Med* 2006;144(3):172-80.
420. Roush GC, Holford TR, Guddati AK. Chlorthalidone compared with hydrochlorothiazide in reducing cardiovascular events: systematic review and network meta-analyses. *Hypertension* 2012;59(6):1110-17.
421. Misini VM, Nazer M, Bassett K, et al. Blood pressure-lowering efficacy of monotherapy with thiazide diuretics for primary hypertension. *Cochrane Database Syst Rev* 2014;5:CD003824.
422. Pottegård A, Hallas J, Olesen M, et al. Hydrochlorothiazide use is strongly associated with risk of lip cancer. *J Intern Med* 2017;282:322-331.
423. Arnsparng S, Gaist D, Johannesdottir Schmidt SA, et al. Hydrochlorothiazide use and risk of non-melanoma skin cancer: A nationwide case-control study from Denmark. *J Am Acad Dermatol* 2017 Dec 4. doi:10.1016/j.jaad.2017.11.042. [Epub ahead of print]
424. Holman RR, Paul SK, Bethel MA, et al. Long term follow-up after tight control of blood pressure in type 2 diabetes. *N Engl J Med* 2008;359:1565-76.
425. Wiliams B, Lacy PS, Thom SM, et al. Differential impact of blood pressure lowering drugs on central blood pressure and clinical outcomes: Principal results of the Conduit Artery Function Evaluation [CAFÉ] study. *Circulation* 2006;113:1213-1225
426. Conlin PR. Four-Year persistence patterns among patients initiating therapy with the angiotensin II receptor antagonist losartan versus other antihypertensive drug classes. *Clin Ther* 2001;23:1999-2010
427. Wang J-G. Chinese Hypertension Guidelines. *Pulse*. 2015;3(1):14-20.
428. Chiang CE, Wang TD, Ueng KC, et al. 2015 Guidelines of the Taiwan Society of Cardiology and the Taiwan Hypertension Society for the management of hypertension. *J Chinese Med Assoc*. 2015;78:1-4
429. Chen GJ, Yang MS. The effects of calcium channel blockers in the prevention of stroke in adults with hypertension: A meta-analysis of data from 273,543 participants in 31 randomized controlled trials. *PLoS One* 2013; 8(3): e57854.
430. Tran KC, Leung AA, Tang KL, et al. Efficacy of calcium channel blockers on major cardiovascular outcomes for the treatment of hypertension in Asian populations: A meta analysis. *Can J Cardiol* 2017;33(5): 635-643.

431. Chi C, Tai C, Bai B, et al. Angiotensin system blockade combined with calcium channel blockers is superior to other combinations in cardiovascular protection with similar blood pressure reduction: A meta-analysis in 20,451 hypertensive patients. *J Clin Hypertens (Greenwich)* 2016;18(8):801-8.
432. Lu Z, Chen Y, Li L, et al. Combination therapy of renin-angiotensin system inhibitors plus calcium channel blockers versus other two-drug combinations for hypertension: a systematic review and meta-analysis. *J Hum Hypertens* 2017;31(1):1-13.
433. Furukawa Y. Angiotensin-converting enzyme inhibitors versus receptor blockers: is one better than the other for cardiovascular prevention? *Heart* 2017;103:1310-12.
434. Ferrari R. Treatment with angiotensin-converting enzyme inhibitors: insight into perindopril cardiovascular protection. *Eur Heart J* 2008;10(suppl G):G13-G20.
435. Thomopoulos C, Parati G, Zanchetti A. Effects of blood pressure lowering on outcome incidence in hypertensives: 5 head-to-head comparisons of various classes of antihypertensive drugs – overview and meta-analyses. *J Hypertens* 2015;33(7):1321-41.
436. The CONSENSUS Trial Study Group. Effects of enalapril on mortality in severe congestive heart failure: results of the Cooperative North Scandinavian Enalapril Survey Study (CONSENSUS). *N Engl J Med* 1988;316:1429-35.
437. The SOLVD Investigators. Effects of enalapril on mortality and the development of heart failure in asymptomatic patients with reduced left ventricular ejection fraction. *N Engl J Med* 1992;327:685-91.
438. Tai C, Gan T, Zou L, et al. Effect of angiotensin-converting enzyme inhibitors and angiotensin II receptor blockers on cardiovascular events in patients with heart failure: a meta-analysis of randomized controlled trials. *BMC Cardiovasc Disord* 2017;17:257. doi:10.1186/s12872-017-0686-z.
439. The Acute Infarction Ramipril Efficacy (AIRE) study investigators. Effect of Ramipril on mortality and morbidity of survivors of acute myocardial with clinical evidence of heart failure. *Lancet* 1993;342:821-8.
440. Gruppo Italiano per lo Studio della Sopravvivenza nell'Infarto Miocardico. GISSI-3: Effects of lisinopril and transdermal glyceryl tritrate singly and together on 6-week mortality and ventricular function after acute myocardial infarction. *Lancet* 1994;343:115-22.
441. ISIS-4 (Fourth International Study of Infarct Survival) Collaborative Group. ISIS-4: A randomised factorial trial assessing early oral captopril, oral mononitrate and intravenous magnesium sulphate in 58,050 patients with suspected acute myocardial infarction. ISIS-4 Collaborative Group. *Lancet* 1995;345:669-85.
442. Ambrosioni E, Borghi C, Magnani B. The effect of the angiotensin-converting-enzyme inhibitor zofenopril on mortality and morbidity after anterior myocardial infarction. The Survival of Myocardial Infarction Long-Term Evaluation (SMILE) Study Investigators. *N Engl J Med* 1995;332:80-5.
443. Pfeffer M, Braunwald E, Moyé L, et al. Effect of captopril on mortality and morbidity in patients with left ventricular dysfunction after myocardial infarction: results of the survival and ventricular enlargement trial. *N Engl J Med* 1992;327(10):669-77.
444. Køber L, Torp-Pedersen C, Carlsen JE, et al. A clinical trial of the angiotensin-converting-enzyme inhibitor trandolapril in patients with left ventricular dysfunction after myocardial infarction. *N Engl J Med* 1995;333:1670-6.

445. Chiurchiu C, Remuzzi G, Ruggenenti P. Angiotensin converting enzyme inhibition and renal protection in nondiabetic patients: the data of the meta-analyses. *J Am Soc Nephrol* 2005;16:S58–S63
446. Lv J, Perkovic V, Foote CV, et al. Antihypertensive agents for preventing diabetic kidney disease. *Cochrane Database Syst Rev* 2012;12:CD004136.
447. Yusuf S, Teo KK, Anderson C, et al. Telmisartan Randomised AssessmeNt Study in ACE iNtolerant subjects with cardiovascular Disease (TRANSCEND) Investigators. Effects of the angiotensin-receptor blocker telmisartan on cardiovascular events in high-risk patients intolerant to angiotensin-converting enzyme inhibitors: A randomised controlled trial. *Lancet* 2008;372(9644):1174-83
448. Bangalore S, Fakheri R, Toklu B, et al. Angiotensin-converting enzyme inhibitors or angiotensin receptor blockers in patients without heart failure? Insights from 254,301 patients from randomized trials. *Mayo Clin Proc* 2016;91(1)51-60.
449. Berl T, Hunsicker LG, Lewis JB, et al. Cardiovascular outcomes in the irbesartan diabetic nephropathy trial of patients with type 2 diabetes and overt nephropathy. *Ann Intern Med* 2003;138:542-549.
450. McMurray JJV, Ostergren J, Swedberg K et al. Effects of candesartan in patients with chronic heart failure and reduced left-ventricular systolic function taking angiotensin-converting-enzyme inhibitors: the CHARM–Added trial. *Lancet* 2003;362:767-71.
451. Cohn JN, Tognori G, Valsartan Heart Failure Trial Investigator. A randomised trial of the angiotensin-receptor blocker valsartan in chronic heart failure. *N Engl J Med* 2001;345:1667-1675.
452. Lindholm LH, Ibsen H, Dahlof B, et al. Cardiovascular morbidity and mortality in patients with diabetes in the Losartan Intervention for Endpoint reduction in hypertension study (LIFE): a randomized trial against atenolol. *Lancet* 2002;359:1004-1010.
453. Pfeffer MA, McMurray JJV, Velazquez EJ, et al. for the Valsartan in Acute Myocardial Infarction Trial investigators (VALIANT). Valsartan, captopril, or both in myocardial infarction complicated by heart failure, left ventricular dysfunction, or both. *N Engl J Med* 2003;349:1893-1906.
454. Strauss MH, Hall AS. Angiotensin receptor blockers do not reduce risk of myocardial infarction, cardiovascular death, or total mortality: Further evidence for the ARB-MI paradox. *Circulation* 2017;135:2088-2090
455. Bangalore S, Kumar S, Wetterslev J, Messerli FH. Angiotensin receptor blockers and risk of myocardial infarction: meta-analyses and trial sequential analyses of 147 020 patients from randomized trials. *BMJ* 2011;342:d2234.
456. Messerli F, Bangalore S. Angiotensin receptor blockers reduce cardiovascular events, including the risk of myocardial infarction. *Circulation* 2017;135:2085-2087.
457. Yusuf S, Teo KK, Pogue J, et al. Telmisartan, ramipril, or both in patients at high risk for vascular events. *N Engl J Med* 2008;358:1547-59.
458. Granger CB, McMurray JJ, Yusuf S, et al. Effects of candesartan in patients with chronic heart failure and reduced left- ventricular systolic function intolerant to angiotensin – converting- enzyme inhibitors: the CHARM–Alternative trial. *Lancet* 2003;362:772-6
459. American Diabetes Association. Standards of medical care in diabetes - 2013. *Diabetes Care* 2013;36(1):S11-S66

460. Xu R, Sun S, Huo Y, et al. Effects of ACEIs versus ARBs on proteinuria or albuminuria in primary hypertension: a meta-analysis of randomized trials. *Medicine* 2015; 94(39): e1560
461. Gillenwater JY, Conn RL, Chrysant SG, et al. Doxazosin for the treatment of benign prostatic hyperplasia in patients with mild to moderate essential hypertension: a double-blind, placebo-controlled, dose-response multicenter study. *J Urol* 1995; 154:110-115.
462. Koch-Weser J, Graham RM, Pettinger WA. Drug therapy: Prazosin. *N Engl J Med* 1979; 300: 232-236.
463. Neaton JD, Grimm RH Jr, Prineas RJ, et al. Treatment of mild hypertension study: Final results treatment of mild hypertension study research group. *JAMA* 1993; 270:713-724.
464. Lund-Johansen P. Short- and long-term (six-year) hemodynamic effects of labetalol in essential hypertension. *Am J Med* 1983; 75:24-31.
465. Packer M, Bristow MR, Cohn JN, et al. The effect of carvedilol on morbidity and mortality in patients with chronic heart failure. U.S. Carvedilol Heart Failure Study Group. *N Engl J Med* 1996; 334:1349-1355.
466. Krum H, Roecker EB, Mohacsi P, et al. Effects of initiating carvedilol in patients with severe chronic heart failure: result from the COPERNICUS Study. *JAMA* 2003; 289:712-718.
467. Poole-Wilson PA, Swedberg K, Cleland JG, et al. Comparison of carvedilol and metoprolol on clinical outcomes in patients with chronic heart failure in the Carvedilol or Metoprolol European Trial (COMET): randomised controlled trial. *Lancet* 2003; 362:7-13.
468. Bakris GL, Fonseca V, Katholi RE, et al. Metabolic effects of carvedilol vs metoprolol in patients with type 2 diabetes and hypertension: a randomized controlled trial. *JAMA* 2004; 292:2227-2236.
469. Salvi RM. Methyl dopa. Available at <http://www.inchem.org/documents/pims/pharm/methyl do.htm>. Accessed Jan 2013.
470. Merck Manual. Clonidine. Available at <http://www.merck.com/mmpe/lexicomp/clonidine.html>. Accessed Jan 2013.
471. Cohn JN, Pfeffer MA, Rouleau J, et al. Adverse mortality effect of central sympathetic inhibition with sustained-release moxonidine in patients with heart failure (MOXCON). *Eur J Heart Fail* 2003; 5:659–67.
472. Oparil S, Schmieder RE. New approaches in the treatment of hypertension. *Circ Res* 2015;116:1074-1095.
473. Tabassum N, Ahmad F. Role of natural herbs in the treatment of hypertension. *Pharmacogn Rev*. 2011;5: 30–40.
474. Wilburn AJ, King DS, Glisson J, et al. The natural treatment of hypertension. *J Clin Hypertens (Greenwich)* 2004;6:242-8. Review published in Medscape Perspective (Available at: https://www.medscape.com/viewarticle/478882_1. Accessed 11 Jan 2018)
475. Patel CH. Yoga and bio-feedback in the management of hypertension. *Lancet* 1973; 302:1053-1055
476. Pickering T, Patel CH. Yoga and bio-feedback in hypertension. *Lancet* 1973; 302:1440-1441
477. Cheung BMY, Lo JLF, Fong DYT, et al. Randomised controlled trial of qigong in the treatment of mild essential hypertension. *Journal of Human Hypertension* 2005; 19: 697-704.

478. Xiong X, Wang P, Li X, Zhang Y. Qigong for hypertension: A systematic review. *Medicine (Baltimore)*. 2015; 94(1):1-14
479. Levine GN, Lange RA, Bairey-Merz CN, et al. Meditation and cardiovascular risk reduction: A scientific statement from the American Heart Association. *J Am Heart Assoc*. 2017;6:e002218
480. Calhoun DA, Jones D, Textor S, et al. Resistant hypertension: diagnosis, evaluation, and treatment: A scientific statement from the American Heart Association Professional Education Committee of the Council for High Blood Pressure Research. *Hypertension* 2008; 51(6):1403–19.
481. Chia YC, Ching SM. Prevalence and predictors of resistant hypertension in a primary care setting: a cross-sectional study. *BMC Fam Pract* 2014;15:131.
482. Persell SD. Prevalence of resistant hypertension in the United States, 2003-2008. *Hypertension* 2011; 57:1076–80.
483. Lazaridis AA, Sarafidis PA, Ruilope LM. Ambulatory blood pressure monitoring in the diagnosis, prognosis, and management of resistant hypertension: Still a matter of our resistance? *Curr Hypertens Rep* 2015;17(10):78. doi: 10.1007/s11906-015-0590-9
484. Akmal HA, Lau GC, Shahrul ZI, et al. The Prevalence of white-coat resistant hypertension (Wc-Rh) amongst patients referred for catheter-based renal denervation (RDN) procedure for true resistant hypertension (TRH) at the National Heart Institute of Malaysia. *J Hypertens* 2012;30(e-suppl):300. doi:10.1097/01.hjh.0000420060.44048.94.
485. Pedrosa RP, Drager LF, Gonzaga CC, et al. Obstructive sleep apnea: the most common secondary cause of hypertension associated with resistant hypertension. *Hypertens* 2011;58(5):811-7.
486. Calhoun DA, Nishizaka MK, Zaman MA, Harding SM. Aldosterone excretion among subjects with resistant hypertension and symptoms of sleep apnea. *Chest* 2004;125:112-117
487. Rossi GP, Bernini G, Caliumi C, et al. A prospective study of the prevalence of primary aldosteronism in 1,125 hypertensive patients. *J Am Coll Cardiol* 2006;48(11):2293-300
488. Douma S, Petidis K, Doumas M, et al. Prevalence of primary hyperaldosteronism in resistant hypertension: a retrospective observational study. *Lancet* 2008;371:1921-1926
489. Liu L, Xu B, Ju Y. Addition of spironolactone in patients with resistant hypertension: A meta-analysis of randomized controlled trials. *Clin Exp Hypertens* 2017;39(3):257-263.
490. Sinnott SJ, Tomlinson LA, Root AA, et al. Comparative effectiveness of fourth-line anti-hypertensive agents in resistant hypertension: A systematic review and meta-analysis. *Eur J Prev Cardiol* 2017;24(3):228-238.
491. Williams B, MacDonald TM, Moran S, et al. Spironolactone versus placebo, bisoprolol, and doxazosin to determine the optimal treatment for drug-resistant hypertension (PATHWAY-2): a randomised, double-blind, crossover trial. *Lancet* 2015; 386(10008):2059–68.
492. Acelajado MC, Pisoni R, Dudenbostel T, et al. Refractory hypertension: definition, prevalence, and patient characteristics. *J Clin Hypertens* 2012; 14(1):7–12.
493. Calhoun DA, Booth JN, Oparil S, et al. Refractory hypertension: determination of prevalence, risk factors, and comorbidities in a large, population-based cohort. *Hypertension* 2014; 63:451–8.

494. Raju NC, Eikelboom JW. The aspirin controversy in primary prevention. *Curr Opin Card* 2012;27(5):499-507.
495. Seshasai SR, Wijesuriya S, Sivakumaran R, et al. Effect of aspirin on vascular and nonvascular outcomes: meta-analysis of randomized controlled trials. *Arch Intern Med*. 2012;172(3):209-16.
496. Zanchetti A, Hansson L, Dahlof B, et al. Benefit and harm of low-dose aspirin in well-treated hypertensive at different baseline cardiovascular risk. *J Hypertens* 2002;20(11):2301-2307.
497. Lee CJ, Oh J, Lee SH, Kang SM et.al. Efficacy of aspirin and statins in primary prevention of cardiovascular mortality in uncomplicated hypertensive participants: a Korean national cohort study. *J Hypertens* 2017 May;35 Suppl 1:S33-S40. doi: 10.1097/HJH.0000000000001279.
498. Esler M, Ferrier C, Lambert G, et al. Biochemical evidence of sympathetic hyperactivity in human hypertension. *Hypertension* 1991;17: III29–III35.
499. Li P, Nader M, Arunagiri K, et al. Device-based therapy for drug-resistant hypertension: an update. *Curr Hypertens Rep* 2016;18(8):64
500. Krum H, Schlaich M, Whitbourn R, et al. Catheter-based renal sympathetic denervation for resistant hypertension: a multicentre safety and proof-of-principle cohort study. *Lancet* 2009; 373:1275–81.
501. Bhatt DL, Kandzari DE, O'Neill WW, et al. A controlled trial of renal denervation for resistant hypertension. *N Engl J Med* 2014; 370:1393-1401.
502. Townsend RR, Mahfoud F, Kandzari DE, et al. Catheter-based renal denervation in patients with uncontrolled hypertension in the absence of antihypertensive medications (SPYRAL HTN-OFF MED): A randomised, sham-controlled, proof-of-concept trial. *Lancet* 2017; 390: 2160–70.
503. Mahfoud F, Schmieder RE, Azizi M, et al. Proceedings from the 2nd European Clinical Consensus Conference for device-based therapies for hypertension: state of the art and considerations for the future. *Eur Heart J* 2017; 38:3272–3281.
504. Lohmeier TE, Irwin ED, Rossing MA, Serdar DJ, Kieval RS. Prolonged activation of the baroreflex produces sustained hypotension. *Hypertension* 2004; 43: 306 –311.
505. Scheffers IJ, Kroon AA, Schmidli J, et al. Novel baroreflex activation therapy in resistant hypertension: results of European multi-center feasibility study. *J Am Coll Cardiol* 2010; 56:1254-58.
506. Jordan J. Device-based approaches for the treatment of arterial hypertension. *Curr Hypertens Rep* 2017; 19:59.

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